

Chapter 9

Inverse Trigonometric Functions

Solutions

SECTION - A

Objective Type Questions (one option is correct)

[Principal Value and Graphical Representation]

1. The principal value of $\sin^{-1}\left(\frac{1}{2}\right)$ is

(1) $\frac{\pi}{6}$

(2) $\frac{5\pi}{6}$

(3) $-\frac{\pi}{6}$

(4) Both (1) & (2)

Sol. Answer (1)

$$\sin^{-1}\left(\frac{1}{2}\right) = \theta$$

$$\Rightarrow \sin \theta = \frac{1}{2}$$

$$\Rightarrow \theta = \frac{\pi}{6}, \text{ as } \theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

2. The principal value of $\cos^{-1}\left(-\frac{1}{2}\right)$ is

(1) $\frac{\pi}{6}$

(2) $\frac{\pi}{3}$

(3) $\frac{2\pi}{3}$

(4) $\frac{5\pi}{6}$

Sol. Answer (3)

$$\cos^{-1}\left(-\frac{1}{2}\right) = \theta, \theta \in [0, \pi]$$

$$\Rightarrow \cos \theta = -\frac{1}{2}$$

$$\Rightarrow \theta = \frac{2\pi}{3}$$

3. The principal value of $\cot^{-1}(-\sqrt{3})$ is

(1) $-\frac{\pi}{3}$

(2) $-\frac{\pi}{6}$

(3) $\frac{2\pi}{3}$

(4) $\frac{5\pi}{6}$

Sol. Answer (4)

$$\cot^{-1}(-\sqrt{3}) = \theta, \theta \in (0, \pi)$$

$$\cot \theta = -\sqrt{3}$$

$$\Rightarrow \theta = \pi - \frac{\pi}{6} = \frac{5\pi}{6}$$

4. The principal value of $\tan^{-1}\left(-\frac{1}{\sqrt{3}}\right)$ is

(1) $-\frac{\pi}{6}$

(2) $-\frac{\pi}{3}$

(3) $\frac{5\pi}{6}$

(4) Both (1) & (3)

Sol. Answer (1)

$$\tan^{-1}\left(-\frac{1}{\sqrt{3}}\right) = \theta, \theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

$$\Rightarrow \tan \theta = -\frac{1}{\sqrt{3}}$$

$$\Rightarrow \theta = -\frac{\pi}{6}$$

5. The principal value of $\operatorname{cosec}^{-1}(-1)$ is

(1) $-\frac{\pi}{2}$

(2) 0

(3) $\frac{\pi}{2}$

(4) $\frac{3\pi}{2}$

Sol. Answer (1)

$$\operatorname{cosec}^{-1}(-1) = \theta, \theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] - \{0\}$$

$$\Rightarrow \operatorname{cosec} \theta = -1$$

$$\Rightarrow \theta = -\frac{\pi}{2}$$

6. Range of $\cot^{-1}x$ is

(1) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

(2) $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

(3) $(0, \pi)$

(4) $[0, \pi]$

Sol. Answer (3)

The principal values of $\cot^{-1}(x)$ lie in $(0, \pi)$.

7. Range of $\sec^{-1}x$ is

- (1) $[0, \pi]$ (2) $[0, \pi] - \left\{\frac{\pi}{2}\right\}$ (3) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ (4) $[\pi, 2\pi] - \left\{\frac{3\pi}{2}\right\}$

Sol. Answer (2)

The principal values of $\sec^{-1}x$ lie in $[0, \pi] - \left\{\frac{\pi}{2}\right\}$

8. The value of $\tan^{-1}(\sqrt{3}) + \cot^{-1}(-1) + \sec^{-1}\left(-\frac{2}{\sqrt{3}}\right)$ is

- (1) $-\frac{\pi}{12}$ (2) $\frac{11\pi}{12}$ (3) $\frac{5\pi}{4}$ (4) $\frac{23\pi}{12}$

Sol. Answer (4)

$$\tan^{-1}\sqrt{3} + \cot^{-1}(-1) + \sec^{-1}\left(-\frac{2}{\sqrt{3}}\right) = ?$$

$$\tan^{-1}\sqrt{3} = \alpha, \alpha \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

$$\Rightarrow \tan \alpha = \sqrt{3}$$

$$\Rightarrow \alpha = \frac{\pi}{3}$$

$$\cot^{-1}(-1) = \beta, \beta \in (0, \pi)$$

$$\Rightarrow \cot \beta = -1$$

$$\Rightarrow \beta = \frac{3\pi}{4}$$

$$\sec^{-1}\left(-\frac{2}{\sqrt{3}}\right) = \gamma, \gamma \in [0, \pi] - \left\{\frac{\pi}{2}\right\}$$

$$\Rightarrow \sec \gamma = \frac{-2}{\sqrt{3}}$$

$$\Rightarrow \gamma = \frac{5\pi}{6}$$

$$\Rightarrow \alpha + \beta + \gamma = \frac{\pi}{3} + \frac{3\pi}{4} + \frac{5\pi}{6} = \frac{23\pi}{12}$$

9. The value of $\cot^{-1}(-\sqrt{3}) + \operatorname{cosec}^{-1}(2) + \tan^{-1}(\sqrt{3})$ is

- (1) $\frac{\pi}{6}$ (2) $\frac{\pi}{3}$ (3) $\frac{5\pi}{6}$ (4) $\frac{4\pi}{3}$

Sol. Answer (4)

$$\cot^{-1}(-\sqrt{3}) + \operatorname{cosec}^{-1}(2) + \tan^{-1}(\sqrt{3}) = ?$$

$$\cot^{-1}(-\sqrt{3}) = \alpha, \alpha \in (0, \pi)$$

$$\Rightarrow \cot \alpha = -\sqrt{3}$$

$$\Rightarrow \alpha = \frac{5\pi}{6}$$

$$\operatorname{cosec}^{-1}(2) = \beta, \beta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] - \{0\}$$

$$\Rightarrow \operatorname{cosec} \beta = 2$$

$$\Rightarrow \beta = \frac{\pi}{6}$$

$$\tan^{-1}(\sqrt{3}) = \gamma, \gamma \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

$$\Rightarrow \tan \gamma = \sqrt{3}$$

$$\Rightarrow \gamma = \frac{\pi}{3}$$

$$\therefore \alpha + \beta + \gamma = \frac{5\pi}{6} + \frac{\pi}{6} + \frac{\pi}{3} = \frac{4\pi}{3}$$

10. The value of $2\cos^{-1}\left(-\frac{1}{2}\right) + 2\sin^{-1}\left(-\frac{1}{2}\right) - \cos^{-1}(-1)$ is

(1) 0

(2) $\frac{\pi}{2}$ (3) π (4) 2π **Sol.** Answer (1)

$$2\cos^{-1}\left(-\frac{1}{2}\right) + 2\sin^{-1}\left(-\frac{1}{2}\right) + \cos^{-1}(-1) = ?$$

$$\cos^{-1}(-1) = \theta, \theta \in [0, \pi]$$

$$\cos \theta = -1$$

$$\Rightarrow \theta = \pi$$

$$\therefore 2\left(\cos^{-1}\left(-\frac{1}{2}\right) + \sin^{-1}\left(-\frac{1}{2}\right)\right) - \cos^{-1}(-1) = \frac{2\pi}{2} - \pi = 0$$

11. The maximum and minimum values of $f(x) = \sin^{-1}x + \cos^{-1}x + \tan^{-1}x$ respectively is

(1) $\frac{3\pi}{4}, \frac{\pi}{2}$ (2) $\frac{3\pi}{4}, \frac{\pi}{4}$ (3) $\frac{\pi}{4}, \frac{-\pi}{4}$ (4) $\pi, 0$ **Sol.** Answer (2)

$$f(x) = \sin^{-1}x + \cos^{-1}x + \tan^{-1}x$$

Common domain of $f(x)$ is $[-1, 1]$

$$\Rightarrow f(x) = \frac{\pi}{2} + \tan^{-1}x$$

$$(f(x))_{\min} = \frac{\pi}{2} + \tan^{-1}(-1) = \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4}$$

$$(f(x))_{\max} = \frac{\pi}{2} + \frac{\pi}{4} = \frac{3\pi}{4}$$

12. Select the wrong option.

$$(1) -1 \leq \sin^{-1}x \leq 1 \Rightarrow -\sin 1 < x \leq \sin 1$$

$$(2) \frac{\pi}{3} \leq \cos^{-1}x \leq \frac{2\pi}{3} \Rightarrow -\frac{1}{2} \leq x \leq \frac{1}{2}$$

$$(3) \frac{\pi}{4} \leq \cot^{-1}x \leq \frac{5\pi}{6} \Rightarrow -\sqrt{3} \leq x \leq 1$$

$$(4) \sec^{-1}x \geq \frac{\pi}{4} \Rightarrow x \leq \sqrt{2}$$

Sol. Answer (4)

Option (1), (2), (3) are true.

But (4) is not true as

$$\sec^{-1}x \geq \frac{\pi}{4}$$

$$\Rightarrow \frac{\pi}{4} \leq \sec^{-1}x \leq \pi$$

$$\Rightarrow x \in (-\infty, -1] \cup [1, \sqrt{2}]$$

13. If x_1 and x_2 are the roots of $15x^2 + 28x + 12 = 0$, then

(1) Both $\cos^{-1}x_1$ and $\cos^{-1}x_2$ are real

(2) Both $\sin^{-1}x_1$ and $\sin^{-1}x_2$ are real

(3) Both $\sec^{-1}x_1$ and $\sec^{-1}x_2$ are real

(4) Both $\cot^{-1}x_1$ and $\cot^{-1}x_2$ are real

Sol. Answer (4)

On solving, $15x^2 + 28x + 12 = 0$

$$x = \frac{-28 \pm \sqrt{520}}{30}$$

One root is greater than -1 and other root is less than -1 .

Hence, option (4) is the only possibility.

14. The expression

$$\tan\left(\frac{\pi}{4} + \frac{1}{2}\cos^{-1}x\right) + \tan\left(\frac{\pi}{4} - \frac{1}{2}\cos^{-1}x\right) \text{ equals}$$

$$(1) \frac{1}{x}$$

$$(2) x$$

$$(3) \frac{2}{x}$$

$$(4) 2x$$

Sol. Answer (3)

$$\begin{aligned} \tan\left(\frac{\pi}{4} + \frac{1}{2}\cos^{-1} x\right) \\ &= \cot\left[\frac{\pi}{2} - \left(\frac{\pi}{4} + \frac{1}{2}\cos^{-1} x\right)\right] \\ &= \cot\left(\frac{\pi}{4} - \frac{1}{2}\cos^{-1} x\right) \end{aligned}$$

$$\text{Let } \frac{\pi}{4} - \frac{1}{2}\cos^{-1} x = \theta$$

$$\begin{aligned} \text{Then L.H.S.} &= \cot\theta + \tan\theta = \frac{1}{\tan\theta} + \tan\theta \\ &= 2\left(\frac{1 + \tan^2\theta}{2\tan\theta}\right) = \frac{2}{\sin 2\theta} \\ &= \frac{2}{\sin\left(\frac{\pi}{2} - \cos^{-1} x\right)} \\ &= \frac{2}{\cos(\cos^{-1} x)} \\ &= \frac{2}{x} \end{aligned}$$

15. If $p > q > 0$ and $pr < -1 < qr$, then $\tan^{-1}\left(\frac{p-q}{1+pq}\right) + \tan^{-1}\left(\frac{q-r}{1+qr}\right) + \tan^{-1}\left(\frac{r-p}{1+rp}\right)$ is equal to

(1) 0

(2) $-\pi$ (3) π (4) $\frac{\pi}{2}$ **Sol.** Answer (3)

$$p > q > 0 \Rightarrow pq > 0$$

$$\therefore \tan^{-1}\left(\frac{p-q}{1+pq}\right) = \tan^{-1} p - \tan^{-1} q \quad \dots(i)$$

Since, $qr > -1$, then

$$\tan^{-1}\frac{q-r}{1+qr} = \tan^{-1} q - \tan^{-1} r \quad \dots(ii)$$

and $pr < -1$, then

$$\tan^{-1}\left(\frac{r-p}{1+rp}\right) = \pi + \tan^{-1} r - \tan^{-1} p \quad \dots(iii)$$

Adding (i), (ii) and (iii), we get

$$\tan^{-1}\left(\frac{p-q}{1+pq}\right) + \tan^{-1}\left(\frac{q-r}{1+qr}\right) + \tan^{-1}\left(\frac{r-p}{1+rp}\right) = \pi$$

16. If k satisfies $k^2 - k - 6 > 0$, then a value exists for

(1) $\sec^{-1}k$

(2) $\sin^{-1}k$

(3) $\cos^{-1}k$

(4) $\sec^{-1}\left(\frac{1}{k}\right)$

Sol. Answer (1)

$$k^2 - k - 6 > 0$$

$$\Rightarrow (k-3)(k+2) > 0$$

$$\Rightarrow k < -2, k > 3$$

$\therefore$ A value exist for $\sec^{-1}k$

17. The domain of the function $f(x) = \cos^{-1}\left(\frac{2-|x|}{4}\right)$ is

(1) $[-6, 6]$

(2) $(-\infty, 2) \cup (2, 3)$

(3) $(2, 3)$

(4) $[-6, 2) \cup (2, 3)$

Sol. Answer (1)

$$f(x) = \cos^{-1}\left(\frac{2-|x|}{4}\right)$$

$$\Rightarrow -1 \leq \frac{2-|x|}{4} \leq 1$$

$$-4 \leq 2 - |x| \leq 4$$

$$-6 \leq -|x| \leq 2$$

$$-2 \leq |x| \leq 6$$

$$\Rightarrow -6 \leq x \leq 6$$

18. The domain of the function given by $f(x) = \sqrt{\sin^{-1}(2x) + \frac{\pi}{6}}$ is

(1) $\left[-\frac{1}{2}, \frac{1}{2}\right]$

(2) $\left[-\frac{1}{4}, \frac{1}{2}\right]$

(3) $[-1, 1]$

(4) $\left[-1, \frac{1}{2}\right]$

Sol. Answer (2)

$$f(x) = \sqrt{\sin^{-1}(2x) + \frac{\pi}{6}}$$

$$-1 \leq 2x \leq 1$$

$$\Rightarrow \frac{-1}{2} \leq x \leq \frac{1}{2} \quad \dots (i)$$

$$\sin^{-1} 2x + \frac{\pi}{6} \geq 0$$

$$\sin^{-1} 2x \geq -\frac{\pi}{6}$$

$$2x \geq -\frac{1}{2}$$

$$\Rightarrow x \geq -\frac{1}{4} \quad \dots (ii)$$

From (i) & (ii), we have

$$\Rightarrow x \in \left[-\frac{1}{4}, \frac{1}{2}\right]$$

19. The domain and range of $f(x) = \sin^{-1}x + \cos^{-1}x + \tan^{-1}x + \cot^{-1}x + \sec^{-1}x + \operatorname{cosec}^{-1}x$ respectively are

(1) $\{-1, 1\}, \frac{3\pi}{2}$

(2) $\{-1, 1\}, \frac{\pi}{2}$

(3) $(-1, 1), \frac{\pi}{2}$

(4) $(-1, 1), 2\pi$

Sol. Answer (1)

$$f(x) = (\sin^{-1}x + \cos^{-1}x) + (\tan^{-1}x + \cot^{-1}x) + (\sec^{-1}x + \operatorname{cosec}^{-1}x)$$

$$\text{Common domain} = [-1, 1]$$

$$\text{But range} = \frac{\pi}{2} + \frac{\pi}{2} + \frac{\pi}{2} = \frac{3\pi}{2}$$

20. Let $f(x) = \sec^{-1}(x-10) + \cos^{-1}(10-x)$. The range of $f(x)$ is

(1) $\left\{0, \frac{\pi}{2}, \pi\right\}$

(2) $\left\{0, \frac{\pi}{2}\right\}$

(3) $\left\{\frac{\pi}{2}\right\}$

(4) $\{\pi\}$

Sol. Answer (4)

$$\sec^{-1}(x-10) \text{ is defined, if}$$

$$x-10 \leq -1 \text{ and } x-10 \geq 1$$

$$\Rightarrow x \leq 9 \text{ and } x \geq 11$$

$$\cos^{-1}(10-x) \text{ is defined, if}$$

$$-1 \leq 10-x \leq 1$$

$$\Rightarrow 9 \leq x \leq 11$$

Hence, $f(x)$ is defined, when $x = 9$ or 11 .

$$f(9) = \sec^{-1}(-1) + \cos^{-1}(1) = \pi + 0 = \pi$$

$$f(11) = \sec^{-1}(1) + \cos^{-1}(-1) = 0 + \pi = \pi$$

$\therefore$ Range of $f(x)$ is $\{\pi\}$.

21. $\sin^{-1}\left(\sin\frac{7\pi}{6}\right)$ is

(1) $-\frac{\pi}{6}$

(2) $\frac{\pi}{6}$

(3) $\frac{7\pi}{6}$

(4) $-\frac{5\pi}{6}$

Sol. Answer (1)

$$\sin^{-1}\left(\sin\frac{7\pi}{6}\right) = \sin^{-1}\sin\left(\pi + \frac{\pi}{6}\right) = \sin^{-1}\left(-\sin\frac{\pi}{6}\right) = -\sin^{-1}\sin\frac{\pi}{6} = -\frac{\pi}{6}$$

22. $\tan^{-1}\left(\tan\frac{3\pi}{4}\right)$ is

(1) $\frac{3\pi}{4}$

(2) $\frac{\pi}{4}$

(3) $-\frac{\pi}{4}$

(4) $-\frac{3\pi}{4}$

Sol. Answer (3)

$$\tan^{-1}\tan\left(\frac{3\pi}{4}\right) = \tan^{-1}\left(\tan\left(\pi - \frac{\pi}{4}\right)\right) = \tan^{-1}\left(-\tan\frac{\pi}{4}\right) = -\tan^{-1}\left(\tan\frac{\pi}{4}\right) = -\frac{\pi}{4}$$

23. Value of $\sec^{-1}\left(\sec\frac{4\pi}{3}\right)$ is

(1) $\frac{\pi}{3}$

(2) $\frac{2\pi}{3}$

(3) $\frac{4\pi}{3}$

(4) $-\frac{\pi}{3}$

Sol. Answer (2)

$$\sec^{-1}\left(\sec\frac{4\pi}{3}\right) = \sec^{-1}\left[\sec\left(2\pi - \frac{4\pi}{3}\right)\right] = \frac{2\pi}{3}$$

24. $\cos^{-1}(-x)$, $|x| \leq 1$, is equal to

(1) $-\cos^{-1}x$

(2) $\cos^{-1}x$

(3) $\frac{\pi}{2} - \cos^{-1}x$

(4) $\frac{\pi}{2} + \sin^{-1}x$

Sol. Answer (4)

$$\cos^{-1}(-x) = \pi - \cos^{-1}x$$

$$= \pi - \left(\frac{\pi}{2} - \sin^{-1}x\right)$$

$$= \frac{\pi}{2} + \sin^{-1}x$$

25. $\operatorname{cosec}^{-1}(-x)$, $x \in R - (-1, 1)$, is equal to

(1) $\operatorname{cosec}^{-1}x$

(2) $-\sin^{-1}x$

(3) $-\sin^{-1}\left(\frac{1}{x}\right)$

(4) $\pi - \operatorname{cosec}^{-1}x$

Sol. Answer (3)

$$\operatorname{cosec}^{-1}(-x) = -\operatorname{cosec}^{-1}x$$

$$= -\sin^{-1}\left(\frac{1}{x}\right)$$

26. $\cot^{-1}(-2)$ is equal to

- (1) $\pi - \tan^{-1}\left(\frac{1}{2}\right)$ (2) $\tan^{-1}\left(\frac{1}{2}\right)$ (3) $-\tan^{-1}\frac{1}{2}$ (4) $\cot^{-1}2$

Sol. Answer (1)

$$\cot^{-1}(-2) = \pi - \cot^{-1}(2)$$

$$= \pi - \tan^{-1}\left(\frac{1}{2}\right)$$

27. The value of $\sin^{-1}\frac{4}{5} + \sin^{-1}\frac{5}{13} + \sin^{-1}\frac{16}{65}$ is

- (1) $\frac{3\pi}{2}$ (2) $\frac{\pi}{2}$ (3) $\pi - \sin^{-1}\frac{3713}{4225}$ (4) $\pi - \tan^{-1}\frac{3713}{2016}$

Sol. Answer (2)

$$\sin^{-1}\frac{4}{5} + \sin^{-1}\frac{5}{13} + \sin^{-1}\frac{16}{65}$$

$$= \sin^{-1}\left(\frac{4}{5}\sqrt{1-\left(\frac{5}{13}\right)^2} + \frac{5}{13}\sqrt{1-\left(\frac{4}{5}\right)^2}\right) + \sin^{-1}\frac{16}{65}$$

$$= \sin^{-1}\left(\frac{48}{65} + \frac{15}{65}\right) + \sin^{-1}\left(\frac{16}{65}\right)$$

$$= \sin^{-1}\left(\frac{63}{65}\right) + \sin^{-1}\left(\frac{16}{65}\right)$$

$$= \cos^{-1}\left(\frac{16}{65}\right) + \sin^{-1}\left(\frac{16}{65}\right) \quad \left(\because \sin^{-1}x = \cos^{-1}\sqrt{1-x^2}\right)$$

$$= \frac{\pi}{2}$$

28. The value of $\cos^{-1}(-1) + \sin^{-1}(1)$ is

- (1) $-\frac{3\pi}{2}$ (2) $\frac{\pi}{2}$ (3) π (4) $\frac{3\pi}{2}$

Sol. Answer (4)

$$\cos^{-1}(-1) + \sin^{-1}(1)$$

$$= \pi + \frac{\pi}{2}$$

$$= \frac{3\pi}{2}$$

29. If $-1 \leq x \leq -\frac{1}{2}$, then $\sin^{-1}(3x - 4x^3)$ equals

- (1) $3 \sin^{-1} x$ (2) $\pi - 3 \sin^{-1} x$ (3) $-\pi - 3 \sin^{-1} x$ (4) $\pi + 3 \sin^{-1} x$

Sol. Answer (3)

$$3 \sin^{-1} x = -\pi - \sin^{-1}(3x - 4x^3), \text{ when } -1 \leq x \leq -\frac{1}{2}$$

$$\Rightarrow \sin^{-1}(3x - 4x^3) = -\pi - 3 \sin^{-1}(x)$$

30. $\sin^{-1}\left(-\left(\frac{1}{2}\right)\right) + \cos^{-1}\left(-\left(\frac{1}{2}\right)\right) + \cot^{-1}(-\sqrt{3}) + \operatorname{cosec}^{-1}(\sqrt{2}) + \tan^{-1}(-1) + \sec^{-1}(\sqrt{2})$ equals

- (1) $\frac{9\pi}{4}$ (2) $\frac{19\pi}{12}$ (3) $\frac{3\pi}{2}$ (4) $\frac{\pi}{2}$

Sol. Answer (2)

$$\text{Sum} = \frac{-\pi}{6} + \frac{2\pi}{3} + \frac{5\pi}{6} + \frac{\pi}{4} - \frac{\pi}{4} + \frac{\pi}{4} = \frac{19\pi}{12}$$

31. The value of $\sin^{-1}\sin(16) + \cos^{-1}\cos(10)$ is

- (1) 26 (2) -26 (3) $6 + \pi$ (4) $9\pi - 26$

Sol. Answer (4)

$$\begin{aligned} \sin^{-1}\sin 16 &= \sin^{-1}\sin(16 - 5\pi + 5\pi) \\ &= 5\pi - 16 \end{aligned}$$

$$\begin{aligned} \cos^{-1}(\cos 10) &= \cos^{-1}\cos(10 - 3\pi + 3\pi) \\ &= \cos^{-1}\cos\{3\pi + (10 - 3\pi)\} \\ &= \cos^{-1}\{-\cos(10 - 3\pi)\} \\ &= \pi - \cos^{-1}\cos(10 - 3\pi) \\ &= \pi - (10 - 3\pi) = 4\pi - 10 \end{aligned}$$

$$\text{Sum} = 5\pi - 16 + 4\pi - 10 = 9\pi - 26$$

32. $\cos^{-1}\left(\cos\frac{5\pi}{3}\right) + \sin^{-1}\left(\sin\frac{5\pi}{3}\right) =$

- (1) 0 (2) $\frac{\pi}{2}$ (3) $10\frac{\pi}{3}$ (4) $\frac{2\pi}{3}$

Sol. Answer (1)

$$\cos^{-1}\left(\cos\frac{5\pi}{3}\right) = \cos^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{3}$$

$$\sin^{-1}\left(\sin\frac{5\pi}{3}\right) = \sin^{-1}\left(-\frac{\sqrt{3}}{2}\right) = -\frac{\pi}{3}$$

$$\text{Adding, } \cos^{-1}\left(\cos\frac{5\pi}{3}\right) + \sin^{-1}\left(\sin\frac{5\pi}{3}\right) = 0$$

33. $4 \tan^{-1}\left(\frac{1}{5}\right) - \tan^{-1}\left(\frac{1}{239}\right)$ is equal to

- (1) π (2) $\frac{\pi}{2}$ (3) $\frac{\pi}{3}$ (4) $\frac{\pi}{4}$

Sol. Answer (4)

$$4 \tan^{-1}\left(\frac{1}{5}\right) = 2 \left(2 \tan^{-1}\frac{1}{5} \right) = 2 \tan^{-1}\left(\frac{\frac{2}{5}}{1 - \frac{1}{25}} \right)$$

$$= 2 \tan^{-1}\left(\frac{5}{12}\right)$$

$$= \tan^{-1}\left(\frac{2\left(\frac{5}{12}\right)}{1 - \frac{25}{144}} \right)$$

$$= \tan^{-1}\left(\frac{120}{119}\right)$$

$$= \tan^{-1}\left(\frac{240}{238}\right)$$

$$= \tan^{-1}\left(\frac{239+1}{239-1}\right)$$

$$= \tan^{-1}\left(\frac{1 + \frac{1}{239}}{1 - \frac{1}{239}} \right)$$

$$= \tan^{-1} 1 + \tan^{-1} \frac{1}{239}$$

$$\Rightarrow 4 \tan^{-1}\left(\frac{1}{5}\right) - \tan^{-1}\left(\frac{1}{239}\right) = \tan^{-1} 1 = \frac{\pi}{4}$$

34. $\tan^{-1}\left(-\frac{2}{5}\right)$ equals

- (1) $\cot^{-1}\left(-\frac{5}{2}\right) + \pi$ (2) $\cot^{-1}\left(\frac{5}{2}\right) - \pi$ (3) $\cot^{-1}\left(\frac{5}{2}\right) + \pi$ (4) $\cot^{-1}\left(-\frac{5}{2}\right) - \pi$

Sol. Answer (4)

$$\cot^{-1}(-x) = \pi - \cot^{-1} x$$

$$\begin{aligned}
 \therefore \cot^{-1}\left(-\frac{5}{2}\right) &= \pi - \cot^{-1}\left(\frac{5}{2}\right) \\
 &= \pi - \tan^{-1}\left(\frac{2}{5}\right) \\
 &= \pi + \tan^{-1}\left(\frac{-2}{5}\right) \quad (\because \tan^{-1}(-x) = -\tan^{-1}x) \\
 \Rightarrow \tan^{-1}\left(\frac{-2}{5}\right) &= \cot^{-1}\left(\frac{-5}{2}\right) - \pi
 \end{aligned}$$

35. If $\sum_{i=1}^{300} \cos^{-1} x_i = 0$. Then find the value of $\sum_{i=1}^{300} \sin^{-1} x_i$
- (1) 0 (2) 150π (3) 150 (4) 151π

Sol. Answer (2)

As we know that

$$0 \leq \cos^{-1} x \leq \pi \quad \forall x \in [-1, 1]$$

If $\cos^{-1} x$ is always non-negative and we given sum of 300, non-negative terms is zero.

$\therefore$ All the terms must be zero.

$$\text{i.e., } \cos^{-1} x_i = 0 \quad \forall x_i$$

$$\Rightarrow x_i = 1 \quad \forall i$$

$$\text{i.e., } \sin^{-1} x_i = \frac{\pi}{2} \quad \forall i$$

$$\Rightarrow \sum_{i=1}^{300} \sin^{-1} x_i = \sum_{i=1}^{300} \frac{\pi}{2} = 150\pi$$

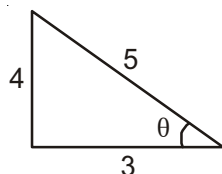
36. $\sin\left(\cos^{-1}\left(\frac{3}{5}\right)\right)$ is equal to

- (1) $\frac{\pi}{2} - \frac{3}{5}$ (2) $\frac{3}{5}$ (3) $-\frac{4}{5}$ (4) $\frac{4}{5}$

Sol. Answer (4)

$$\sin\left(\cos^{-1}\left(\frac{3}{5}\right)\right) = ?$$

$$\text{Let } \cos^{-1}\left(\frac{3}{5}\right) = \theta$$



$$\Rightarrow \cos \theta = \frac{3}{5}$$

$$\Rightarrow \sin \theta = \frac{4}{5}$$

$$\Rightarrow \sin \cos^{-1}\left(\frac{3}{5}\right) = \sin \sin^{-1} \frac{4}{5} = \frac{4}{5}$$

37. $\tan\left(\frac{1}{2}\cos^{-1}\left(\frac{\sqrt{5}}{3}\right)\right)$ is equal to

(1) $\frac{3-\sqrt{5}}{2}$

(2) $\frac{3+\sqrt{5}}{2}$

(3) $\frac{\sqrt{5}-3}{2}$

(4) $\frac{-\sqrt{5}-3}{2}$

Sol. Answer (1)

$$\tan\left(\frac{1}{2}\cos^{-1}\frac{\sqrt{5}}{3}\right) = ?$$

$$\text{Let } \cos^{-1}\left(\frac{\sqrt{5}}{3}\right) = \theta$$

$$\Rightarrow \cos\theta = \frac{\sqrt{5}}{3}$$

$$\therefore \tan^2 \frac{\theta}{2} = \frac{1-\cos\theta}{1+\cos\theta}$$

$$= \frac{1-\frac{\sqrt{5}}{3}}{1+\frac{\sqrt{5}}{3}} = \frac{3-\sqrt{5}}{3+\sqrt{5}}$$

$$= \frac{(3-\sqrt{5})^2}{4}$$

$$\Rightarrow \tan \frac{\theta}{2} = \pm \frac{3-\sqrt{5}}{2}$$

But $\tan \frac{\theta}{2} \neq -\left(\frac{3-\sqrt{5}}{2}\right)$ as $\frac{\theta}{2}$ is an acute angle.

38. $\sin^{-1}\left(\frac{4}{5}\right) + 2\tan^{-1}\left(\frac{1}{3}\right)$ is equal to

(1) $\tan^{-1}\left(\frac{3}{4}\right)$

(2) $\tan^{-1}\left(\frac{4}{3}\right)$

(3) $\tan^{-1}\left(\frac{3}{\sqrt{10}}\right)$

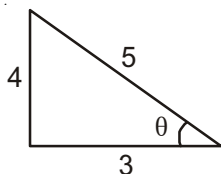
(4) $\frac{\pi}{2}$

Sol. Answer (4)

$$\sin^{-1}\frac{4}{5} + 2\tan^{-1}\left(\frac{1}{3}\right)$$

$$\text{Let } \sin^{-1}\left(\frac{4}{5}\right) = \theta$$

$$\Rightarrow \sin\theta = \frac{4}{5}$$



$$\Rightarrow \tan \theta = \frac{4}{3}$$

$$\Rightarrow \sin^{-1}\left(\frac{4}{5}\right) + 2\tan^{-1}\left(\frac{1}{3}\right)$$

$$= \tan^{-1}\left(\frac{4}{3}\right) + \tan^{-1}\left(\frac{\frac{2}{3}}{1 - \frac{1}{9}}\right)$$

$$= \tan^{-1}\left(\frac{4}{3}\right) + \tan^{-1}\left(\frac{2}{3} \times \frac{9}{8}\right)$$

$$= \tan^{-1}\left(\frac{4}{3}\right) + \tan^{-1}\left(\frac{3}{4}\right)$$

$$= \tan^{-1}\frac{4}{3} + \cot^{-1}\left(\frac{4}{3}\right)$$

$$= \tan^{-1}\frac{4}{3} + \cot^{-1}\left(\frac{4}{3}\right)$$

$$= \frac{\pi}{2}$$

39. $\sin\left(2\tan^{-1}\left(\frac{1}{3}\right)\right) + \cos\left(\tan^{-1}(2\sqrt{2})\right)$ is equal to

(1) $\frac{14}{15}$

(2) $\frac{3}{4}$

(3) $\frac{2\sqrt{2}}{7}$

(4) $\frac{2\sqrt{2}}{15}$

Sol. Answer (1)

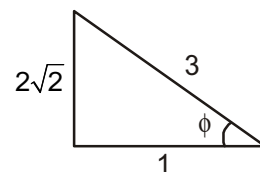
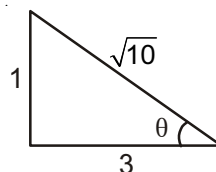
$$\sin\left(2\tan^{-1}\left(\frac{1}{3}\right)\right) + \cos\left(\tan^{-1}(2\sqrt{2})\right) = ?$$

$$\tan^{-1}\left(\frac{1}{3}\right) = \theta, \quad \tan^{-1}(2\sqrt{2}) = \phi$$

$$\Rightarrow \sin 2\theta + \cos \phi = \frac{2\tan \theta}{1 + \tan^2 \theta} + \cos \phi$$

$$= \frac{2\left(\frac{1}{3}\right)}{1 + \left(\frac{1}{3}\right)^2} + \frac{1}{3} = \frac{3}{5} + \frac{1}{3}$$

$$= \frac{14}{15}$$



40. If $\sin(\cot^{-1}(1+x)) = \cos(\tan^{-1}x)$, then x is

- (1) 1 (2) $\frac{1}{2}$ (3) 0 (4) $-\frac{1}{2}$

Sol. Answer (4)

$$\sin(\cot^{-1}(1+x)) = \cos(\tan^{-1}x)$$

$$\cot^{-1}(1+x) = \theta$$

$$\Rightarrow \cot \theta = (1+x)$$

$$\Rightarrow \sin \theta = \frac{1}{\sqrt{2+2x+x^2}}$$

$$\Rightarrow \tan^{-1} x = \phi$$

$$\Rightarrow \tan \phi = x$$

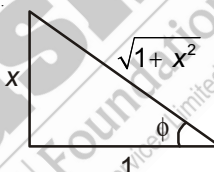
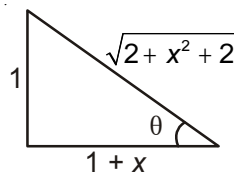
$$\cos \phi = \frac{1}{\sqrt{1+x^2}}$$

$$\therefore \sin \theta = \cos \phi$$

$$\Rightarrow \frac{1}{\sqrt{2+2x+x^2}} = \frac{1}{\sqrt{1+x^2}}$$

$$\Rightarrow 1+x^2 = 2+2x+x^2$$

$$\Rightarrow x = -\frac{1}{2}$$



41. $\tan^{-1}\left(\frac{1}{2}\right) + \tan^{-1}\left(\frac{1}{3}\right)$ is equal to

- (1) $\frac{\pi}{6}$ (2) $\frac{\pi}{4}$ (3) $\frac{\pi}{3}$ (4) $\frac{5\pi}{12}$

Sol. Answer (2)

$$\tan^{-1}\left(\frac{1}{2}\right) + \tan^{-1}\left(\frac{1}{3}\right) = \tan^{-1}\left(\frac{\left(\frac{1}{2} + \frac{1}{3}\right)}{1 - \frac{1}{6}}\right) = \tan^{-1}1 = \frac{\pi}{4}$$

42. If $x_1 = \cos^{-1}\left(\frac{3}{5}\right) + \cos^{-1}\left(\frac{2\sqrt{2}}{3}\right)$ and $x_2 = \sin^{-1}\left(\frac{3}{5}\right) + \sin^{-1}\left(\frac{2\sqrt{2}}{3}\right)$, then

- (1) $x_1 < x_2$ (2) $x_1 = x_2$ (3) $x_1 > x_2$ (4) Can't be determined

Sol. Answer (1)

$$x_1 = \cos^{-1}\left(\frac{3}{5}\right) + \cos^{-1}\left(\frac{2\sqrt{2}}{3}\right)$$

$$= \tan^{-1}\left(\frac{4}{3}\right) + \tan^{-1}\left(\frac{1}{2\sqrt{2}}\right)$$

$$= \tan^{-1}\left(\frac{\frac{4}{3} + \frac{1}{2\sqrt{2}}}{1 - \frac{4}{3} \times \frac{1}{2\sqrt{2}}}\right)$$

$$= \tan^{-1}\left(\frac{8\sqrt{2} + 3}{6\sqrt{2} - 4}\right) > 1$$

$$\Rightarrow x_1 \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$$

$$x_2 = \sin^{-1}\left(\frac{3}{5}\right) + \sin^{-1}\left(\frac{2\sqrt{2}}{3}\right)$$

$$\sin^{-1}\left(\frac{3}{5}\right) = \theta$$

$$\sin \theta = \frac{3}{5}$$

$$\Rightarrow \tan \theta = \frac{3}{4}$$

$$\sin^{-1}\left(\frac{2\sqrt{2}}{3}\right) = \phi$$

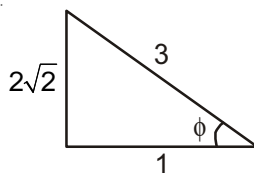
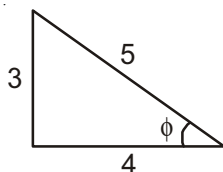
$$\Rightarrow \sin \phi = \frac{2\sqrt{2}}{3}$$

$$\Rightarrow \tan \phi = 2\sqrt{2}$$

$$\Rightarrow \tan \theta \tan \phi = \frac{3}{4} \times 2\sqrt{2} > 1$$

$$\Rightarrow x_2 \in \left(\frac{\pi}{2}, \pi\right)$$

$$\Rightarrow x_2 > x_1$$



43. The value of $\tan^{-1}1 + \tan^{-1}2 + \tan^{-1}3$ is

(1) 0

(2) π

(3) $-\pi$

(4) $-2\tan^{-1}3$

Sol. Answer (2)

$$\tan^{-1}1 + \tan^{-1}2 + \tan^{-1}3$$

$$= \tan^{-1}1 + \left(\pi + \tan^{-1} \frac{2+3}{1-2 \cdot 3} \right) \quad \left\{ \because \text{when } 1-xy < 0, x > 0, y > 0, \tan^{-1}x + \tan^{-1}y = \pi + \tan^{-1} \left(\frac{x+y}{1-xy} \right) \right\}$$

$$= \tan^{-1}1 + \{ \pi + \tan^{-1}(-1) \}$$

$$= \tan^{-1}1 + (\pi - \tan^{-1}1)$$

$$= \pi$$

44. $\cos\left(\tan^{-1}\frac{3}{4}\right) + \cos(\tan^{-1}x)$ is equal to

(1) $\frac{4}{5} + \frac{x}{\sqrt{1+x^2}}$

(2) $\frac{3}{5} + \frac{1}{\sqrt{1+x^2}}$

(3) $\frac{4}{5} + \frac{1}{\sqrt{1+x^2}}$

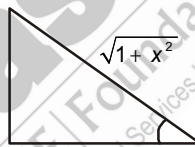
(4) $\frac{3}{5} + \frac{x}{\sqrt{1+x^2}}$

Sol. Answer (3)

$$\cos\left(\tan^{-1}\frac{3}{4}\right) + \cos(\tan^{-1}x)$$

$$= \cos \cos^{-1}\left(\frac{4}{5}\right) + \cos \cos^{-1}\left(\frac{1}{\sqrt{1+x^2}}\right)$$

$$= \frac{4}{5} + \frac{1}{\sqrt{1+x^2}}$$



45. The value of $\sin \cot^{-1} \tan \cos^{-1}x$ is equal to

(1) x

(2) $\frac{x}{2}$

(3) $-x$

(4) x^2

Sol. Answer (1)

$$\sin \cot^{-1} \tan \cos^{-1}x$$

$$= \sin \left\{ \frac{\pi}{2} - \tan^{-1} \tan \cos^{-1}x \right\}$$

$$= \sin \left(\frac{\pi}{2} - \cos^{-1}x \right)$$

$$= \sin(\sin^{-1}x)$$

$$= x$$

Second Method

We have,

$$\operatorname{sincot}^{-1} \tan \cos^{-1} x$$

$$= \operatorname{sincot}^{-1} \tan \theta, \text{ where } \cos^{-1} x = \theta$$

$$= \operatorname{sincot}^{-1} \left(\frac{\sqrt{1-x^2}}{x} \right)$$

$$= \sin \phi, \text{ where } \cot^{-1} \left(\frac{\sqrt{1-x^2}}{x} \right) = \phi$$

$$= x$$

46. $\sin^{-1}(\cos \sin^{-1} x) + \cos^{-1}(\sin \cos^{-1} x)$ is equal to

- (1) 0 (2) $\frac{\pi}{4}$ (3) $\frac{\pi}{2}$ (4) $\frac{3\pi}{4}$

Sol. Answer (3)

$$\begin{aligned} \sin^{-1}(\cos \sin^{-1} x) &= \frac{\pi}{2} - \cos^{-1}(\cos \sin^{-1} x) \\ &= \frac{\pi}{2} - \sin^{-1} x \end{aligned}$$

$$\text{Similarly, } \cos^{-1}(\sin \cos^{-1} x) = \frac{\pi}{2} - \cos^{-1} x$$

$$\begin{aligned} \text{Adding, } \sin^{-1}(\cos \sin^{-1} x) + \cos^{-1}(\sin \cos^{-1} x) &= \pi - (\sin^{-1} x + \cos^{-1} x) \\ &= \frac{\pi}{2} \end{aligned}$$

47. The value of $\cot \left[\frac{1}{2} \sin^{-1} \frac{\sqrt{3}}{2} \right]$ is

- (1) 1 (2) $\frac{1}{\sqrt{3}}$ (3) $\sqrt{3}$ (4) 0

Sol. Answer (3)

$$\text{Let } \sin^{-1} \frac{\sqrt{3}}{2} = 2\theta$$

$$\Rightarrow \frac{\pi}{3} = 2\theta$$

$$\Rightarrow \theta = \frac{\pi}{6}$$

$$\therefore \cot \theta = \cot \frac{\pi}{6} = \sqrt{3}$$

48. The value $\tan\{\sin^{-1}(\cos(\sin^{-1} x))\} \tan\{\cos^{-1}(\sin(\cos^{-1} x))\}$ is equal to (where $0 < x < 1$)

(1) 0

(2) 1

(3) -1

(4) 2

Sol. Answer (2)

$$\tan\{\sin^{-1}(\cos \sin^{-1} x)\}$$

$$= \tan\left\{\frac{\pi}{2} - \cos^{-1}(\cos \sin^{-1} x)\right\}$$

$$= \tan\left(\frac{\pi}{2} - \sin^{-1} x\right)$$

$$= \cot(\sin^{-1} x)$$

Similarly,

$$\tan\{\cos^{-1}(\sin \cos^{-1} x)\} = \tan\left\{\frac{\pi}{2} - \sin^{-1}(\sin \cos^{-1} x)\right\}$$

$$= \tan\left(\frac{\pi}{2} - \cos^{-1} x\right)$$

$$= \tan(\sin^{-1} x)$$

$$\therefore \tan\{\sin^{-1} \cos \sin^{-1} x\} \cdot \tan\{\cos^{-1} \sin \cos^{-1} x\} = 1$$

49. $\tan\left(2\tan^{-1}\left(\frac{1}{5}\right) - \frac{\pi}{4}\right)$ is equal to

(1) $\frac{5}{12}$ (2) $-\frac{5}{12}$ (3) $\frac{7}{17}$ (4) $-\frac{7}{17}$

Sol. Answer (4)

$$\tan\left(2\tan^{-1}\left(\frac{1}{5}\right) - \frac{\pi}{4}\right)$$

$$\text{Let } \tan^{-1}\left(\frac{1}{5}\right) = \theta$$

$$\Rightarrow \tan \theta = \frac{1}{5}$$

$$\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta} = \frac{\frac{2}{5}}{1 - \frac{1}{25}} = \frac{2}{5} \times \frac{25}{24} = \frac{5}{12}$$

$$\therefore \tan\left(2\theta - \frac{\pi}{4}\right) = \frac{\tan 2\theta - 1}{1 + \tan 2\theta} = \frac{\frac{5}{12} - 1}{1 + \frac{5}{12}} = \frac{-7}{17}$$

50. $\frac{1}{2} \tan^{-1}\left(\frac{12}{5}\right)$ is equal to

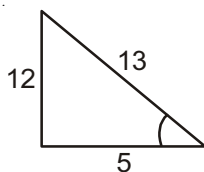
(1) $\tan^{-1}\left(\frac{3}{2}\right)$

(2) $\tan^{-1}\left(\frac{2}{3}\right)$

(3) $\tan^{-1}\left(\frac{3}{4}\right)$

(4) $\tan^{-1}\left(\frac{7}{17}\right)$

Sol. Answer (2)



$$\frac{1}{2} \tan^{-1}\left(\frac{12}{5}\right) = ?$$

$$\text{Let } \tan^{-1}\left(\frac{12}{5}\right) = \theta$$

$$\Rightarrow \tan \theta = \frac{12}{5}$$

$$\Rightarrow \cos \theta = \frac{5}{13} = \frac{1 - \tan^2 \frac{\theta}{2}}{1 + \tan^2 \frac{\theta}{2}}$$

$$\Rightarrow \frac{1 - \tan^2 \frac{\theta}{2} + 1 + \tan^2 \frac{\theta}{2}}{1 - \tan^2 \frac{\theta}{2} - 1 - \tan^2 \frac{\theta}{2}} = \frac{5 + 13}{5 - 13}$$

$$\tan^2 \frac{\theta}{2} = \frac{8}{18}$$

$$\Rightarrow \tan \frac{\theta}{2} = \pm \frac{2}{3}$$

But $\tan \frac{\theta}{2} \neq -\frac{2}{3}$ as it is positive

$$\therefore \tan \frac{\theta}{2} = \frac{2}{3}$$

$$\Rightarrow \frac{\theta}{2} = \tan^{-1}\left(\frac{2}{3}\right)$$

51. The value of $\tan^{-1}\left(\frac{1}{2}\right) + \tan^{-1}\left(\frac{1}{5}\right) + \tan^{-1}\left(\frac{1}{8}\right)$ is

(1) $\frac{\pi}{4}$

(2) $\frac{\pi}{3}$

(3) $\tan^{-1}\left(\frac{12}{5}\right)$

(4) $\tan^{-1}\left(\frac{4}{3}\right)$

Sol. Answer (1)

$$\begin{aligned}
 & \tan^{-1}\left(\frac{1}{2}\right) + \tan^{-1}\left(\frac{1}{5}\right) + \tan^{-1}\left(\frac{1}{8}\right) \\
 &= \tan^{-1}\left(\frac{\frac{1}{2} + \frac{1}{5}}{1 - \frac{1}{2} \times \frac{1}{5}}\right) + \tan^{-1}\left(\frac{1}{8}\right) \\
 &= \tan^{-1}\left(\frac{7}{9}\right) + \tan^{-1}\left(\frac{1}{8}\right) \\
 &= \tan^{-1}\left(\frac{\frac{7}{9} + \frac{1}{8}}{1 - \frac{7}{9} \times \frac{1}{8}}\right) \\
 &= \tan^{-1}\left(\frac{65}{65}\right) \\
 &= \frac{\pi}{4}
 \end{aligned}$$

[Solution of Inverse Trigonometric Equations]

52. If $\sin^{-1}\left(\frac{3}{x}\right) + \sin^{-1}\left(\frac{4}{x}\right) = \frac{\pi}{2}$, then x equals

(1) -5

(2) 5

(3) 7

(4) 12

Sol. Answer (2)

Given, $\sin^{-1}\left(\frac{3}{x}\right) + \sin^{-1}\left(\frac{4}{x}\right) = \frac{\pi}{2}$... (i)

Also, $\sin^{-1}\left(\frac{3}{x}\right) + \cos^{-1}\left(\frac{3}{x}\right) = \frac{\pi}{2}$... (ii)

From (i) and (ii),

$$\sin^{-1}\left(\frac{4}{x}\right) = \cos^{-1}\left(\frac{3}{x}\right)$$

$$\sin^{-1}\left(\frac{4}{x}\right) = \sin^{-1}\sqrt{1 - \left(\frac{3}{x}\right)^2}$$

$$\Rightarrow \frac{4}{x} = \sqrt{1 - \left(\frac{3}{x}\right)^2}$$

$$\Rightarrow \frac{4^2}{x^2} + \frac{3^2}{x^2} = 1$$

$$\Rightarrow x = \pm 5$$

$$\Rightarrow x = 5 \text{ is a solution}$$

53. If $\tan^{-1} x + \tan^{-1} y = \frac{\pi}{4}$, then $(1+x)(1+y)$ is equal to

(1) 1

(2) 2

(3) 3

(4) 4

Sol. Answer (2)

$$\tan^{-1} x + \tan^{-1} y = \frac{\pi}{4}$$

$$\Rightarrow \tan^{-1} \left(\frac{x+y}{1-xy} \right) = \tan^{-1} 1$$

$$\Rightarrow \frac{x+y}{1-xy} = 1$$

$$\Rightarrow x+y = 1-xy$$

$$\Rightarrow x+y+xy+1 = 2$$

$$\Rightarrow (1+x)(1+y) = 2$$

54. If $\tan^{-1} \left(\frac{1}{x} \right) + \cos^{-1} \left(\frac{2}{\sqrt{5}} \right) = \frac{\pi}{4}$, then x equals

(1) $\frac{1}{\sqrt{5}}$ (2) $\frac{4}{\sqrt{5}}$

(3) 1

(4) 3

Sol. Answer (4)

$$\tan^{-1} \left(\frac{1}{x} \right) + \cos^{-1} \left(\frac{2}{\sqrt{5}} \right) = \frac{\pi}{4}$$

$$\cos^{-1} \left(\frac{2}{\sqrt{5}} \right) = \theta$$

$$\Rightarrow \cos \theta = \frac{2}{\sqrt{5}}$$

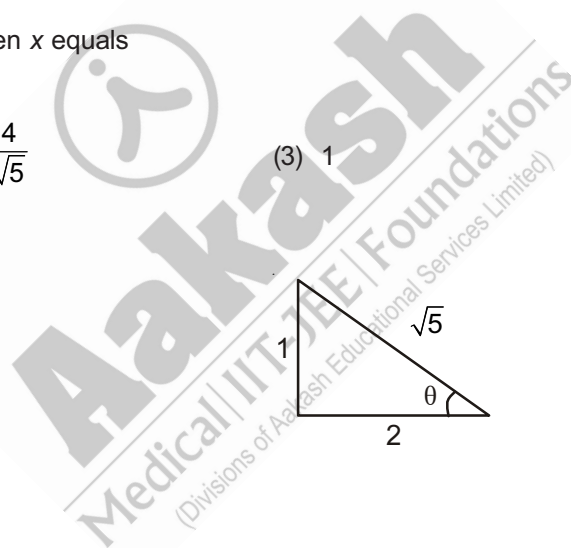
$$\Rightarrow \tan \theta = \frac{1}{2}$$

$$\tan^{-1} \left(\frac{1}{x} \right) + \tan^{-1} \left(\frac{1}{2} \right) = \frac{\pi}{4}$$

$$\Rightarrow \tan^{-1} \frac{\left(\frac{1}{x} + \frac{1}{2} \right)}{1 - \frac{1}{2x}} = \frac{\pi}{4}$$

$$\Rightarrow \frac{2+x}{2x-1} = 1$$

$$\Rightarrow x = 3$$



55. If $\cot^{-1} x + \tan^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{4}$, then x is

(1) $\frac{1}{3}$

(2) $\frac{2}{3}$

(3) 2

(4) 3

Sol. Answer (4)

$$\cot^{-1} x + \tan^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{4}$$

$$\tan^{-1} \frac{1}{x} + \tan^{-1} \frac{1}{2} = \frac{\pi}{4}$$

$$\tan^{-1} \left(\frac{\frac{1}{x} + \frac{1}{2}}{1 - \frac{1}{2x}} \right) = \frac{\pi}{4}$$

$$\Rightarrow 2 + x = 2x - 1$$

$$\Rightarrow x = 3$$

56. $\tan^{-1} x - \tan^{-1} y = \tan^{-1} \left(\frac{x-y}{1+xy} \right)$ holds good for

(1) All $x, y \in \mathbb{R}$

(2) $|x| < 1, |y| < 1$

(3) $|x| > 1, |y| > 1$

(4) $xy > -1$

Sol. Answer (4)

$$\tan^{-1} x - \tan^{-1} y = \tan^{-1} \left(\frac{x-y}{1+xy} \right), \text{ if } xy > -1$$

57. If $\sin^{-1} x + \sin^{-1}(1-x) + \cos^{-1} x = 0$, then x is

(1) 0

(2) 1

(3) 2

(4) No real value

Sol. Answer (4)

$$\sin^{-1} x + \sin^{-1}(1-x) + \cos^{-1} x = 0$$

$$\sin^{-1}(1-x) = -\frac{\pi}{2}$$

$$1-x = -1$$

$$\Rightarrow x = 2$$

But $x = 2$ is not possible for $\cos^{-1} x$ and $\sin^{-1} x$

$\Rightarrow$ No value of x .

58. $\tan^{-1} \left(\frac{x}{y} \right) - \tan^{-1} \left(\frac{x-y}{x+y} \right)$ equals

(1) $\frac{\pi}{2}$

(2) $\frac{\pi}{3}$

(3) $\frac{\pi}{4}$

(4) $\frac{\pi}{4}$ or $-\frac{3\pi}{4}$

Sol. Answer (3)

$$\begin{aligned}
 & \tan^{-1}\left(\frac{x}{y}\right) - \tan^{-1}\left(\frac{x-y}{x+y}\right) \\
 &= \tan^{-1}\left(\frac{x}{y}\right) - \tan^{-1}\left(\frac{\frac{x}{y} - 1}{1 + \frac{x}{y} \cdot 1}\right) \\
 &= \tan^{-1}\left(\frac{x}{y}\right) - \left\{ \tan^{-1}\left(\frac{x}{y}\right) - \tan^{-1} 1 \right\} \\
 &= \tan^{-1} 1 \\
 &= \frac{\pi}{4}
 \end{aligned}$$

59. If $\tan^{-1}x + \tan^{-1}y + \tan^{-1}z = \pi$, then $\frac{1}{xy} + \frac{1}{yz} + \frac{1}{zx} =$

(1) 0

(2) 1

(3) $\frac{1}{xyz}$ (4) xyz **Sol.** Answer (2)Let $\tan^{-1}x = A, \tan^{-1}y = B, \tan^{-1}z = C$ Then $A + B + C = \pi$

$$\frac{1}{xy} + \frac{1}{yz} + \frac{1}{zx} = \frac{x+y+z}{xyz}$$

$$= \frac{\tan A + \tan B + \tan C}{\tan A \tan B \tan C}$$

$$= 1 \quad (\because A + B + C = \pi \Rightarrow \tan A + \tan B + \tan C = \tan A \tan B \tan C)$$

60. If $x + y + z = xyz$, then $\tan^{-1}x + \tan^{-1}y + \tan^{-1}z$

(1) π (2) $\frac{\pi}{2}$

(3) 1

(4) $\frac{\pi}{3}$ **Sol.** Answer (1)Let $x = \tan A, y = \tan B, z = \tan C$ Then, $x + y + z = xyz$

$$\Rightarrow \tan A + \tan B + \tan C = \tan A \tan B \tan C$$

$$\Rightarrow \tan A + \tan B + \tan C - \tan A \tan B \tan C = 0$$

$$\Rightarrow \tan(A + B + C) = 0$$

$$\Rightarrow A + B + C = n\pi$$

$$\therefore A + B + C = \pi$$

61. If $\tan(\sin^{-1}\sqrt{1-x^2}) = \sin(\tan^{-1}2)$, then x is

(1) $-\frac{\sqrt{5}}{3}$

(2) $\frac{3}{\sqrt{10}}$

(3) $\frac{\sqrt{5}}{3}$

(4) Both (1) & (3)

Sol. Answer (4)

$$\tan(\sin^{-1}(\sqrt{1-x^2})) = \sin(\tan^{-1}2)$$

$$\sin^{-1}(\sqrt{1-x^2}) = \theta$$

$$\Rightarrow \sin\theta = \sqrt{1-x^2}$$

$$\tan\theta = \frac{\sqrt{1-x^2}}{x}$$

$$\text{Let } \tan^{-1}(2) = \phi$$

$$\tan\phi = 2$$

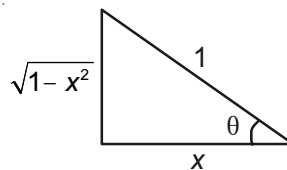
$$\Rightarrow \sin\phi = \frac{2}{\sqrt{5}}$$

$$\therefore \tan\theta = \sin\phi$$

$$\Rightarrow \frac{\sqrt{1-x^2}}{x} = \frac{2}{\sqrt{5}}$$

$$\Rightarrow 5(1-x^2) = 4x^2$$

$$\Rightarrow x = \pm \frac{\sqrt{5}}{3}$$



62. $\sin^{-1}(\sin x) = x$ if

(1) $x \in \mathbb{R}$

(2) $x \in [-1, 1]$

(3) $x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

(4) $x \in [-\pi, \pi]$

Sol. Answer (3)

$$\sin^{-1}(\sin x) = x$$

$$\Rightarrow x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

63. If $\sin^{-1}x + \sin^{-1}y = \frac{2\pi}{3}$, then $\cos^{-1}x + \cos^{-1}y$ equals

(1) $\frac{2\pi}{3}$

(2) $\frac{\pi}{3}$

(3) $\frac{\pi}{6}$

(4) π

Sol. Answer (2)

$$\begin{aligned}
 \cos^{-1} x + \cos^{-1} y &= \left(\frac{\pi}{2} - \sin^{-1} x \right) + \left(\frac{\pi}{2} - \sin^{-1} y \right) \\
 &= \pi - (\sin^{-1} x + \sin^{-1} y) \\
 &= \pi - \frac{2\pi}{3} \\
 &= \frac{\pi}{3}
 \end{aligned}$$

64. If $\sin^{-1}\left(x - \frac{x^2}{2} + \frac{x^3}{4} - \dots\right) + \cos^{-1}\left(x^2 - \frac{x^4}{2} + \frac{x^6}{4} - \dots\right) = \frac{\pi}{2}$ there $0 < |x| < \sqrt{2}$, then x equals

- (1) -1 (2) $-\frac{1}{2}$ (3) $\frac{1}{2}$ (4) 1

Sol. Answer (4)

$$\sin^{-1}\left(x - \frac{x^2}{2} + \frac{x^4}{4} - \dots\right) + \cos^{-1}\left(x^2 - \frac{x^4}{2} + \frac{x^6}{4} - \dots\right) = \frac{\pi}{2}$$

$$\sin^{-1} x + \cos^{-1} y = \frac{\pi}{2}$$

$$\Rightarrow x = y$$

$$\therefore x - \frac{x^2}{2} + \frac{x^4}{4} - \dots = x^2 - \frac{x^4}{2} + \frac{x^6}{4} - \dots$$

$$\Rightarrow \frac{x}{1 - \left(-\frac{x}{2}\right)} = \frac{x^2}{1 - \left(-\frac{x^2}{2}\right)}$$

$$\Rightarrow \frac{2x}{2+x} = \frac{2x^2}{2+x^2}$$

$$\Rightarrow x \neq 0$$

$$\Rightarrow 2 + x^2 = 2x + x^2$$

$$\Rightarrow x = 1$$

65. If $\sin\left(\sin^{-1}\frac{3\pi}{11} + \cos^{-1} x\right) = 1$, then x is

- (1) $-\frac{3\pi}{11}$ (2) $\frac{\pi}{2} - \frac{3\pi}{11}$ (3) $\frac{3\pi}{11}$ (4) $\frac{8\pi}{11}$

Sol. Answer (3)

$$\sin\left(\sin^{-1}\frac{3\pi}{11} + \cos^{-1} x\right) = 1$$

$$\Rightarrow \sin^{-1} \frac{3\pi}{11} + \cos^{-1} x = \sin^{-1} 1 = \frac{\pi}{2}$$

$$\Rightarrow \sin^{-1} \frac{3\pi}{11} = \frac{\pi}{2} - \cos^{-1} x = \sin^{-1} x$$

$$\Rightarrow x = \frac{3\pi}{11}$$

66. If $\sin^{-1}x + \sin^{-1}y + \sin^{-1}z = \frac{3\pi}{2}$, the value of $x^{100} + y^{100} + z^{100} - \frac{9}{x^{101} + y^{101} + z^{101}}$ is

(1) 0

(2) 1

(3) 2

(4) 3

Sol. Answer (1)

Since $-\frac{\pi}{2} \leq \sin^{-1} x \leq \frac{\pi}{2}$, etc.

Hence, $\sin^{-1} x + \sin^{-1} y + \sin^{-1} z = \frac{3\pi}{2}$, only when

$$\sin^{-1} x = \sin^{-1} y = \sin^{-1} z = \frac{\pi}{2}$$

$$\Rightarrow x = y = z = 1$$

Hence,

$$x^{100} + y^{100} + z^{100} - \frac{9}{x^{101} + y^{101} + z^{101}}$$

$$= 1 + 1 + 1 - \frac{9}{1 + 1 + 1}$$

$$= 3 - \frac{9}{3}$$

$$= 0$$

67. Let $x_i \in [-1, 1]$ for $i = 1, 2, 3, \dots, 24$, such that $\sin^{-1}x_1 + \sin^{-1}x_2 + \dots + \sin^{-1}x_{24} = 12\pi$ then the value of $x_1 + 2x_2 + 3x_3 + \dots + 24x_{24}$ is

(1) 276

(2) 300

(3) 325

(4) 351

Sol. Answer (2)

Since, $-\frac{\pi}{2} \leq \sin^{-1} x_i \leq \frac{\pi}{2}$

Therefore, $\sin^{-1} x_1 + \sin^{-1} x_2 + \dots + \sin^{-1} x_{24} \leq 12\pi$

Equality holds, when $\sin^{-1} x_1 = \sin^{-1} x_2 = \dots = \sin^{-1} x_{24} = \frac{\pi}{2}$

$$\Rightarrow x_1 = x_2 = \dots = x_{24} = 1$$

$$\begin{aligned}
 \therefore x_1 + 2x_2 + 3x_3 + \dots + 24x_{24} \\
 &= 1 + 2 + 3 + \dots + 24 \\
 &= \frac{24}{2}(24+1) \\
 &= 300
 \end{aligned}$$

68. How many solutions does the equation $5\tan^{-1}x + 3\cot^{-1}x = 2\pi$ have?

- (1) Zero (2) Exactly one (3) Exactly two (4) Infinite

Sol. Answer (2)

$$\begin{aligned}
 5\tan^{-1}x + 3\cot^{-1}x &= 2\pi \\
 \Rightarrow 2\tan^{-1}x + 3(\tan^{-1}x + \cot^{-1}x) &= 2\pi \\
 \Rightarrow 2\tan^{-1}x &= 2\pi - \frac{3\pi}{2} = \frac{\pi}{2} \\
 \Rightarrow \tan^{-1}x &= \frac{\pi}{4} \\
 \Rightarrow x &= 1 \\
 \therefore \text{Exactly one solution}
 \end{aligned}$$

69. The equation $\sin^{-1}x - \cos^{-1}x = \cos^{-1}\left(\frac{\sqrt{3}}{2}\right)$ has

- (1) No solutions (2) Unique solution
(3) Infinite number of solution (4) Two solutions

Sol. Answer (2)

$$\begin{aligned}
 \sin^{-1}x - \cos^{-1}x &= \cos^{-1}\left(\frac{\sqrt{3}}{2}\right) \\
 \Rightarrow 2\sin^{-1}x - \frac{\pi}{2} &= \frac{\pi}{6} \\
 \Rightarrow \sin^{-1}x &= \frac{\pi}{3} \\
 \Rightarrow x &= \frac{\sqrt{3}}{2} \\
 \therefore \text{Unique solution exists.}
 \end{aligned}$$

70. $\cot^{-1}(\sqrt{\cos\alpha}) - \tan^{-1}(\sqrt{\cos\alpha}) = x$, $x \geq 0$, then $\sin x$ equals

- (1) $\tan^2 \frac{\alpha}{2}$ (2) $\cot^2 \frac{\alpha}{2}$ (3) $\tan \alpha$ (4) $\cot \frac{\alpha}{2}$

Sol. Answer (1)

$$\cot^{-1} \sqrt{\cos \alpha} - \tan^{-1} \sqrt{\cos \alpha} = x, x \geq 0$$

$$\Rightarrow \frac{\pi}{2} - 2 \tan^{-1} \sqrt{\cos \alpha} = x \left(\because \cot^{-1} x = \frac{\pi}{2} - \tan^{-1} x \right)$$

$$\Rightarrow \frac{\pi}{2} - x = 2 \tan^{-1} \sqrt{\cos \alpha}$$

$$\therefore \sin x = \cos \left(\frac{\pi}{2} - x \right) = \cos (2 \tan^{-1} \sqrt{\cos \alpha})$$

$$= \cos \left\{ \cos^{-1} \frac{1 - \cos \alpha}{1 + \cos \alpha} \right\} \quad \left(\because 2 \tan^{-1} x = \cos^{-1} \frac{1 - x^2}{1 + x^2} \right)$$

$$= \tan^2 \frac{\alpha}{2}$$

71. The equation $\sin^{-1} x = 2 \sin^{-1} \alpha$ has a solution

(1) For all values of α (2) $|\alpha| < \frac{1}{2}$ (3) $|\alpha| \geq \frac{1}{\sqrt{2}}$ (4) $-\frac{1}{\sqrt{2}} \leq \alpha \leq \frac{1}{\sqrt{2}}$

Sol. Answer (4)

Since, $-\frac{\pi}{2} \leq \sin^{-1} x \leq \frac{\pi}{2}$, the equation

$$\sin^{-1} x = 2 \sin^{-1} \alpha \text{ has a solution, if } -\frac{\pi}{2} \leq 2 \sin^{-1} \alpha \leq \frac{\pi}{2}$$

$$\Rightarrow -\frac{\pi}{4} \leq \sin^{-1} \alpha \leq \frac{\pi}{4}$$

$$\Rightarrow -\frac{1}{\sqrt{2}} \leq \alpha \leq \frac{1}{\sqrt{2}}$$

72. If $\left| \cos^{-1} \left(\frac{1 - x^2}{1 + x^2} \right) \right| < \frac{\pi}{3}$, then x belongs to the interval

(1) $\left[-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right]$

(2) $\left(-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right)$

(3) $\left(0, \frac{1}{\sqrt{3}} \right)$

(4) $[1, 2]$

Sol. Answer (2)

$$\left| \cos^{-1} \left(\frac{1 - x^2}{1 + x^2} \right) \right| < \frac{\pi}{3}$$

$$\Rightarrow |2 \tan^{-1} x| < \frac{\pi}{3}$$

$$\Rightarrow \frac{-\pi}{3} < 2 \tan^{-1} x < \frac{\pi}{3}$$

$$\Rightarrow \frac{-\pi}{6} < \tan^{-1} x < \frac{\pi}{6}$$

$$\Rightarrow \frac{-1}{\sqrt{3}} < x < \frac{1}{\sqrt{3}}$$

73. The set of values of x satisfying

$$|\sin^{-1} x| \leq |\cos^{-1} x| \text{ is}$$

$$(1) \left[-1, \frac{1}{\sqrt{2}}\right]$$

$$(2) \left[-1, -\frac{1}{\sqrt{2}}\right] \cup \left[\frac{1}{\sqrt{2}}, 1\right]$$

$$(3) \left(-1, \frac{1}{\sqrt{2}}\right)$$

$$(4) \left[\frac{1}{\sqrt{2}}, 1\right] \text{ only}$$

Sol. Answer (1)

$$\text{Since, } \cos^{-1} x \in [0, \pi]$$

$$\text{Therefore, } |\cos^{-1} x| = \cos^{-1} x$$

$$\sin^{-1} x = \begin{cases} \sin^{-1} x & , 0 \leq x \leq 1 \\ -\sin^{-1} x & , -1 \leq x < 0 \end{cases}$$

If $-1 \leq x < 0$, then

$$-\sin^{-1} x \leq \cos^{-1} x$$

$$\Rightarrow 0 \leq \sin^{-1} x + \cos^{-1} x$$

$$\Rightarrow 0 \leq \frac{\pi}{2}$$

which is always true.

$$\text{Hence, } x \in [-1, 0) \quad \dots(1)$$

If $0 \leq x \leq 1$, then

$$\sin^{-1} x \leq \cos^{-1} x \Rightarrow \sin^{-1} x \leq \frac{\pi}{2} - \sin^{-1} x$$

$$\Rightarrow \sin^{-1} x \leq \frac{\pi}{4}$$

$$\Rightarrow x \leq \frac{1}{\sqrt{2}}$$

$$\Rightarrow 0 \leq x \leq \frac{1}{\sqrt{2}} \quad \dots(2)$$

Combining (1) & (2)

$$-1 \leq x \leq \frac{1}{\sqrt{2}}$$

74. The solution of the equation $\cos^{-1}(x\sqrt{3}) + \cos^{-1}x = \frac{\pi}{2}$ is given by

(1) $x = \frac{1}{2}$

(2) $x = -\frac{1}{2}$

(3) $x = \pm \frac{1}{2}$

(4) $x = \frac{1}{4}$

Sol. Answer (1)

$$\cos^{-1}(x\sqrt{3}) + \cos^{-1}x = \frac{\pi}{2}$$

$$\Rightarrow \cos^{-1}(x\sqrt{3}) = \frac{\pi}{2} - \cos^{-1}x = \sin^{-1}x$$

$$\Rightarrow x\sqrt{3} = \sqrt{1-x^2}$$

$$\text{Squaring, } 3x^2 = 1 - x^2$$

$$\Rightarrow 4x^2 = 1 \Rightarrow x = \pm \frac{1}{2}$$

But, $x = -\frac{1}{2}$ does not satisfy the given equation

$\therefore x = \frac{1}{2}$ is the only solution.

75. The solution of $\sin^{-1}\left(\frac{2a}{1+a^2}\right) - \cos^{-1}\left(\frac{1-b^2}{1+b^2}\right) = \tan^{-1}\left(\frac{2x}{1-x^2}\right)$ is

(1) $\frac{a-b}{1-ab}$

(2) $\frac{1+ab}{a-b}$

(3) $\frac{ab-1}{a+b}$

(4) $\frac{a-b}{1+ab}$

Sol. Answer (4)

$$\sin^{-1}\left(\frac{2a}{1+a^2}\right) - \cos^{-1}\left(\frac{1-b^2}{1+b^2}\right) = \tan^{-1}\left(\frac{2x}{1-x^2}\right)$$

$$\Rightarrow 2\tan^{-1}a - 2\tan^{-1}b = 2\tan^{-1}x$$

$$\Rightarrow x = \frac{a-b}{1+ab}$$

76. Solution of $\sin^{-1}x = \cos^{-1}x$ is $\sin^{-1}(1-x) - 2\sin^{-1}x = \frac{\pi}{2}$

- (1) Less than $\frac{1}{2}$ (2) Less than $\frac{1}{3}$ (3) Greater than $\frac{1}{3}$ (4) Greater than $\frac{4}{5}$

Sol. Answer (3)

$$\sin^{-1}x = \cos^{-1}x$$

$$\Rightarrow \sin^{-1}x = \sin^{-1}\sqrt{1-x^2}$$

$$\Rightarrow x = \sqrt{1-x^2}$$

$$\Rightarrow 2x^2 = 1 \Rightarrow x^2 = \frac{1}{2}$$

$$\Rightarrow x = \pm \frac{1}{\sqrt{2}}$$

$$\therefore x = \frac{1}{\sqrt{2}} \text{ is a solution.}$$

77. Number of solution of the equation $\sin\left(\frac{1}{5}\cos^{-1}x\right) = 1$ is

- (1) 1 (2) 0 (3) 2 (4) Infinitely many

Sol. Answer (2)

Given equation is

$$\sin\left(\frac{1}{5}\cos^{-1}x\right) = 1$$

$$\text{i.e., } \frac{1}{5}\cos^{-1}x = (4n+1)\cdot\frac{\pi}{2} \quad n \in I$$

$$\text{i.e., } \cos^{-1}x = \frac{5}{2}\cdot(4n+1)\pi$$

which is not possible as $0 \leq \cos^{-1}x \leq \pi$.

$\therefore$ No solution of given equation is possible.

[Miscellaneous]

78. Minimum value of $n \in N$ such that $\cot^{-1}\left(\frac{n^2 - 10n + 21.6}{\pi}\right) > \frac{\pi}{6}$ is

- (1) 4 (2) 3 (3) 5 (4) 2

Sol. Answer (2)

We have,

$$\cot^{-1}\left(\frac{n^2 - 10n + 21.6}{\pi}\right) > \frac{\pi}{6}$$

$$\Rightarrow \frac{n^2 - 10n + 21.6}{\pi} < \cot\left(\frac{\pi}{6}\right)$$

$$\Rightarrow n^2 - 10n + 21.6 < \pi\sqrt{3}$$

$$\Rightarrow (n - 5)^2 < \pi\sqrt{3} + 3.4$$

$$\Rightarrow -\sqrt{\pi\sqrt{3} + 3.4} < n - 5 < \sqrt{\pi\sqrt{3} + 3.4}$$

$$\Rightarrow 5 - \sqrt{\pi\sqrt{3} + 3.4} < n < \sqrt{\pi\sqrt{3} + 3.4} + 5$$

Since, $\sqrt{3} \cdot \pi = 5.5$ nearly

$$\sqrt{\sqrt{3}\pi + 3.4} \sim \sqrt{8.9} \sim 2.9$$

$$2.4 < n < 7.9$$

$$\Rightarrow n \in \{3, 4, 5, 6, 7\}$$

Clearly, least value of n is 3.

79. The set of values of k for which $x^2 - kx + \sin^{-1}(\sin 4) > 0$ for all values of x

(1) No values of k is possible

(2) $[-4(\pi - 4), 4(\pi - 4)]$

(3) $[0, 1]$

(4) All real values of k

Sol. Answer (1)

We know that

$$\sin^{-1}(\sin 4) = \sin^{-1}(\sin(\pi - 4)) = \pi - 4$$

We have, $x^2 - kx + \pi - 4 > 0 \quad x \in \mathbb{R}$

$$\Rightarrow D < 0 \Rightarrow k^2 - 4(\pi - 4) < 0$$

$$\Rightarrow k^2 + 4(4 - \pi) < 0$$

which is not true for any real value of k .

80. Number of solutions of the equation $2(\sin^{-1} x)^2 - \sin^{-1} x - 6 = 0$ is

(1) 2

(2) 1

(3) 0

(4) 3

Sol. Answer (2)

$$\text{Solving } 2(\sin^{-1} x)^2 - \sin^{-1} x - 6 = 0$$

$$\text{either } \sin^{-1} x = 2 \text{ or } \sin^{-1} x = -1.5$$

$$\sin^{-1} x \text{ cannot be greater than } \frac{\pi}{2}$$

$$\therefore \sin^{-1} x \neq 2$$

Only one solution is there

$$x = \sin(-1.5)$$

81. $\cos^{-1}(2x^2 - 1)$, $0 \leq x \leq 1$, is equal to

- (1) $2\cos^{-1}x$ (2) $2\sin^{-1}x$ (3) $\pi - 2\cos^{-1}x$ (4) $\pi + 2\cos^{-1}x$

Sol. Answer (1)

Let $x = \cos\theta$

$$\begin{aligned}\cos^{-1}(2x^2 - 1) &= \cos^{-1}(\cos 2\theta) \\ &= 2\theta \\ &= 2\cos^{-1}x\end{aligned}$$

if $0 \leq 2\theta \leq \pi$

$$0 \leq \theta \leq \frac{\pi}{2}$$

$$0 \leq x \leq 1$$

82. $\sin^{-1}(2x\sqrt{1-x^2})$, $x \in \left[\frac{1}{\sqrt{2}}, 1\right]$ is equal to

- (1) $2\sin^{-1}x$ (2) $2\cos^{-1}x$ (3) $-2\sin^{-1}x$ (4) $-2\cos^{-1}x$

Sol. Answer (2)

Let $x = \cos\theta$

$$\begin{aligned}\sin^{-1}(2\cos\theta\sin\theta) &= \sin^{-1}(\sin 2\theta) \\ &= 2\theta, \\ &= 2\cos^{-1}x\end{aligned}$$

$$\text{if } -\frac{\pi}{2} \leq 2\theta \leq \frac{\pi}{2}$$

$$-\frac{\pi}{4} \leq \theta \leq \frac{\pi}{4}$$

$$\frac{1}{\sqrt{2}} \leq x \leq 1$$

83. $\tan^{-1}\left(\frac{\sqrt{1+x^2} + \sqrt{1-x^2}}{\sqrt{1+x^2} - \sqrt{1-x^2}}\right)$, $|x| \leq \frac{1}{\sqrt{2}}$, is equal to

- (1) $\frac{1}{2}\cos^{-1}x^2$ (2) $\frac{\pi}{4} - \frac{1}{2}\cos^{-1}x^2$ (3) $\frac{\pi}{4} + \frac{1}{2}\cos^{-1}x^2$ (4) $\frac{\pi}{2} - \frac{1}{2}\cos^{-1}x^2$

Sol. Answer (3)

Let $x^2 = \cos 2\theta$

$$\Rightarrow \tan^{-1}\left(\frac{\sqrt{1+\cos 2\theta} + \sqrt{1-\cos 2\theta}}{\sqrt{1+\cos 2\theta} - \sqrt{1-\cos 2\theta}}\right)$$

$$= \tan^{-1}\left(\frac{\cos\theta + \sin\theta}{\cos\theta - \sin\theta}\right) \quad \left(\because x^2 \leq \frac{1}{2} \Rightarrow \cos 2\theta \leq \frac{1}{2} \Rightarrow \frac{\pi}{4} \leq 2\theta \leq \frac{3\pi}{4}\right)$$

$$= \tan^{-1} \left(\frac{1 + \tan \theta}{1 - \tan \theta} \right)$$

$$= \tan^{-1} \left(\tan \left(\frac{\pi}{4} + \theta \right) \right) \quad \left(\because \frac{\pi}{8} \leq \theta \leq \frac{\pi}{4} \therefore \frac{3\pi}{8} \leq \theta + \frac{\pi}{4} \leq \frac{\pi}{2} \right)$$

$$= \frac{\pi}{4} + \theta$$

$$= \frac{\pi}{4} + \frac{1}{2} \cos^{-1} x^2$$

84. If a, b, c are real positive numbers and $\theta = \tan^{-1} \left[\frac{a(a+b+c)}{bc} \right]^{\frac{1}{2}} + \tan^{-1} \left[\frac{b(a+b+c)}{ca} \right]^{\frac{1}{2}} + \tan^{-1} \left[\frac{c(a+b+c)}{ab} \right]^{\frac{1}{2}}$,

then $\tan \theta$ equals

- (1) 0 (2) 1 (3) $\frac{a+b+c}{abc}$ (4) 2

Sol. Answer (1)

Let $\left(\frac{a+b+c}{abc} \right)^{\frac{1}{2}} = k$

Then $\theta = \tan^{-1}(ak) + \tan^{-1}(bk) + \tan^{-1}(ck)$

If $\tan^{-1}(ak) = A, \tan^{-1}(bk) = B, \tan^{-1}(ck) = C$, then

$\theta = A + B + C$

Hence, $\tan \theta = \tan(A + B + C)$

$$= \frac{\tan A + \tan B + \tan C - \tan A \tan B \tan C}{1 - \tan A \tan B - \tan B \tan C - \tan C \tan A}$$

$$= 0$$

$\therefore \tan A + \tan B + \tan C - \tan A \tan B \tan C$

$= ak + bk + ck - (abc)k^3$

$= (a+b+c)k - abc \left(\frac{a+b+c}{abc} \right) k, \quad k^2 = \frac{a+b+c}{abc}$

$= (a+b+c)k - (a+b+c)k$

$= 0$

85. Let $\cos^{-1} \frac{x}{2} + \cos^{-1} \frac{y}{3} = \theta$ and denote by $f(x, y, \theta) = 0$ the rational integral expression in x and y . Then for $\theta = \frac{\pi}{2}$ the

locus represented by $f\left(x, y, \frac{\pi}{2}\right) = 0$ is

- (1) An ellipse (2) A parabola (3) A hyperbola (4) A pair of lines

Sol. Answer (1)

$$\cos^{-1}\left(\frac{x}{2}\right) + \cos^{-1}\left(\frac{y}{3}\right) = \theta$$

$$\Rightarrow 9x^2 - 12xy \cos \theta + 4y^2 = 36 \sin^2 \theta$$

for $\theta = \frac{\pi}{2}$, we have

$$9x^2 + 4y^2 = 36$$

$$\Rightarrow \frac{x^2}{4} + \frac{y^2}{9} = 1, \text{ which is an ellipse.}$$

86. $2 \tan^{-1} \left[\left(\frac{a-b}{a+b} \right)^{\frac{1}{2}} \tan \left(\frac{\theta}{2} \right) \right]$ equals

(1) $\cos^{-1} \left(\frac{a \cos \theta + b}{a + b \cos \theta} \right)$ (2) $\cos^{-1} \left(\frac{a + b \cos \theta}{a \cos \theta + b} \right)$ (3) $\cos^{-1} \left(\frac{a \cos \theta}{a + b \cos \theta} \right)$ (4) $\cos^{-1} \left(\frac{b \cos \theta}{a \cos \theta + b} \right)$

Sol. Answer (1)

Let $2 \tan^{-1} \left[\sqrt{\frac{a-b}{a+b}} \tan \frac{\theta}{2} \right] = \phi$

Then, $\sqrt{\frac{a-b}{a+b}} \tan \frac{\theta}{2} = \tan \frac{\phi}{2}$

$$\cos \phi = \frac{1 - \tan^2 \frac{\phi}{2}}{1 + \tan^2 \frac{\phi}{2}} = \frac{1 - \left(\frac{a-b}{a+b} \right) \tan^2 \frac{\theta}{2}}{1 + \left(\frac{a-b}{a+b} \right) \tan^2 \frac{\theta}{2}}$$

$$= \frac{(a+b) - (a-b) \tan^2 \frac{\theta}{2}}{(a+b) + (a-b) \tan^2 \frac{\theta}{2}}$$

$$= \frac{a \left(1 - \tan^2 \frac{\theta}{2} \right) + \left(1 + \tan^2 \frac{\theta}{2} \right)}{a \left(1 + \tan^2 \frac{\theta}{2} \right) + b \left(1 - \tan^2 \frac{\theta}{2} \right)}$$

$$= \frac{a \left(\frac{1 - \tan^2 \frac{\theta}{2}}{1 + \tan^2 \frac{\theta}{2}} \right) + b}{a + b \left(\frac{1 - \tan^2 \frac{\theta}{2}}{1 + \tan^2 \frac{\theta}{2}} \right)} = \frac{a \cos \theta + b}{a + b \cos \theta}$$

$$\therefore \phi = \cos^{-1} \left(\frac{a \cos \theta + b}{a + b \cos \theta} \right)$$

87. The value of $\cos(2 \cos^{-1} 0.8)$ is

(1) 0.48

(2) 0.96

(3) 0.6

(4) 0.28

Sol. Answer (4)

$$\begin{aligned}\cos(2 \cos^{-1} 0.8) &= \cos(\cos^{-1}(2(0.8)^2 - 1)) \\ &= 2(0.8)^2 - 1 \\ &= 2(0.64) - 1 \\ &= 1.28 - 1 \\ &= 0.28\end{aligned}$$

88. The sum $\sum_{n=1}^{\infty} \tan^{-1}\left(\frac{1}{2^n + 2^{1-n}}\right)$ equals

(1) $\frac{\pi}{2}$

(2) $\frac{\pi}{4}$

(3) $\frac{\pi}{6}$

(4) $\frac{\pi}{3}$

Sol. Answer (2)

$$\frac{1}{2^n + 2^{1-n}} = \frac{2^{n-1}}{1 + 2^n \cdot 2^{n-1}} = \frac{(2-1) \cdot 2^{n-1}}{1 + 2^n \cdot 2^{n-1}} = \frac{2^n - 2^{n-1}}{1 + 2^n \cdot 2^{n-1}}$$

$$\therefore \sum_{n=1}^{\infty} \tan^{-1}\left(\frac{1}{2^n + 2^{1-n}}\right)$$

$$= \sum_{n=1}^{\infty} \{\tan^{-1}(2^n) - \tan^{-1}(2^{n-1})\}$$

$$= \lim_{n \rightarrow \infty} \sum_{n=1}^n \{\tan^{-1}(2^n) - \tan^{-1}(2^{n-1})\}$$

$$= \lim_{n \rightarrow \infty} \{\tan^{-1}(2^n) - \tan^{-1}(1)\}$$

$$= \frac{\pi}{2} - \frac{\pi}{4}$$

$$= \frac{\pi}{4}$$

89. The sum $\tan^{-1} \frac{1}{1+x+x^2} + \tan^{-1} \left(\frac{1}{3+3x+x^2} \right) + \tan^{-1} \left(\frac{1}{7+5x+x^2} \right) + \tan^{-1} \frac{1}{13+7x+x^2} + \dots$ of first 100 terms of the series is

(1) $\tan^{-1} \left(\frac{100}{1+x^2+100x} \right)$

(2) $\tan^{-1} \left(\frac{2x-100}{1-x^2-100x} \right)$

(3) $\tan^{-1} \left(\frac{100}{1+x^2-100x} \right)$

(4) $\tan^{-1} \left(\frac{2x+100}{1+x^2-100x} \right)$

Sol. Answer (1)

Consider the series

$$\begin{aligned}
 & \tan^{-1} \frac{1}{1+x+x^2} + \tan^{-1} \frac{1}{3+3x+x^2} + \tan^{-1} \frac{1}{7+5x+x^2} + \dots + 100 \text{ term} \\
 &= \tan^{-1} \left[\frac{(x+1)-x}{1+(x+1) \cdot x} \right] + \tan^{-1} \left[\frac{(x+2)-(x+1)}{1+(x+2)(x+1)} \right] + \tan^{-1} \left[\frac{(x+3)-(x+2)}{1+(x+3)(x+2)} \right] + \dots + \tan^{-1} \left[\frac{(x+100)-(x+99)}{1+(x+99)(x+100)} \right] \\
 &= [\tan^{-1}(x+1) - \tan^{-1}x] + [\tan^{-1}(x+2) - \tan^{-1}(x+1)] + [\tan^{-1}(x+3) - \tan^{-1}(x+2)] + \dots + \tan^{-1}(x+100) - \tan^{-1}(x+99) \\
 &= \tan^{-1}(x+100) - \tan^{-1}x \\
 &= \tan^{-1} \left(\frac{100}{1+x^2+100x} \right)
 \end{aligned}$$

90. Sum of infinite series $\sin^{-1} \frac{1}{\sqrt{2}} + \sin^{-1} \left(\frac{\sqrt{2}-1}{\sqrt{6}} \right) + \sin^{-1} \left(\frac{\sqrt{3}-\sqrt{2}}{2\sqrt{3}} \right) + \dots + \sin^{-1} \left(\frac{\sqrt{n}-\sqrt{n-1}}{\sqrt{n(n+1)}} \right) + \dots$ is

- (1) $\frac{\pi}{3}$ (2) $\frac{\pi}{2}$ (3) $\frac{\pi}{4}$ (4) $\frac{\pi}{6}$

Sol. Answer (2)

Here, $T_n = \sin^{-1} \left[\frac{\sqrt{n}-\sqrt{n-1}}{\sqrt{n(n+1)}} \right]$

$$= \sin^{-1} \left[\frac{1}{\sqrt{n}} \left(\sqrt{1-\frac{1}{1+n}} \right) - \frac{1}{\sqrt{1+n}} \cdot \sqrt{1-\frac{1}{n}} \right]$$

$$= \sin^{-1} \frac{1}{\sqrt{n}} - \sin^{-1} \frac{1}{\sqrt{1+n}}$$

Putting $n = 1, 3, 5, \dots$ and addition.

Clearly sum is

$$T_1 + T_2 + T_3 + T_4 + \dots$$

$$= \sin^{-1} 1 - \sin^{-1} \frac{1}{\sqrt{2}} + \sin^{-1} \frac{1}{\sqrt{2}} - \sin^{-1} \frac{1}{\sqrt{3}} + \dots$$

$$= \sin^{-1} 1 = \frac{\pi}{2}$$

91. If x_1, x_2, x_3, x_4 are the roots of the equation $x^4 - x^3 \sin 2\beta + x^2 \cos 2\beta - x \cos \beta - \sin \beta = 0$, then $\tan^{-1} x_1 + \tan^{-1} x_2 + \tan^{-1} x_3 + \tan^{-1} x_4$ is equal to

- (1) $\frac{\pi}{2} - \beta$ (2) $\frac{\pi}{2} + \beta$ (3) $\frac{\pi}{3} - \beta$ (4) $\frac{\pi}{6} - 2\beta$

Sol. Answer (1)

The given equation is

$$x^4 - x^3 \sin 2\beta + x^2 \cos 2\beta - x \cos \beta - \sin \beta = 0$$

Here we have,

$$\sum x_i = \sin 2\beta \quad \sum x_1 x_2 = -\cos 2\beta$$

$$\sum x_1 x_2 x_3 = \cos \beta$$

$$x_1 x_2 x_3 x_4 = -\sin \beta$$

$$\tan \alpha_1 = x_1, \tan \alpha_2 = x_2, \tan \alpha_3 = x_3, \tan \alpha_4 = x_4$$

$$\begin{aligned} \text{then } \tan(\alpha_1 + \alpha_2 + \alpha_3 + \alpha_4) &= \frac{\sum x_i - \sum x_1 x_2 x_3}{1 - \sum x_1 x_2 + \sum x_1 x_2 x_3 x_4} \\ &= \frac{\sin 2\beta - \cos \beta}{1 - \cos 2\beta - \sin \beta} = \cot \beta = \tan\left(\frac{\pi}{2} - \beta\right) \end{aligned}$$

$$\Rightarrow \alpha_1 + \alpha_2 + \alpha_3 + \alpha_4 = \frac{\pi}{2} - \beta$$

$$\Rightarrow \tan^{-1} x_1 + \tan^{-1} x_2 + \tan^{-1} x_3 + \tan^{-1} x_4 = \frac{\pi}{2} - \beta$$

92. The number of solutions of (x, y) which satisfy $|y| = \sin x$ and $y = \cos^{-1}(\cos x)$, when $-2\pi \leq x \leq 2\pi$

(1) 1

(2) 2

(3) 3

(4) Infinite

Sol. Answer (3)

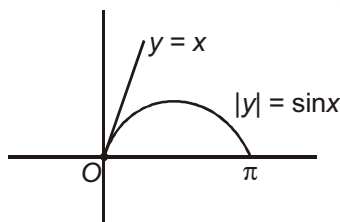
$$\text{In } [0, \pi], |y| = \sin x, y = \cos^{-1}(\cos x) = x$$

$$\text{In } [\pi, 2\pi], |y| = \sin x, y = \cos^{-1}(\cos x) = 2\pi - x$$

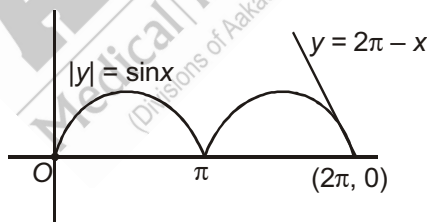
$$\text{In } [-\pi, 0], |y| = \sin x, y = \cos^{-1}(\cos x) = -x$$

$$\text{In } [-2\pi, -\pi], |y| = \sin x, y = \cos^{-1}(\cos x) = 2\pi + x$$

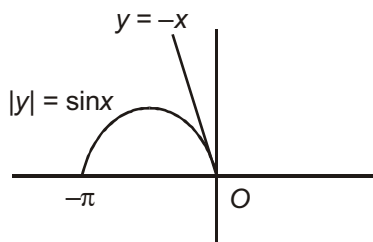
Plotting the graph



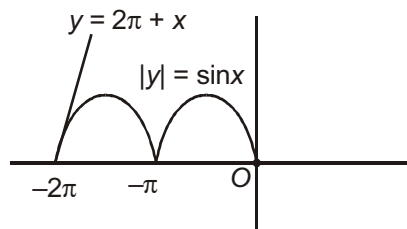
One solution $(0, 0)$



One solution $(2\pi, 0)$



One solution $(0, 0)$



Solution $(-2\pi, 0)$

Clearly there are three solutions.

93. The sum of the maximum and minimum values of the function $f(x) = \sin^{-1}4x + \cos^{-1}4x + \sec^{-1}4x$ is

(1) 2π

(2) π

(3) $\frac{3\pi}{2}$

(4) $\frac{\pi}{2}$

Sol. Answer (1)

Domain of $f(x) \Rightarrow x \in \left\{-\frac{1}{4}, \frac{1}{4}\right\}$ only

$$\therefore f(x) \text{ is min when } x = \frac{1}{4} \text{ i.e., } f_{\min}\left(\frac{1}{4}\right) = \frac{\pi}{2}.$$

$$\text{and } f(x) \text{ is max when } x = -\frac{1}{4} \text{ i.e., } f_{\max}\left(-\frac{1}{4}\right) = \frac{3\pi}{2}$$

$\therefore$ Sum of maximum and minimum value of function is 2π .

94. If $0 < x < 1$, then $\sqrt{1+x^2} \cdot [x \cos(\cot^{-1} x) + \sin(\cot^{-1} x)]^2 - 1]^{\frac{1}{2}} =$

[IIT-JEE 2008]

(1) $\frac{x}{\sqrt{1+x^2}}$

(2) x

(3) $x\sqrt{1+x^2}$

(4) $\sqrt{1+x^2}$

Sol. Answer (3)

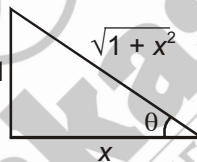
We have $0 < x < 1$,

Let $\cot^{-1} x = \theta$

$$\Rightarrow \cot \theta = x$$

$$\Rightarrow \sin \theta = \frac{1}{\sqrt{1+x^2}} = \sin(\cot^{-1} x)$$

$$\text{and } \cos \theta = \frac{x}{\sqrt{1+x^2}} = \cos(\cot^{-1} x)$$



$$\text{Now } \sqrt{1+x^2} [x \cos(\cot^{-1} x) + \sin(\cot^{-1} x)]^2 - 1]^{\frac{1}{2}}$$

$$= \sqrt{1+x^2} \left[\left\{ x \cdot \frac{x}{\sqrt{1+x^2}} + \frac{1}{\sqrt{1+x^2}} \right\}^2 - 1 \right]^{\frac{1}{2}}$$

$$= \sqrt{1+x^2} \left[\left(\frac{1+x^2}{\sqrt{1+x^2}} \right)^2 - 1 \right]^{\frac{1}{2}}$$

$$= \sqrt{1+x^2} [1+x^2 - 1]^{\frac{1}{2}}$$

$$= x\sqrt{1+x^2}$$

95. The value of $\cot\left(\sum_{n=1}^{23} \cot^{-1}\left(1 + \sum_{k=1}^n 2k\right)\right)$ is [JEE(Advanced)-2013]

- (1) $\frac{23}{25}$ (2) $\frac{25}{23}$ (3) $\frac{23}{24}$ (4) $\frac{24}{23}$

Sol. Answer (2)

$$\sum_{n=1}^{23} \cot^{-1}(1 + k(k+1)) = \sum_{n=1}^{23} \tan^{-1}(k+1) - \tan^{-1}(k) = \tan^{-1}(24) - \tan^{-1}(1)$$

$$\text{Now, } \cot\left(\tan^{-1}(24) - \tan^{-1}(1)\right) = \cot\left(\tan^{-1}\left(\frac{24-1}{1+24 \cdot 1}\right)\right) = \cot\left(\tan^{-1}\left(\frac{23}{25}\right)\right) = \frac{25}{23}$$

96. If $x_1, x_2, x_3, x_4, x_5, x_6$ all are independent then the maximum and minimum values of $[\sin^{-1}x_1] + [\cos^{-1}x_2] + [\tan^{-1}x_3] + [\cot^{-1}x_4] + [\sec^{-1}x_5] + [\operatorname{cosec}^{-1}x_6]$, where $[\]$ represents greatest integer function, respectively are

- (1) 9, 3 (2) 11, 5 (3) 12, -6 (4) 12, -3

Sol. Answer (3)

$$[\sin^{-1}x_1] = -2, -1, 0, 1$$

$$[\cos^{-1}x_2] = 0, 1, 2, 3$$

$$[\tan^{-1}x_3] = -2, -1, 0, 1$$

$$[\cot^{-1}x_4] = +0, 1, 2, 3$$

$$[\sec^{-1}x_5] = 0, 1, 2, 3$$

$$[\operatorname{cosec}^{-1}x_6] = -2, -1, 0, 1$$

$$\text{Hence minimum value} = -2 + 0 - 2 + 0 - 2 + 0 = -6$$

$$\text{Maximum value} = 1 + 3 + 1 + 3 + 1 + 3 = 12$$

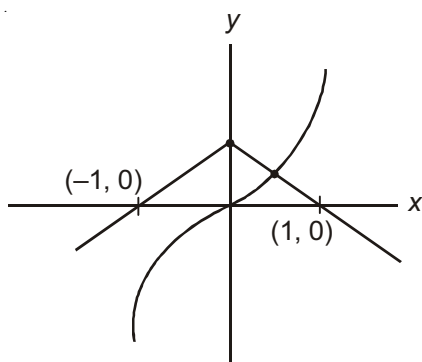
97. If the number of solutions of $\sin^{-1}x + |x| = 1$, $\cos^{-1}x + |x| = 1$, $\tan^{-1}x + |x| = 1$, $\cot^{-1}x + |x| = 1$, $\sec^{-1}x + |x| = 1$ and $\operatorname{cosec}^{-1}x + |x| = 1$ are $n_1, n_2, n_3, n_4, n_5, n_6$ respectively, then the value of $n_1 + n_2 + n_3 + n_4 + n_5 + n_6$ is

- (1) 6 (2) 7 (3) 8 (4) 9

Sol. Answer (1)

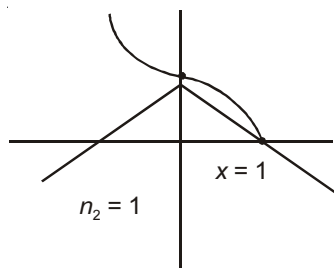
$$(i) \sin^{-1}x + |x| = 1 \Rightarrow \sin^{-1}x = 1 - |x|$$

To find the number of solution we draw the graph of $y = \sin^{-1}x$ and $y = 1 - |x|$

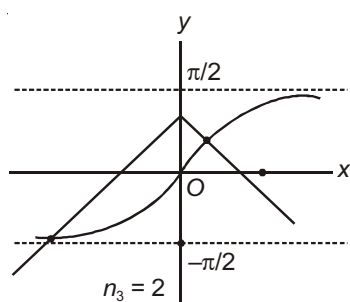


Only one point of intersection is there. Hence $n_1 = 1$

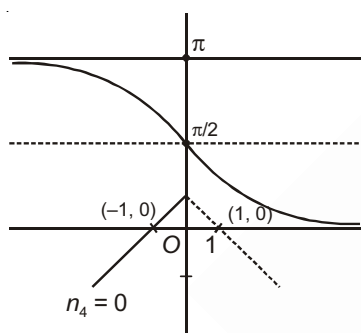
(ii) $\cos^{-1}x = 1 - |x|$



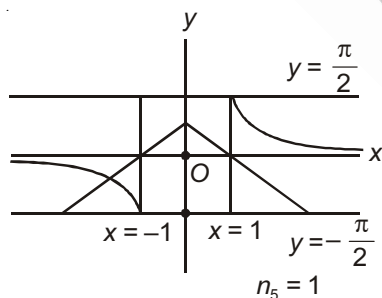
(iii)



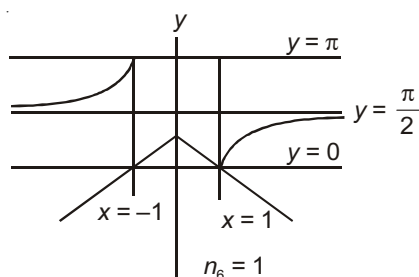
(iv) $y = \cot^{-1}x, y = 1 - |x|$



(v) $y = \operatorname{cosec}^{-1}x, y = 1 - |x|$



(vi) $y = \sec^{-1}x, y = 1 - |x|$



$$n_1 + n_2 + n_3 + n_4 + n_5 + n_6 = 6$$

98. If $x + \frac{1}{x} = 2$ and $[\sin^{-1}y] = -2$, where $[]$ represents the greatest integer function, then the minimum value of $\sin^{-1}x + \sin^{-1}y$ is

- (1) $\frac{\pi}{4}$ (2) 0 (3) $-\pi$ (4) $\frac{3\pi}{2}$

Sol. Answer (2)

$$x + \frac{1}{x} = 2$$

$$\Rightarrow x = 1$$

$$\therefore \sin^{-1}x = \frac{\pi}{2}$$

$$\text{and } -\frac{\pi}{2} \leq \sin^{-1}y \leq -1$$

$$\text{Hence minimum value} = \frac{\pi}{2} - \frac{\pi}{2} = 0.$$

99. Let $[]$ represents the greatest integer function and $[\cos^{-1} \sin^{-1} \tan^{-1}x] = 1$ then 'x' lies in the interval.

- (1) $[\sin \tan \cos 2, \tan \sin \cos 1]$ (2) $[\tan \sin \cos 2, \tan \sin \cos 1]$
 (3) $[\tan \sin \cos 1, \tan \sin \cos 2]$ (4) $(\tan \sin \cos 2, \tan \sin \cos 1]$

Sol. Answer (4)

$$1 \leq \cos^{-1} \sin^{-1} \tan^{-1}x < 2$$

$$\Rightarrow \cos 2 < \sin^{-1} \tan^{-1}x \leq \cos 1$$

$$\Rightarrow \sin(\cos 2) < \tan^{-1}x \leq \sin \cos 1$$

$$\Rightarrow \tan \sin(\cos 2) < x \leq \tan \sin \cos 1$$

100. Let $f(x) = \operatorname{cosec}^{-1}[1 + \sin^2x]$, where $[\cdot]$ denotes the greatest integer function, then the range of f

- (1) Is discrete and has two members (2) Is discrete and has four members
 (3) Is continuous (4) Is $\left(\frac{\pi}{2}, \frac{\pi}{6}\right)$

Sol. Answer (1)

$$0 \leq \sin^2x \leq 1 \Rightarrow 1 \leq 1 + \sin^2x \leq 2$$

$$\therefore [1 + \sin^2x] \text{ has two values 1 and 2}$$

$$\therefore f(x) = \operatorname{cosec}^{-1}(1) \text{ and } \operatorname{cosec}^{-1}(2)$$

$$= \frac{\pi}{2} \text{ and } \frac{\pi}{6}$$

$$\therefore f(x) \text{ is discrete and has two members.}$$

101. Let $f(x) = \sin^{-1} \left(\frac{1}{|x^2 - 1|} \right) + \cos^{-1} \left(\frac{1 - 2|x|}{3} \right)$, then D_f is...

(1) $[-2, 2]$

(2) $[-2, -\sqrt{2}]$

(3) $(-\infty, -\sqrt{2}] \cup [\sqrt{2}, \infty)$

(4) $[-2, -\sqrt{2}] \cup [\sqrt{2}, 2] \cup \{0\}$

Sol. Answer (4)

$$\text{Let } f(x) = \sin^{-1} \left(\frac{1}{|x^2 - 1|} \right) + \cos^{-1} \left(\frac{1 - 2|x|}{3} \right)$$

$\downarrow$ $g(x)$ (say) $\downarrow$ $h(x)$ (say)

$$\therefore D_g : 0 < \frac{1}{|x^2 - 1|} \leq 1$$

$$\Rightarrow x^2 - 1 \geq 1$$

$$\Rightarrow x^2 - 1 \geq 1, x^2 - 1 \leq -1$$

$$\Rightarrow x^2 \geq 2, x^2 \leq 0$$

$$\Rightarrow x \geq 0, |x| \geq \sqrt{2}$$

$$\Rightarrow x \in (-\infty, -\sqrt{2}] \cup [\sqrt{2}, \infty) \cup \{0\} \quad \dots(1)$$

$$D_h : -1 \leq \frac{1 - 2|x|}{3} \leq 1$$

$$\Rightarrow -3 \leq 1 - 2|x| \leq 3$$

$$\Rightarrow -4 \leq -2|x| \leq 2$$

$$\Rightarrow -1 \leq |x| \leq 2$$

$$\Rightarrow |x| \leq 2$$

$$\Rightarrow -2 \leq x \leq 2 \quad \dots(2)$$

From (1) & (2), we have

$$x \in [-2, -\sqrt{2}] \cup [\sqrt{2}, 2] \cup \{0\}$$

SECTION - B

Objective Type Questions (More than one options are correct)

1. If $6 \sin^{-1} (x^2 - 6x + 8.5) = \pi$ then

(1) $x = 1$

(2) $x = 2$

(3) $x = 3$

(4) $x = 4$

Sol. Answer (2, 4)

$$\sin^{-1} \left[(x^2 - 6x + 9) - \frac{1}{2} \right] = \frac{\pi}{6}$$

$$\Rightarrow \sin^{-1} \left[(x - 3)^2 - \frac{1}{2} \right] = \frac{\pi}{6}$$

$$\Rightarrow (x-3)^2 - \frac{1}{2} = \frac{1}{2}$$

$$\Rightarrow (x-3)^2 = 1$$

$$\Rightarrow x-3 = 1, -1$$

$$x = 2, 4$$

2. Let $f(x) = \sin^{-1}x$ and $g(x) = \cos^{-1}x$, then which of the following statements are correct?

$$(1) f(x) > g(x) \text{ if } x \in \left(\frac{1}{\sqrt{2}}, 1 \right]$$

$$(2) f(x) < g(x) \text{ if } x \in \left[-1, \frac{1}{\sqrt{2}} \right)$$

$$(3) f(x) > g(x) \text{ if } x \in \left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right)$$

$$(4) f(x) < g(x) \text{ if } x \in \left[-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right]$$

Sol. Answer (1, 2)

$$\sin^{-1}x > \cos^{-1}x$$

$$\Rightarrow \sin^{-1}x > \frac{\pi}{2} - \sin^{-1}x$$

$$\Rightarrow 2\sin^{-1}x > \frac{\pi}{2}$$

$$\Rightarrow \sin^{-1}x > \frac{\pi}{4}$$

$$\Rightarrow x \in \left(\frac{1}{\sqrt{2}}, 1 \right]$$

$$\sin^{-1}x < \cos^{-1}x$$

$$\Rightarrow \sin^{-1}x < \frac{\pi}{4}$$

$$\Rightarrow x \in \left[-1, \frac{1}{\sqrt{2}} \right)$$

3. If $\cos^{-1}\left(\frac{1-n^2}{1+\pi^2}\right) > \frac{\pi}{2}$ and $n \in \mathbb{N}$, then possible values of n , are

$$(1) 2$$

$$(2) 3$$

$$(3) 4$$

$$(4) 5$$

Sol. Answer (1, 2)

$$\frac{\pi}{2} < \cos^{-1}\left(\frac{1-n^2}{1+\pi^2}\right) \leq \pi$$

$$\Rightarrow -1 \leq \frac{1-n^2}{1+\pi^2} < 0$$

$$\Rightarrow -(1+\pi^2) \leq 1-n^2$$

$$\Rightarrow n^2 \leq \pi^2 + 2$$

$$\therefore n = 2, 3$$

4. If the equation $\sin^{-1}(x^2 + x + 1) + \cos^{-1}(\lambda x + 1) = \frac{\pi}{2}$ has exactly two solutions, then λ may be equal to

(1) -1

(2) 0

(3) 1

(4) 2

Sol. Answer (2)

$$\sin^{-1}(x^2 + x + 1) + \cos^{-1}(\lambda x + 1) = \frac{\pi}{2}$$

$$\Rightarrow \cos^{-1}(\lambda x + 1) = \frac{\pi}{2} - \sin^{-1}(x^2 + x + 1) = \cos^{-1}(x^2 + x + 1)$$

$$\Rightarrow x^2 + (1 - \lambda)x = 0$$

$$x[x + (1 - \lambda)] = 0$$

$$\Rightarrow x = 0, \lambda - 1$$

If x has exactly two solutions, $\lambda \neq 1$.

$$\therefore \lambda = -1, 0, 2$$

If $\lambda = -1, 2$, then $\sin^{-1}(x^2 + x + 1)$ is not defined.

5. The value(s) of x satisfying the equation $\sin^{-1}|\sin x| = \sqrt{\sin^{-1}|\sin x|}$ is/are given by ($n \in \mathbb{I}$)

(1) $n\pi - 1$ (2) $n\pi$ (3) $n\pi + 1$ (4) $2n\pi + 1$

Sol. Answer (1, 2, 3, 4)

$$\sin^{-1}|\sin x| = \sqrt{\sin^{-1}|\sin x|}$$

$$\Rightarrow \sin^{-1}|\sin x| = 0, 1$$

$$\Rightarrow |\sin x| = \sin 0, \sin 1$$

$$\Rightarrow x = n\pi, n\pi \pm 1$$

6. If $\cos^{-1} x = \tan^{-1} x$, then

$$(1) x^2 = \frac{\sqrt{5}-1}{2}$$

$$(2) x^2 = \frac{\sqrt{5}+1}{2}$$

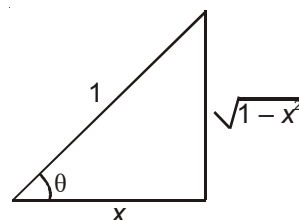
$$(3) \sin(\cos^{-1} x) = \frac{\sqrt{5}-1}{2} \quad (4) \tan(\cos^{-1} x) = \frac{\sqrt{5}-1}{2}$$

Sol. Answer (1, 3)

$$\cos^{-1} x = \tan^{-1} x \Rightarrow \tan(\cos^{-1} x) = x$$

$$\text{Let } \cos^{-1} x = \theta \Rightarrow x = \cos \theta$$

$$\text{Then, } \tan \theta = \frac{\sqrt{1-x^2}}{x}$$



$$\Rightarrow \frac{\sqrt{1-x^2}}{x} = x$$

$$\Rightarrow \sqrt{1-x^2} = x^2 \quad \dots(1)$$

$$\Rightarrow 1 - x^2 = x^4$$

$$\Rightarrow x^4 + x^2 - 1 = 0$$

$$\Rightarrow x^2 = \frac{-1 \pm \sqrt{1+4}}{2}$$

$$\Rightarrow x^2 = \frac{-1 \pm \sqrt{5}}{2}$$

Rejecting $x^2 = \frac{-1 - \sqrt{5}}{2} < 0$, we have $x^2 = \frac{\sqrt{5} - 1}{2}$

Next, $\sin(\cos^{-1} x) = \sin \theta = \sqrt{1-x^2} = x^2$ (by eq. (1))

$$= \frac{\sqrt{5} - 1}{2}$$

7. Which of the following is/are true?

(1) $\tan|\tan^{-1}x| = |x|$ (2) $\cot|\cot^{-1}x| = x$ (3) $\tan^{-1}|\tan x| = |x|$ (4) $\sin|\sin^{-1}x| = |x|$

Sol. Answer (1, 2, 4)

$\tan^{-1}|\tan x| = |x|$, only when $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

Hence, $\tan^{-1}|\tan x| \neq |x|$, $\forall x \in R$

8. If $\sin [2 \cos^{-1} \{\cot (2 \tan^{-1} x)\}] = 0$ then x equals

(1) ± 1 (2) $1 \pm \sqrt{2}$ (3) $-1 \pm \sqrt{2}$ (4) 2

Sol. Answer (1, 2, 3)

Let $2 \tan^{-1} x = \theta$, then $x = \tan\left(\frac{\theta}{2}\right)$

$$\cot \theta = \frac{1 - \tan^2\left(\frac{\theta}{2}\right)}{2 \tan \frac{\theta}{2}} = \frac{1 - x^2}{2x}$$

$$\therefore \cot(2 \tan^{-1} x) = \frac{1 - x^2}{2x} \quad \dots(1)$$

$$\begin{aligned} \therefore \sin [2 \cos^{-1} \{\cot (2 \tan^{-1} x)\}] \\ = \sin \left[2 \cos^{-1} \left(\frac{1 - x^2}{2x} \right) \right] \end{aligned} \quad \dots(2)$$

Next, let $2 \cos^{-1} \left(\frac{1 - x^2}{2x} \right) = \theta$

$$\text{Then } \frac{1-x^2}{2x} = \cos\left(\frac{\theta}{2}\right)$$

$$\Rightarrow \cos \theta = 2\cos^2\left(\frac{\theta}{2}\right) - 1$$

$$= 2\left(\frac{1-x^2}{2x}\right)^2 - 1 \quad \dots(3)$$

$$\text{But, } \sin \theta = 0 \Rightarrow \cos \theta = 1, -1$$

$$\therefore 2\left(\frac{1-x^2}{2x}\right)^2 - 1 = 1 \text{ or } -1$$

$$\Rightarrow \left(\frac{1-x^2}{2x}\right)^2 = 1 \text{ or } 0$$

$$\Rightarrow x = \pm 1 \text{ or } \frac{1-x^2}{2x} = \pm 1$$

$$\Rightarrow 1 - x^2 = \pm 2x$$

$$\Rightarrow x^2 + 2x - 1 = 0 \text{ or } x^2 - 2x - 1 = 0$$

$$\Rightarrow x = -1 \pm \sqrt{2} \text{ or } x = 1 \pm \sqrt{2}$$

$$\therefore x = \pm 1, -1 \pm \sqrt{2}, 1 \pm \sqrt{2}$$

9. If the numerical value of $\tan\left\{\cos^{-1}\left(\frac{4}{5}\right) + \tan^{-1}\left(\frac{2}{3}\right)\right\}$ is $\frac{a}{b}$ (where a and b are relatively prime), then

$$(1) a + b = 23$$

$$(2) a - b = 11$$

$$(3) 3b = a + 1$$

$$(4) 2a = 3b$$

Sol. Answer (1, 2, 3)

$$\cos^{-1}\left(\frac{4}{5}\right) = \tan^{-1}\left(\frac{3}{4}\right)$$

$$\therefore \frac{a}{b} = \tan\left\{\cos^{-1}\frac{4}{5} + \tan^{-1}\frac{2}{3}\right\} = \tan\left\{\tan^{-1}\left(\frac{3}{4}\right) + \tan^{-1}\frac{2}{3}\right\} = \frac{\frac{3}{4} + \frac{2}{3}}{1 - \frac{3}{4} \cdot \frac{2}{3}} = \frac{17}{6}$$

$$\Rightarrow 6a = 17b$$

$$\text{Taking } a = 17, b = 6$$

$$a + b = 23, a - b = 11, 3b = a + 1$$

10. If $\theta = \tan^{-1}(2 \tan^2 \theta) - \tan^{-1}\left\{\frac{1}{3} \tan \theta\right\}$, then

$$(1) \tan \theta = -2$$

$$(2) \tan \theta = 0$$

$$(3) \tan \theta = 1$$

$$(4) \tan \theta = 2$$

Sol. Answer (1, 2, 3)

$$\tan \theta = \frac{2 \tan^2 \theta - \frac{\tan \theta}{3}}{1 + \frac{2}{3} \tan^3 \theta}$$

$$\Rightarrow \tan \theta \left(1 + \frac{2}{3} \tan^3 \theta \right) = \left(2 \tan \theta - \frac{1}{3} \right) \tan \theta$$

$$\Rightarrow \tan \theta \left(\frac{2}{3} \tan^3 \theta - 2 \tan \theta + \frac{4}{3} \right) = 0$$

$$\Rightarrow \tan \theta (2 \tan^3 \theta - 6 \tan \theta + 4) = 0$$

$$\Rightarrow \tan \theta (\tan \theta - 1) (2 \tan^2 \theta + 2 \tan \theta - 4) = 0$$

$$\Rightarrow \tan \theta (\tan \theta - 1) (\tan^2 \theta + \tan \theta - 2) = 0$$

$$\Rightarrow \tan \theta (\tan \theta - 1) (\tan \theta + 2) (\tan \theta - 1) = 0$$

$$\Rightarrow \tan \theta = 0, 1, -2$$

11. The number of integral values of x satisfying $\tan^{-1}(x+1) + \tan^{-1}(x-1) = \tan^{-1}\left(\frac{2x}{2-x^2}\right)$ will be

(1) 1

(2) 2

(3) 3

(4) 4

Sol. Answer (3)

The identity is valid only when $(x+1)(x-1) < 1$

$$\Rightarrow -\sqrt{2} < x < \sqrt{2}$$

Hence possible integral values are $-1, 0, 1$.

12. Let $f(x) = e^{\cos^{-1} \sin(x + \frac{\pi}{3})}$ then

(1) $f\left(\frac{8\pi}{9}\right) = e^{\frac{5\pi}{18}}$

(2) $f\left(\frac{8\pi}{9}\right) = e^{\frac{13\pi}{18}}$

(3) $f\left(-\frac{7\pi}{4}\right) = e^{\frac{\pi}{12}}$

(4) $f\left(-\frac{7\pi}{4}\right) = e^{\frac{11\pi}{12}}$

Sol. Answer (2, 3)

$$\cos^{-1} \sin\left(\frac{8\pi}{9} + \frac{\pi}{3}\right) = \cos^{-1} \sin\left(\frac{11\pi}{9}\right)$$

$$= \cos^{-1} \sin\left(\pi + \frac{2\pi}{9}\right)$$

$$= \cos^{-1} \left(-\sin \frac{2\pi}{9}\right)$$

$$= \pi - \cos^{-1} \sin \frac{2\pi}{9}$$

$$= \pi - \left[\frac{\pi}{2} - \sin^{-1} \sin \frac{2\pi}{9} \right]$$

$$= \frac{\pi}{2} + \frac{2\pi}{9}$$

$$= \frac{13\pi}{18}$$

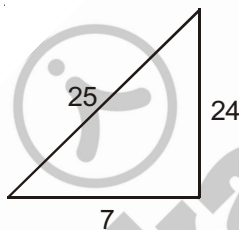
$$\begin{aligned}
 \cos^{-1} \sin\left(\frac{-7\pi}{4} + \frac{\pi}{3}\right) &= \cos^{-1} \sin\left(\frac{-17\pi}{12}\right) \\
 &= \cos^{-1} \sin\left(-\pi - \frac{5\pi}{12}\right) \\
 &= \cos^{-1} \sin\left(\frac{5\pi}{12}\right) \\
 &= \frac{\pi}{2} - \sin^{-1} \sin\left(\frac{5\pi}{12}\right) \\
 &= \frac{\pi}{2} - \frac{5\pi}{12} \\
 &= \frac{\pi}{12}
 \end{aligned}$$

13. $\cot^{-1}\left(-\frac{7}{24}\right)$ equals

- (1) $\pi - \sin^{-1} \frac{24}{25}$ (2) $\pi - \cos^{-1} \frac{7}{25}$ (3) $\pi - \tan^{-1} \frac{24}{7}$ (4) $\cos^{-1}\left(-\frac{7}{25}\right)$

Sol. Answer (1, 2, 3, 4)

$$\begin{aligned}
 \cot^{-1}\left(-\frac{7}{24}\right) &= \pi - \cot^{-1}\left(\frac{7}{24}\right) \\
 &= \pi - \tan^{-1}\left(\frac{24}{7}\right) \\
 &= \pi - \sin^{-1}\left(\frac{24}{25}\right) \\
 &= \pi - \cos^{-1}\left(\frac{7}{25}\right) \\
 &= \cos^{-1}\left(-\frac{7}{25}\right)
 \end{aligned}$$



14. $2\tan^{-1}(-3)$ equals

- (1) $-\frac{\pi}{2} + \cot^{-1}\left(-\frac{4}{3}\right)$ (2) $-\cos^{-1}\left(-\frac{4}{5}\right)$ (3) $\frac{\pi}{2} - \cot^{-1}\left(-\frac{4}{3}\right)$ (4) $-\pi + \cos^{-1}\left(\frac{4}{5}\right)$

Sol. Answer (2, 4)

$$\begin{aligned}
 2 \tan^{-1}(-3) &= -2 \tan^{-1}(3) \\
 &= -\tan^{-1}\left[\frac{2(3)}{1-3^2}\right] \\
 &= -\tan^{-1}\left(-\frac{3}{4}\right) \\
 &= -\cos^{-1}\left(-\frac{4}{5}\right) \\
 &= -\pi + \cos^{-1}\frac{4}{5}
 \end{aligned}$$

15. A value of x satisfying

$$\tan(\sec^{-1} x) = \sin\left(\cos^{-1} \frac{1}{\sqrt{5}}\right) \text{ is}$$

$$(1) \quad x = -\frac{3}{\sqrt{5}}$$

$$(2) \quad x = \frac{\sqrt{5}}{2}$$

$$(3) \quad x = \frac{3}{\sqrt{5}}$$

$$(4) \quad x = -\frac{\sqrt{5}}{2}$$

Sol. Answer (3)

$$\cos^{-1} \frac{1}{\sqrt{5}} = \sin^{-1} \frac{2}{\sqrt{5}}$$

$$\sec^{-1} x = \tan^{-1} \sqrt{1 - x^2}$$

$$\therefore \sqrt{x^2 - 1} = \frac{2}{\sqrt{5}}$$

$$\Rightarrow x^2 = \frac{9}{5}$$

$$\Rightarrow x = \frac{3}{\sqrt{5}}$$

16. $\frac{1}{2} \cos^{-1}\left(\frac{3}{5}\right)$ equals

$$(1) \quad \frac{\pi}{2} - \cos^{-1} \frac{4}{5}$$

$$(2) \quad \tan^{-1} \frac{1}{2}$$

$$(3) \quad \frac{\pi}{4} - \frac{1}{2} \cos^{-1} \frac{4}{5}$$

$$(4) \quad \frac{\pi}{2} - \tan^{-1} \frac{1}{2}$$

Sol. Answer (2, 3)

$$\frac{1}{2} \cos^{-1}\left(\frac{3}{5}\right)$$

$$= \frac{1}{2} \cos^{-1}\left(\frac{4-1}{4+1}\right)$$

$$= \frac{1}{2} \cos^{-1}\left(\frac{1 - \frac{1}{4}}{1 + \frac{1}{4}}\right)$$

$$= \frac{1}{2} \cdot 2 \tan^{-1}\left(\frac{1}{2}\right)$$

$$= \tan^{-1}\left(\frac{1}{2}\right)$$

$$\text{Also, } \frac{1}{2} \cos^{-1} \frac{3}{5} = \frac{1}{2} \sin^{-1} \frac{4}{5} = \frac{1}{2} \left[\frac{\pi}{2} - \cos^{-1} \frac{4}{5} \right]$$

$$= \frac{\pi}{4} - \frac{1}{2} \cos^{-1} \frac{4}{5}$$

17. If $x, y, z \in (-1, 1)$ satisfy $\cos^{-1}x + \cos^{-1}y + \cos^{-1}z = \pi$, then x, y, z satisfy

$$(1) (xy + z)^2 = (1 - x^2)(1 - y^2)$$

$$(2) (xy - z)^2 = (1 + x^2)(1 + y^2)$$

$$(3) x^2 + y^2 + z^2 + 2xyz = 1$$

$$(4) (yz + x)^2(1 - x^2) = (zx + y)^2(1 - y^2) = (xy + z)^2(1 - z^2)$$

Sol. Answer (1, 3, 4)

$$\cos^{-1}x + \cos^{-1}y = \pi - \cos^{-1}z$$

$$\Rightarrow \cos(\cos^{-1}x + \cos^{-1}y) = \cos(\pi - \cos^{-1}z)$$

$$\Rightarrow xy - \sqrt{1-x^2}\sqrt{1-y^2} = -z$$

$$\Rightarrow xy + z = \sqrt{1-x^2}\sqrt{1-y^2}$$

$$\Rightarrow (xy + z)^2 = (1 - x^2)(1 - y^2) \quad \dots(1)$$

$$\text{Next, } x^2y^2 + z^2 + 2xyz = 1 - x^2 - y^2 + x^2y^2$$

$$\Rightarrow x^2 + y^2 + z^2 + 2xyz = 1 \quad \dots(2)$$

By (1), we can also get

$$(yz + x)^2 = (1 - y^2)(1 - z^2) \quad \dots(3)$$

$$\text{and, } (zx + y)^2 = (1 - z^2)(1 - x^2) \quad \dots(4)$$

It becomes obvious, that

$$(xy + z)^2(1 - z^2) = (yz + x)^2(1 - x^2) = (zx + y)^2(1 - y^2)$$

18. Let $f(x) = \tan^{-1}\left(\frac{1}{2}\tan 2x\right) + \tan^{-1}(\cot x) + \tan^{-1}(\cot^3 x)$. Then

$$(1) f\left(\frac{3\pi}{8}\right) = \pi$$

$$(2) f\left(\frac{\pi}{8}\right) = 0$$

$$(3) f\left(\frac{\pi}{8}\right) = \pi$$

$$(4) f\left(\frac{3\pi}{8}\right) = 0$$

Sol. Answer (3, 4)

$$\text{If } \cot x < 1 \Rightarrow x \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$$

$$\therefore \text{ if } \cot x < 1$$

$$\tan^{-1}(\cot x) + \tan^{-1}(\cot^3 x)$$

$$= \tan^{-1}\left(\frac{\cot x + \cot^3 x}{1 - \cot^4 x}\right)$$

$$= \tan^{-1}\left(\frac{\cot x}{1 - \cot^2 x}\right)$$

$$= \tan^{-1}\left(\frac{\tan x}{\tan^2 x - 1}\right)$$

$$= \tan^{-1}\left(\frac{1}{2} \cdot \frac{2 \tan x}{\tan^2 x - 1}\right)$$

$$= \tan^{-1}\left(-\frac{1}{2}\tan 2x\right)$$

$$= -\tan^{-1}\left(\frac{1}{2}\tan 2x\right)$$

$$\Rightarrow f(x) = 0$$

$$\text{If } \cot x > 1 \Rightarrow x \in \left(0, \frac{\pi}{4}\right)$$

$$\therefore \tan^{-1}(\cot x) + \tan^{-1}(\cot^3 x) = \pi + \tan^{-1}\left(\frac{\cot x + \cot^3 x}{1 - \cot^4 x}\right) = \pi - \tan^{-1}\left(\frac{1}{2} \tan 2x\right)$$

$$\therefore f(x) = \pi$$

19. If $\tan^{-1}x + \tan^{-1}y + \tan^{-1}z = \pi$, $x, y, z > 0$ then which of the following are true?

$$(1) \ x + y + z = xyz \quad (2) \ xy + yz + zx = 1 \quad (3) \ xyz \geq 3\sqrt{3} \quad (4) \ xyz \leq 3\sqrt{3}$$

Sol. Answer (1, 3)

$$\text{If } \tan^{-1}x = A, \tan^{-1}y = B, \tan^{-1}z = C$$

$$\Rightarrow A + B + C = \pi$$

Hence

$$\tan A + \tan B + \tan C = \tan A \tan B \tan C$$

$$x + y + z = xyz$$

$$\frac{x + y + z}{3} \geq (xyz)^{1/3}$$

$$\frac{xyz}{3} \geq (xyz)^{1/3}$$

$$xyz \geq 3\sqrt{3}$$

20. If $\sin^{-1}x + \sin^{-1}y + \sin^{-1}z = \pi$, then which of the following are true?

$$(1) \ x\sqrt{1-x^2} + y\sqrt{1-y^2} + z\sqrt{1-z^2} = 2xyz$$

$$(2) \ \sqrt{1-x^2} + \sqrt{1-y^2} + \sqrt{1-z^2} = xyz$$

$$(3) \ x^2 + y^2 + z^2 - 2\sqrt{1-x^2}\sqrt{1-y^2}\sqrt{1-z^2} = 2$$

$$(4) \ x^2 + y^2 + z^2 + 2xyz = 2$$

Sol. Answer (1, 3)

$$\sin^{-1}x = A, \sin^{-1}y = B, \sin^{-1}z = C$$

$$A + B + C = \pi$$

$$\sin 2A + \sin 2B + \sin 2C = 4\sin A \sin B \sin C$$

$$\cos 2A + \cos 2B + \cos 2C = -1 - 4\cos A \cos B \cos C$$

21. If $x^3 - x^2 + \tan^2 \alpha x - \tan^2 \alpha = 0$, has positive roots x_1, x_2, x_3 then which of following are true?

$$(1) \ \tan^{-1} x_1 + \tan^{-1} x_2 + \tan^{-1} x_3 = \frac{\pi}{4}$$

$$(2) \ \tan^{-1} x_1 + \tan^{-1} x_2 + \tan^{-1} x_3 = \frac{\pi}{2}$$

$$(3) \ \tan \alpha \leq \frac{1}{3\sqrt{3}}$$

$$(4) \ \tan \alpha \geq \frac{1}{3\sqrt{3}}$$

Sol. Answer (1, 3)

$$\begin{aligned} & \tan^{-1}x_1 + \tan^{-1}x_2 + \tan^{-1}x_3 \\ &= \tan^{-1}\left(\frac{x_1 + x_2 + x_3 - (x_1x_2x_3)}{1 - (x_1x_2 + x_2x_3 + x_3x_1)}\right) \\ &= \tan^{-1}\left(\frac{1 - \tan^2\alpha}{1 + \tan^2\alpha}\right) = \frac{\pi}{4} \\ &\frac{x_1 + x_2 + x_3}{3} \geq (x_1x_2x_3)^{1/3} \end{aligned}$$

22. $S = \tan^{-1}\frac{1}{3} + \tan^{-1}\frac{2}{9} + \tan^{-1}\frac{4}{33} + \dots$ then correct among following is

- (1) Sum upto n terms are $\tan^{-1}2^n - \frac{\pi}{4}$ (2) Sum upto n terms are $\tan^{-1}\left(\frac{1}{2^{-n}}\right) - \frac{\pi}{4}$
 (3) Sum upto infinity will be $\frac{\pi}{4}$ (4) Sum upto infinity will be $-\frac{\pi}{4}$

Sol. Answer (1, 2, 3)

$$\tan^{-1}\frac{2-1}{1+2.1} + \tan^{-1}\frac{2^2-2}{1+2^2.2} + \tan^{-1}\frac{2^3-2^2}{1+2^3.2^2} + \dots + \tan^{-1}\frac{2^n-2^{n-1}}{1+2^n.2^{n-1}} = \tan^{-1}2^n - \tan^{-1}1$$

23. Solution(s) of the equation $\sin\left(-\frac{\pi}{3} + \tan^{-1}x + \cot^{-1}x\right) = \frac{1}{2}$ is/are

- (1) $\frac{1}{2}$ (2) $-\frac{1}{2}$ (3) $\sqrt{3}$ (4) $-\sqrt{3}$

Sol. Answer (1, 2, 3, 4)

$$\text{Since, } \tan^{-1}x + \cot^{-1}x = \frac{\pi}{2}$$

$$\sin\left(-\frac{\pi}{3} + \frac{\pi}{2}\right) = \frac{1}{2}$$

$$\cos\frac{\pi}{3} = \frac{1}{2} \quad \forall x$$

24. Let (a, b) satisfy following equation

$$\tan^{-1}a + \tan^{-1}b = \frac{\pi}{4}$$

$$\tan^{-1}\frac{1}{a} + \tan^{-1}b = \tan^{-1}(7)$$

then

- (1) $a = \frac{1}{2}$ (2) $b = \frac{1}{3}$ (3) $b = \frac{1}{2}$ (4) $b = -\frac{11}{7}$

Sol. Answer (1, 2)

$$\tan^{-1} a + \tan^{-1} b = \frac{\pi}{4}$$

$$\frac{a+b}{1-ab} = 1$$

$$a + b + ab = 1 \quad \dots (1)$$

$$\tan^{-1} \left(\frac{\frac{1}{a} + b}{1 - \frac{b}{a}} \right) = \tan^{-1} \frac{7}{4}$$

$$\frac{1+ab}{a-b} = \frac{7}{4} \quad \dots (2)$$

Solving (1) and (2)

$$a = \frac{1}{2}$$

$$b = \frac{1}{3}$$

25. If $\tan^{-1}x + \tan^{-1}y - \sin^{-1}x = \theta$, then which of the following are true?

(1) $x^2((x+y)^2 + (1-xy)^2)\sec^2\theta = ((x+y) - (1-xy)\tan\theta)^2$

(2) $x^2((x-y)^2 + (1-xy)^2)\sec^2\theta = ((x-y) - (1-xy)\tan\theta)^2$

(3) $x^2((x+y)^2 + (1+xy)^2)\sec^2\theta = ((x-y) - (1-xy)\tan\theta)^2$

(4) $x^2((x+y)^2 + (1+xy)^2)\sec^2\theta = ((x-y) - (1-xy)\tan\theta)^2$

Sol. Answer (1)

$$\tan^{-1}x + \tan^{-1}y = \theta + \sin^{-1}x$$

$$\frac{x+y}{1-xy} = \tan(\theta + \sin^{-1}x)$$

$$\frac{x+y}{1-xy} = \frac{\tan\theta + \frac{x}{\sqrt{1-x^2}}}{1 - \tan\theta \frac{x}{\sqrt{1-x^2}}}$$

$$\frac{x+y}{1-xy} = \frac{\sqrt{1-x^2} \tan\theta + x}{\sqrt{1-x^2} - x \tan\theta}$$

$$(x+y)\sqrt{1-x^2} - x(x+y)\tan\theta$$

$$= (1-xy)\tan\theta\sqrt{1-x^2} + x(1-xy)$$

$$\sqrt{1-x^2}((x+y) - (1-xy)\tan\theta) = x(1-xy) - x(x+y)\tan\theta$$

$$(1 - x^2)[(x + b)^2 + (1 - xy)^2 \tan^2 \theta - 2(x + y)(1 - xy) \tan \theta] = x^2(1 - xy)^2 + x^2(x + y)^2 \tan^2 \theta - 2x^2(1 - xy)(x + y) \tan \theta$$

$$(x + b)^2 + (1 - xy)^2 \tan^2 \theta - 2(x + y)(1 - xy) \tan \theta$$

$$-x^2(x + y)^2 - x^2(1 - xy)^2 \tan^2 \theta + 2x^2(x + y)(1 - xy) \tan \theta = x^2(1 - xy)^2 + x^2(x + y)^2 \tan^2 \theta - 2x^2(1 - xy)(x + y) \tan \theta$$

$$x^2(x + b)^2 \sec^2 \theta + x^2(1 - xy)^2 \sec^2 \theta = ((x + y) - (1 - xy) \tan \theta)^2$$

26. If $\alpha = 3 \sin^{-1} \left(\frac{6}{11} \right)$ and $\beta = 3 \cos^{-1} \left(\frac{4}{9} \right)$, where the inverse trigonometric functions take only the principal values, then the correct option(s) is (are) [JEE(Advanced)-2015]

(1) $\cos \beta > 0$

(2) $\sin \beta < 0$

(3) $\cos(\alpha + \beta) > 0$

(4) $\cos \alpha < 0$

Sol. Answer (2, 3, 4)

$$\alpha = 3 \sin^{-1} \frac{6}{11}$$

$$\text{since } \frac{\pi}{4} > \sin^{-1} \frac{6}{11} > \frac{\pi}{6}$$

$$\frac{3\pi}{4} > 3 \sin^{-1} \frac{6}{11} > \frac{\pi}{2}$$

$$\cos \alpha < 0$$

$$\cos \left(3 \cos^{-1} \frac{4}{9} \right) = 4 \times \left(\frac{4}{9} \right)^3 - 3 \times \frac{4}{9} = \frac{4 \times 64}{729} - \frac{12}{9} = \frac{256 - 12 \times 81}{729} < 0$$

$$\frac{5\pi}{12} > \cos^{-1} \frac{4}{9} > \frac{\pi}{3}$$

$$\frac{5\pi}{4} > 3 \cos^{-1} \frac{4}{9} > \pi$$

$$\sin \left(3 \cos^{-1} \frac{4}{9} \right) < 0$$

$$\frac{3\pi}{4} > 3 \sin^{-1} \frac{6}{11} > \frac{\pi}{2}$$

$$\frac{5\pi}{4} > 3 \cos^{-1} \frac{4}{9} > \pi$$

$$2\pi > 3 \sin^{-1} \frac{6}{11} + 3 \cos^{-1} \frac{4}{9} > \frac{3\pi}{2}$$

$$\cos \left(3 \sin^{-1} \frac{6}{11} + 3 \cos^{-1} \frac{4}{9} \right) > 0$$

27. For any positive integer n , define $f_n : (0, \infty) \rightarrow \mathbb{R}$ as $f_n(x) = \sum_{j=1}^n \tan^{-1} \left(\frac{1}{1+(x+j)(x+j-1)} \right)$ for all $x \in (0, \infty)$

(Here, the inverse trigonometric function $\tan^{-1}x$ assumes values in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.)

Then, which of the following statement(s) is (are) TRUE?

[JEE(Advanced)-2018]

- (1) $\sum_{j=1}^5 \tan^2(f_j(0)) = 55$
- (2) $\sum_{j=1}^{10} (1 + f'_j(0)) \sec^2(f_j(0)) = 10$
- (3) For any fixed positive integer n , $\lim_{x \rightarrow \infty} \tan(f_n(x)) = \frac{1}{n}$
- (4) For any fixed positive integer n , $\lim_{x \rightarrow \infty} \sec^2(f_n(x)) = 1$

Sol. Answer (4)

$$f_n(x) = \tan^{-1}(n+x) - \tan^{-1}(x) = \tan^{-1} \left(\frac{n}{1+(n+x)x} \right)$$

$$f'_n(x) = \frac{1}{1+(n+x)^2} - \frac{1}{1+x^2}$$

(1) $x = 0$ is not in domain

(2) $x = 0$ is not in domain

$$(3) \lim_{x \rightarrow \infty} \tan(f_n(x)) = \lim_{x \rightarrow \infty} \frac{n}{1+(n+x)x} = 0$$

$$(4) \lim_{x \rightarrow \infty} (1 + \tan^2(f_n(x))) = 1 + 0 = 1$$

SECTION - C

Linked Comprehension Type Questions

Comprehension-I

Transformations of inverse trigonometric functions need to be handled with care. Consider the identity

$$\sin 2\theta = \frac{2 \tan \theta}{1 + \tan^2 \theta} \text{ in its domain of definition. Suppose we set } \tan \theta = x, \text{ then we have } \sin 2\theta = \frac{2x}{1+x^2}.$$

Taking $\sin^{-1}$ of both sides yields $2\theta = \sin^{-1} \frac{2x}{1+x^2}$ i.e., $2 \tan^{-1} x = \sin^{-1} \frac{2x}{1+x^2}$. But we will discover that the

above identity is not valid for all x . Choose $x = \sqrt{3}$, LHS = $2 \tan^{-1} \sqrt{3} = 2 \times \frac{\pi}{3} = \frac{2\pi}{3}$, RHS = $\sin^{-1} \frac{2\sqrt{3}}{1+3} =$

$\sin^{-1} \frac{\sqrt{3}}{2} = \frac{\pi}{3}$. And so left hand and right hand side don't match. The reason is that we have disregarded the

principal values of inverse functions. So it is well to remember that the inverse trigonometric formulae have restrictions attached to the argument. When the values of x lie outside the interval of validity then the formula needs to be corrected.

1. Let $f(x) = \sin^{-1} \frac{2x}{1+x^2}$, $g(x) = 2\tan^{-1}x$. Then the largest interval in R on which f and g both are agree

(1) $[-1, 1]$

(2) $(-\infty, -1]$

(3) $[1, \infty)$

(4) $(-\infty, 1]$

Sol. Answer (1)

$$\text{If } \sin^{-1} \frac{2x}{1+x^2} = 2\tan^{-1}x$$

$$\text{Then } \frac{-\pi}{2} \leq 2\tan^{-1}x \leq \frac{\pi}{2}$$

$$\Rightarrow \frac{-\pi}{4} \leq \tan^{-1}x \leq \frac{\pi}{4}$$

$$\Rightarrow -1 \leq x \leq 1$$

2. $\tan^{-1}\left(\frac{\pi}{2}\right)$ equals

(1) $\pi + \sin^{-1}\left(\frac{4\pi}{4+\pi^2}\right)$

(2) $\frac{\pi}{2} - \frac{1}{2}\sin^{-1}\left(\frac{4\pi}{4+\pi^2}\right)$

(3) $\frac{\pi}{2} + \frac{1}{2}\sin^{-1}\left(\frac{4\pi}{4+\pi^2}\right)$

(4) $-\pi + \sin^{-1}\left(\frac{4\pi}{4+\pi^2}\right)$

Sol. Answer (2)

$$\tan^{-1}\left(\frac{\pi}{2}\right)$$

$$= \frac{1}{2} \left[2\tan^{-1}\left(\frac{\pi}{2}\right) \right]$$

$$= \frac{1}{2} \left[\pi - \sin^{-1} \left\{ \frac{2\left(\frac{\pi}{2}\right)}{1+\left(\frac{\pi}{2}\right)^2} \right\} \right] \quad \left(\because 2\tan^{-1}x = \pi - \sin^{-1}\left(\frac{2x}{1+x^2}\right), \text{ if } x > 1 \right)$$

$$= \frac{\pi}{2} - \frac{1}{2}\sin^{-1}\left(\frac{4\pi}{4+\pi^2}\right)$$

3. If $x \in (-\infty, -1)$, then which of the relation holds?

$$(1) 2 \tan^{-1} x = \sin^{-1} \frac{2x}{1+x^2} - \frac{\pi}{2}$$

$$(2) 2 \tan^{-1} x = -\pi - \sin^{-1} \frac{2x}{1+x^2}$$

$$(3) 2 \tan^{-1} x = \pi - \sin^{-1} \frac{2x}{1+x^2}$$

$$(4) 2 \tan^{-1} x = \frac{\pi}{2} + \sin^{-1} \frac{2x}{1+x^2}$$

Sol. Answer (2)

When $x < -1$, then

$$2 \tan^{-1} x = -\pi - \sin^{-1} \left(\frac{2x}{1+x^2} \right)$$

Comprehension-II

To find the maximum and minimum values of different expressions we may use perfect square method or may take the help of quadratic equations. Now give the answer of the following question by using the relation in different trigonometric inverse function.

1. If the maximum possible value of

$(\sin^{-1} x)^2 + (\tan^{-1} y)^2 + \pi \sin^{-1} x + \pi \tan^{-1} y + \frac{\pi^2}{2}$ is k , and $[]$ represents the greatest integer function then the value

of $\left[\frac{k}{2\pi^2} \right]$ is

$$(1) 0$$

$$(2) 1$$

$$(3) 2$$

$$(4) 3$$

Sol. Answer (1)

The given expression is $\left(\sin^{-1} x + \frac{\pi}{2} \right)^2 + \left(\sin^{-1} y + \frac{\pi}{2} \right)^2$

Hence maximum $k < 2\pi \Rightarrow \left[\frac{k}{2\pi^2} \right] = 0$

2. If the maximum value of $(\sec^{-1} x)^2 + (\operatorname{cosec}^{-1} x)^2$ approaches a , the minimum value of $(\tan^{-1} x)^3 + (\cot^{-1} x)^3$ approaches b then $\left(a + \frac{b}{\pi} \right)$ is equal to

$$(1) \pi^2$$

$$(2) \frac{41\pi^2}{32}$$

$$(3) \frac{\pi^2}{32}$$

$$(4) \frac{43\pi^2}{32}$$

Sol. Answer (2)

$$a = \frac{5\pi^2}{4}, \frac{b}{\pi} = \frac{\pi^2}{32} \Rightarrow a + \frac{b}{\pi} = \frac{41\pi^2}{32}$$

3. $(\cos^{-1} x + \sec^{-1} x + \tan^{-1} x + \cot^{-1} x)_{\max} + 6((\sin^{-1} x)^2 + \pi \sin^{-1} x)_{\min}$ equals

$$(1) \frac{5\pi^2 + 3\pi}{2}$$

$$(2) \frac{\pi^2 + 3\pi}{2}$$

$$(3) \frac{5\pi - 3\pi^2}{2}$$

$$(4) \frac{\pi + 3\pi^3}{2}$$

Sol. Answer (3)

$$\text{Value} = \frac{5\pi}{2} - \frac{3\pi^2}{2} = \frac{5\pi - 3\pi^2}{2}$$

Comprehension-III

Sometimes we are just concerned with finding integral solutions to equations. Consider the equation

$$\tan^{-1} \frac{1}{m} + \tan^{-1} \frac{1}{n} = \tan^{-1} \frac{1}{\lambda}, \text{ where } m, n, \lambda \in N$$

1. How many positive integral solutions (m, n) exist for the equation if $\lambda = 3$?

- (1) Two (2) Four (3) Six (4) Eight

Sol. Answer (2)

$$\tan^{-1} \frac{1}{\lambda} = \tan^{-1} \frac{1}{m} + \tan^{-1} \frac{1}{n}$$

$$\Rightarrow \frac{1}{\lambda} = \frac{\frac{1}{m} + \frac{1}{n}}{1 - \frac{1}{mn}}$$

$$\Rightarrow \lambda = \frac{mn-1}{m+n} = 3 \text{ given}$$

$$\Rightarrow m = \frac{3n+1}{n-3} = \frac{3(n-3)+10}{n-3} \\ = 3 + \frac{10}{n-3}$$

$$\text{Clearly, } \frac{10}{n-3} \in N$$

$$\text{if } n-3 = 1, 2, 5, 10$$

$$\text{or } n = 4, 5, 8, 13$$

$\therefore$ Four solutions for (m, n) exist.

2. For $\lambda = 11$, an integral pair (m, n) satisfying the equation is

- (1) (12, 72) (2) (12, 133) (3) (13, 74) (4) (13, 136)

Sol. Answer (2)

$$\text{If } \lambda = 11, \text{ then } m = \frac{11n+1}{n-11} = \frac{11(n-11)+122}{n-11}$$

$$\Rightarrow m = 11 + \frac{122}{n-11}$$

$$\frac{122}{n-11} \in N,$$

$$\text{if } n-11 = 1, 122$$

$$n = 12, 133$$

$$\text{when } n = 133, m = 12$$

3. If λ (any fixed value) is such that $\lambda^2 + 1$ is a prime, then how many solutions (m, n) are there for the equation?

- (1) One (2) Two (3) Four (4) Infinite

Sol. Answer (2)

If $\lambda^2 + 1$ is a prime number, then λ must be even

$$\therefore \frac{mn-1}{m+n} = 2p \quad (p \in \mathbb{N})$$

$$\Rightarrow m = \frac{(2p)n+1}{n-2p}$$

$$= \frac{2p(n-2p) + (1+4p^2)}{n-2p}$$

$$= 2p + \frac{4p^2+1}{n-2p}$$

$m \in \mathbb{N}$, if $n-2p = 1$ or $4p^2+1$ ($\because 4p^2+1 = \lambda^2+1$ is prime)

$$n = 2p+1 \text{ or } 4p^2+2p+1$$

$\therefore$ Only two solutions

Comprehension-IV

If $|x| \leq 1$ and $|y| \leq 1$, then system of equations are defined as

$$\cos^{-1} x + (\sin^{-1} y)^2 = \frac{a\pi^2}{4} \quad \dots(i)$$

$$\{\cos^{-1} x\} \cdot \{\sin^{-1} y\}^2 = \frac{\pi^4}{16} \quad \dots(ii)$$

then

1. The set of values of a for which equations (i) holds true

- (1) $\left(0, \frac{4}{\pi} + 1\right)$ (2) $\left[0, \frac{4}{\pi} + 1\right]$ (3) R (4) ϕ

Sol. Answer (2)

We know that,

$$0 \leq \cos^{-1} x \leq \pi; \quad 0 \leq \{\sin^{-1} y\}^2 \leq \frac{\pi^2}{4}$$

$$\therefore 0 \leq \cos^{-1} x + \{\sin^{-1}(y)\}^2 \leq \pi + \frac{\pi^2}{4}$$

$$\Rightarrow 0 \leq \frac{a\pi^2}{4} \leq \pi + \frac{\pi^2}{4}$$

$$\Rightarrow 0 \leq a \leq \left(\frac{4}{\pi} + 1\right) \Rightarrow a \in \left[0, \frac{4}{\pi} + 1\right] \quad \dots(\text{iii})$$

2. The integral values of a for which equations (i) and (ii) possesses solutions are

- (1) $\{2\}$ (2) $\{x : x = n, n \in \mathbb{Z}\}$ (3) $\{-1, 0, 1\}$ (4) Both (1) & (2)

Sol. Answer (1)

Eliminating $\{\sin^{-1}y\}^2$ from equations (i) and (ii), we get

$$\cos^{-1}x + \frac{\pi^4}{16\cos^{-1}x} - \frac{a\pi^4}{4} = 0$$

$$\Rightarrow 16[\cos^{-1}x]^2 - \{4a\pi^2\}\cos^{-1}x + \pi^4 = 0$$

$$\because \cos^{-1}x \text{ is real; } D \geq 0$$

$$\therefore 16a^2\pi^4 - 64\pi^4 \geq 0 \Rightarrow a^2 - 4 \geq 0$$

$$\Rightarrow a \in (-\infty, -2] \cup [2, \infty) \quad \dots(\text{iv})$$

$\therefore$ from equations (iii) & (iv)

$$a \in \left[2, \frac{4}{\pi} + 1\right]$$

$\Rightarrow$ Integral solution of a is 2

3. The values x and y for which system of equation (i) and (ii) possesses solutions for integral values of a ,

$$(1) \left\{ \cos \frac{\pi^2}{4}, 1 \right\}$$

$$(2) \left\{ \cos \frac{\pi^2}{4}, -1 \right\}$$

$$(3) \left\{ \cos \frac{\pi^2}{4}, \pm 1 \right\}$$

$$(4) \{(x, y) : x \in \mathbb{R}, y \in \mathbb{R}\}$$

Sol. Answer (3)

Substitutes $a = 2$ in equation (iv), we get

$$16\{\cos^{-1}x\}^2 - 8\pi^2\{\cos^{-1}x\} + \pi^4 = 0$$

$$\therefore [4\cos^{-1}x - \pi^2]^2 = 0$$

$$\Rightarrow \cos^{-1}x = \frac{\pi^2}{4} \Rightarrow x = \cos \frac{\pi^2}{4}$$

$$\Rightarrow [\sin^{-1}y]^2 = \frac{\pi^4}{16} \times \frac{4}{\pi^2}$$

$$\Rightarrow \{\sin^{-1} y\}^2 = \frac{\pi^2}{4} \Rightarrow \sin^{-1} y = \pm \frac{\pi}{2}$$

$$\Rightarrow y = \pm 1$$

$$\text{Hence, } (x, y) = \left(\cos \frac{\pi^2}{4}, \pm 1 \right)$$

Comprehension-V

As we know, inverse trigonometric functions have restricted domain and range and $[\cdot]$ and $|\cdot|$ denotes greatest integer function and modulus function respectively, then

1. The solution set for $[\cot^{-1}x]^2 - 6[\cot^{-1}x] + 9 \leq 0$, is

- (1) $(-\infty, \cot 3]$ (2) $[\cot 3, \cot 2]$ (3) $[\cot 3, \infty)$ (4) $[\cot 2, \infty)$

Sol. Answer (1)

$$[\cot^{-1}x]^2 - 6[\cot^{-1}x] + 9 \leq 0$$

$$\Rightarrow \{[\cot^{-1}x] - 3\}^2 \leq 0$$

$$\Rightarrow [\cot^{-1}x] = 3$$

$$\Rightarrow 3 \leq \cot^{-1}x < 4 \Rightarrow x \in (-\infty, \cot 3]$$

2. If $[\sin^{-1}x] + [\cos^{-1}x] = 0$, (where x is a non-negative real number), the solution set for x is,

- (1) $(\cos 1, 1)$ (2) $(-1, \cos 1)$ (3) $(\sin 1, 1)$ (4) $(\cos 1, \sin 1)$

Sol. Answer (4)

$$[\sin^{-1}x] + [\cos^{-1}x] = 0 \text{ and } x \geq 0$$

$$\Rightarrow [\sin^{-1}x] = 0, [\cos^{-1}x] = 0$$

$$\Rightarrow x \in [0, \sin 1), x \in (\cos 1, 1]$$

$$\Rightarrow x \in (\cos 1, \sin 1)$$

3. The equation $\sin^{-1}x = |x - a|$ will have at least one solution; if

- (1) $a \in [-1, 1]$ (2) $a \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ (3) $a \in \left[-1, 1 + \frac{\pi}{2}\right]$ (4) $a \in \left[1 - \frac{\pi}{2}, 1 + \frac{\pi}{2}\right]$

Sol. Answer (4)

$$|x| \leq 1 \quad \dots(i)$$

$$|x - a| \leq \frac{\pi}{2} \quad \dots(ii)$$

Add (i) and (ii), we get

$$|x| + |x - a| \leq 1 + \frac{\pi}{2}$$

$$|x| + |a - x| \leq 1 + \frac{\pi}{2} \quad \dots(iii)$$

$$|a| \leq |x| + |a - x| \quad \dots(iv)$$

From (iii) and (iv), we get

$$|a| \leq 1 + \frac{\pi}{2}$$

$$a \in \left[-1 - \frac{\pi}{2}, 1 + \frac{\pi}{2}\right]$$

SECTION - D

Assertion-Reason Type Questions

1. STATEMENT-1 : $\sin^{-1} x + \cos^{-1} y = \frac{\pi}{2} \Rightarrow x = y \forall x, y \in [-1, 1]$

and

STATEMENT-2 : $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2} \forall x \in [-1, 1]$ and the function $f : [-1, 1] \rightarrow \left[-\frac{\pi}{2}, \frac{\pi}{2}\right], f(x) = \sin^{-1} x$ is one-one.

Sol. Answer (1)

Statement 2 is obviously true.

So if, $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}, \forall x \in [-1, 1]$

Then, $\sin^{-1} x + \cos^{-1} y = \frac{\pi}{2}$

$\Rightarrow x = y \forall x, y \in [-1, 1]$

Both statements correct, and statement 2 explains statement 1.

2. STATEMENT-1 : $\tan^{-1} x = \sin^{-1} y \Rightarrow y \in (-1, 1)$

and

STATEMENT-2 : $-\frac{\pi}{2} < \tan^{-1} x < \frac{\pi}{2}$

Sol. Answer (1)

Statement 2 is obviously true.

Now if, $y \in (-1, 1)$

Then, $\sin^{-1} y \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

Therefore, for some x , such that $-\frac{\pi}{2} < \tan^{-1} x < \frac{\pi}{2}$

$\sin^{-1} y = \tan^{-1} x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

Both statements correct, and statement 2 explains statement 1.

3. STATEMENT-1 : If $[\sin^{-1}x] > [\cos^{-1}x]$, where $[]$ represents the greatest integer function, then $x \in [\sin 1, 1]$ is
and

STATEMENT-2 : $\cos^{-1}(\cos x) = x, x \in [-1, 1]$

Sol. Answer (3)

Statement 2 is false, since

$$\cos^{-1}(\cos x) = x, x \in [0, \pi]$$

4. STATEMENT-1 : The function $f : [-1, 1] \rightarrow [0, \pi], f(x) = \cos^{-1} x$ is not one-one.

and

STATEMENT-2 : The function $f : (-\infty, \infty) \rightarrow [-1, 1], f(x) = \cos x$ is not one-one.

Sol. Answer (4)

Statement 1 is false, since

$$f : [-1, 1] \rightarrow [0, \pi], f(x) = \cos^{-1}(x) \text{ is one-one}$$

5. STATEMENT-1 : The value of $\tan^{-1} x + \tan^{-1} \frac{1}{x} = \frac{\pi}{2}, \forall x \in \mathbb{R} - \{0\}$.

and

$$\text{STATEMENT-2 : The value of } \tan^{-1}\left(\frac{1}{x}\right) = \begin{cases} \cot^{-1} x; & x > 0 \\ -\pi + \cot^{-1} x; & x < 0 \end{cases}$$

Sol. Answer (4)

We have,

$$\tan^{-1}\left(\frac{1}{x}\right) = \begin{cases} \cot^{-1} x; & x > 0 \\ -\pi + \cot^{-1} x; & x < 0 \end{cases}$$

$$\therefore \tan^{-1} x + \tan^{-1}\left(\frac{1}{x}\right) = \begin{cases} \tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}, & x > 0 \\ \tan^{-1} x + \cot^{-1} x - \pi = -\frac{\pi}{2}; & x < 0 \end{cases}$$

6. STATEMENT-1 : The value of $\cos^{-1} x + \cos^{-1} \left\{ \frac{x}{2} + \frac{\sqrt{3-3x^2}}{2} \right\} = \frac{\pi}{3}$, when $\frac{1}{2} \leq x \leq 1$.

and

STATEMENT-2 : As $\cos^{-1} x$ is increasing function for $0 \leq x \leq 1$.

Sol. Answer (3)As $\cos^{-1}x$ is decreasing function for $0 \leq x \leq 1$, shown as $\Rightarrow$ Statement-2 false.Let $\cos^{-1}x = y$, then $x = \cos y$

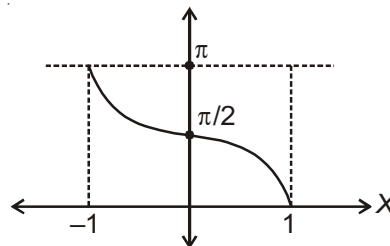
{Statement-1 True}

Now, $\frac{1}{2} \leq x \leq 1$

$$\Rightarrow \cos^{-1}\frac{1}{2} \geq \cos^{-1}x \geq \cos^{-1}1$$

{ $\therefore \cos^{-1}x$ is decreasing function for $-1 \leq x \leq 1$ }

$$\Rightarrow \frac{\pi}{3} \geq y \geq 0 \quad \Rightarrow 0 \leq y \leq \frac{\pi}{3}$$



$$\therefore \cos^{-1}x + \cos^{-1}\left\{\frac{x}{2} + \frac{\sqrt{3-3x^2}}{2}\right\}$$

$$= y + \cos^{-1}\left\{\frac{\cos y}{2} + \frac{\sqrt{3}}{2}\sin y\right\}$$

$$= y + \cos^{-1}\left\{\cos\left(\frac{\pi}{3} - y\right)\right\}$$

$$= y + \frac{\pi}{3} - y = \frac{\pi}{3}$$

7. STATEMENT-1 : The solution of $\sin^{-1}6x + \sin^{-1}6\sqrt{3}x = \frac{\pi}{2}$ is, $x = \pm \frac{1}{12}$.**and**STATEMENT-2 : As, $\sin^{-1}x$ is defined for $|x| \leq 1$.**Sol.** Answer (4)

Statement-1 is false.

$$\sin^{-1}6x + \sin^{-1}6\sqrt{3}x = -\frac{\pi}{2}$$

$$\therefore \sin^{-1}6x = -\frac{\pi}{2} - \sin^{-1}6\sqrt{3}x$$

$$\Rightarrow \sin(\sin^{-1}6x) = -\sin\left\{\frac{\pi}{2} + \sin^{-1}6\sqrt{3}x\right\}$$

$$\therefore 6x = -\cos\{\sin^{-1}6\sqrt{3}x\}$$

$$6x = -\cos\{\cos^{-1}\sqrt{1-108x^2}\}$$

$$\therefore 36x^2 = 1 - 108x^2 \Rightarrow x^2 = \frac{1}{144} \Rightarrow x = \pm \frac{1}{12}$$

But, $\sin^{-1}6x$ and $\sin^{-1}6\sqrt{3}x$ are defined for $-1 \leq 6x \leq 1$ and $-1 \leq 6\sqrt{3}x \leq 1$

$$\Rightarrow x \in \left[\frac{-1}{6\sqrt{3}}, \frac{1}{6\sqrt{3}} \right]$$

$$\Rightarrow x \in \left[-\frac{1}{6\sqrt{3}}, 0 \right) \quad \{\text{as RHS is negative so } x < 0\}.$$

Clearly, $x = -\frac{1}{12}$ is the solution.

Statement-2 is true.

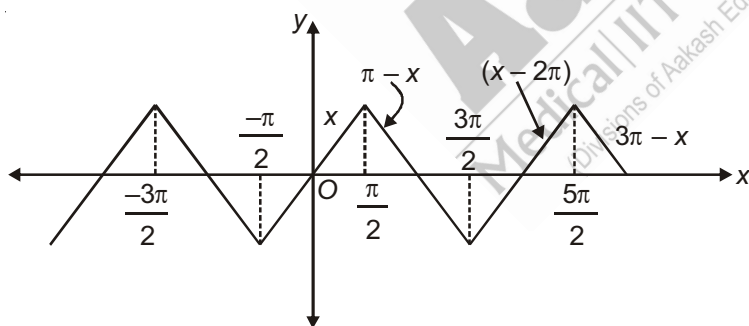
8. STATEMENT-1 : The solution set for, $\sin^{-1}(\sin 5) > x^2 - 4x$, is $(2 - \sqrt{9 - 2\pi}, 2 + \sqrt{9 - 2\pi})$.

and

STATEMENT-2 : As $\sin^{-1}(\sin x) = x - 2\pi, \frac{3\pi}{2} \leq x \leq \frac{5\pi}{2}$.

Sol. Answer (1)

$\sin^{-1}(\sin x)$ is shown below



$$\Rightarrow \sin^{-1}(\sin x) = x - 2\pi, \frac{3\pi}{2} \leq x \leq \frac{5\pi}{2}$$

$$\therefore \sin^{-1}(\sin 5) = 5 - 2\pi$$

Now, $\sin^{-1}(\sin 5) > x^2 - 4x$

$$\Rightarrow 5 - 2\pi > x^2 - 4x$$

$$\Rightarrow x^2 - 4x - 5 + 2\pi < 0$$

$$\Rightarrow x \in (2 - \sqrt{9 - 2\pi}, 2 + \sqrt{9 - 2\pi})$$

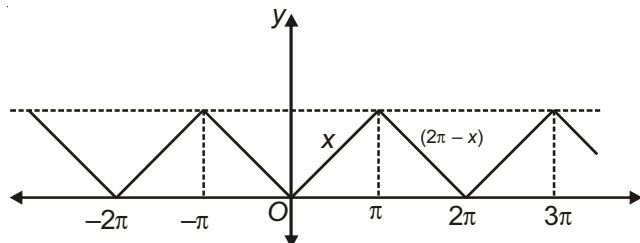
9. STATEMENT-1 : The solution set for $\cos^{-1}(\cos 4) > 3x^2 - 4x$ is ϕ .

and

STATEMENT-2 : The value of $\cos^{-1}(\cos x) = 2\pi - x$ for $x \in (\pi, 2\pi)$

Sol. Answer (4)

$\cos^{-1}(\cos x)$ is shown as



If $\pi < x < 2\pi \Rightarrow \cos^{-1}(\cos x) = 2\pi - x$

Again, $\cos^{-1}(\cos 4) > 3x^2 - 4x$

$$\Rightarrow 2\pi - 4 > 3x^2 - 4x$$

$$\Rightarrow 3x^2 - 4x + (4 - 2\pi) < 0$$

$$\Rightarrow x \in \left(\frac{2 - \sqrt{6\pi - 8}}{3}, \frac{2 + \sqrt{6\pi - 8}}{3} \right)$$

SECTION - E

Matrix-Match Type Questions

1. Match the following.

Column I (Expressions)

Column II (Value)

(A) $\cot\left(\cos^{-1}\left(-\frac{1}{3}\right)\right)$

(p) $\frac{1}{9}$

(B) $\sin^{-1}\frac{1}{3} + \sin^{-1}\frac{3}{4}$

(q) $-\frac{1}{2\sqrt{2}}$

(C) $\cos\left(2\sin^{-1}\frac{2}{3}\right)$

(r) $\cos^{-1}\frac{2\sqrt{14}-3}{12}$

(D) $\cos^{-1}\frac{4}{5} - \cos^{-1}\frac{1}{4}$

(s) $-\sin^{-1}\frac{4\sqrt{15}-3}{20}$

Sol. Answer A(q), B(r), C(p), D(s)

$$(A) \cot\left(\cos^{-1}\left(-\frac{1}{3}\right)\right) = \cot\left(\pi - \cos^{-1}\frac{1}{3}\right)$$

$$= -\cot\left(\cos^{-1}\frac{1}{3}\right)$$

$$= -\frac{1}{2\sqrt{2}}$$

$$\begin{aligned}
 \text{(B)} \quad \sin^{-1} \frac{1}{3} + \sin^{-1} \frac{3}{4} &= \sin^{-1} \left(\frac{1}{3} \sqrt{1 - \left(\frac{3}{4}\right)^2} + \frac{3}{4} \sqrt{1 - \left(\frac{1}{3}\right)^2} \right) \\
 &= \sin^{-1} \left(\frac{\sqrt{7}}{12} + \frac{6\sqrt{2}}{12} \right) \\
 &= \cos^{-1} \sqrt{1 - \left(\frac{\sqrt{7} + 6\sqrt{2}}{12} \right)^2} \\
 &= \cos^{-1} \sqrt{\frac{144 - (79 + 12\sqrt{14})}{144}} \\
 &= \cos^{-1} \sqrt{\frac{65 - 12\sqrt{14}}{144}} \\
 &= \cos^{-1} \left(\frac{2\sqrt{14} - 3}{12} \right)
 \end{aligned}$$

$$\text{(C)} \quad \text{Let } \sin^{-1} \frac{2}{3} = \theta$$

$$\begin{aligned}
 \text{Then, } \cos \left(2 \sin^{-1} \frac{2}{3} \right) &= \cos 2\theta \\
 &= 1 - 2\sin^2 \theta \\
 &= 1 - 2 \left(\frac{2}{3} \right)^2 \\
 &= 1 - \frac{8}{9} = \frac{1}{9}
 \end{aligned}$$

$$\begin{aligned}
 \text{(D)} \quad \cos^{-1} \frac{4}{5} - \cos^{-1} \frac{1}{4} &= - \left(\cos^{-1} \frac{1}{4} - \cos^{-1} \frac{4}{5} \right) \\
 &= -\cos^{-1} \left[\left(\frac{4}{5} \right) \left(\frac{1}{4} \right) + \sqrt{1 - \left(\frac{4}{5} \right)^2} \sqrt{1 - \left(\frac{1}{4} \right)^2} \right] \\
 &= -\cos^{-1} \left[\frac{4}{20} + \frac{3\sqrt{15}}{20} \right] \\
 &= -\sin^{-1} \sqrt{\frac{(20)^2 - (4 + 3\sqrt{15})^2}{20^2}} \\
 &= -\sin^{-1} \sqrt{\frac{400 - (151 + 24\sqrt{15})}{20^2}} \\
 &= -\sin^{-1} \sqrt{\frac{249 - 24\sqrt{15}}{20^2}} \\
 &= -\sin^{-1} \left(\frac{4\sqrt{15} - 3}{20} \right)
 \end{aligned}$$

2. Let $P_1 = (\sin^{-1} x)^{\sin^{-1} x}$, $P_2 = (\sin^{-1} x)^{\cos^{-1} x}$, $P_3 = (\cos^{-1} x)^{\sin^{-1} x}$, $P_4 = (\cos^{-1} x)^{\cos^{-1} x}$ then match the entries of Column I to the entries of Column II.

Column I

- (A) If $x \in (0, \cos 1)$ then
- (B) If $x \in \left(\cos 1, \frac{1}{\sqrt{2}}\right)$ then
- (C) If $x \in \left(\frac{1}{\sqrt{2}}, \sin 1\right)$ then
- (D) If $x \in (\sin 1, 1)$ then

Column II

- (p) $P_1 > P_2 > P_4 > P_3$
- (q) $P_4 > P_3 > P_1 > P_2$
- (r) $P_2 > P_1 > P_4 > P_3$
- (s) $P_3 > P_4 > P_1 > P_2$

Sol. Answer A(q), B(s), C(r), D(p)

- (A) $x \in (0, \cos 1)$ a(ii), b(i), c(iv), d(iii)

$$0 < x < \cos 1$$

Taking $\cos^{-1}$ both sides

$$\cos^{-1} \cos 1 < \cos^{-1} x < \cos^{-1} 0$$

$$1 < \cos^{-1} x < \frac{\pi}{2}$$

$$1 < \frac{\pi}{2} - \sin^{-1} x < \frac{\pi}{2}$$

$$0 < \sin^{-1} x < \frac{\pi}{2} - 1$$

$$\therefore P_4 > P_3 > P_1 > P_2$$

- (B) $\cos 1 < x < \frac{1}{\sqrt{2}}$

Taking $\cos^{-1}$ both sides

$$\cos^{-1} \frac{1}{\sqrt{2}} < \cos^{-1} x < 1$$

$$\frac{\pi}{4} < \cos^{-1} x < 1$$

$$\frac{\pi}{4} < \frac{\pi}{2} - \sin^{-1} x < 1$$

$$\frac{\pi}{2} - 1 < \sin^{-1} x < \frac{\pi}{4}$$

$$\therefore P_3 > P_4 > P_1 > P_2$$

(C) $x \in \left(\frac{1}{\sqrt{2}}, \sin 1\right)$

$$\frac{1}{\sqrt{2}} < x < \sin 1$$

Taking $\sin^{-1}$ both sides

$$\sin^{-1} \frac{1}{\sqrt{2}} < \sin^{-1} x < \sin^{-1} \sin 1$$

$$\frac{\pi}{4} < \sin^{-1} x < 1$$

$$\frac{\pi}{4} < \frac{\pi}{2} - \cos^{-1} x < 1$$

$$\frac{\pi}{2} - 1 < \cos^{-1} x < \frac{\pi}{4}$$

$$\therefore P_2 > P_1 > P_4 > P_3$$

(D) $x \in (\sin 1, 1)$

$$\sin 1 < x < 1$$

Taking $\sin^{-1}$ both sides

$$1 < \sin^{-1} x < \frac{\pi}{2}$$

$$0 < \cos^{-1} x < \frac{\pi}{2} - 1$$

$$\therefore P_1 > P_2 > P_4 > P_3$$

3. Match the following according to the number of solutions of the equation.

Column I (Equations)

Column II (Number of solutions)

(A) $x + \cos^{-1} x = 1$, when $x \in [-1, 1]$ is

(p) 2

(B) $\sin^{-1} \frac{36x}{85} + \sin^{-1} \frac{77x}{85} = \sin^{-1} x$, is

(q) 0

(C) $\tan^{-1} \sqrt{x(x+1)} + \sin^{-1} \sqrt{x^2 + x + 1} = \frac{\pi}{2}$, is

(r) 1

(D) $\sin^{-1}(2x\sqrt{1-x^2}) + \sin^{-1}(3x-4x^3) = \frac{7\pi}{9}$ in $\left(\frac{1}{2}, \frac{1}{\sqrt{2}}\right)$ is

(s) 3

Sol. Answer A(r), B(s), C(p), D(r)

(A) $x + \cos^{-1}x = 1$

$$\Rightarrow \cos^{-1}x = 1 - x$$

From graph, we can say that the equation have 1 solution.

(B) Clearly $36^2 + 77^2 = 85^2$

Hence, $x = 1$ satisfies the equation,

$$\text{Since } \sin^{-1}\frac{36}{85} + \sin^{-1}\frac{77}{85} = \sin^{-1}\frac{36}{85} + \cos^{-1}\frac{36}{85} = \frac{\pi}{2} = \sin^{-1}1.$$

Also $x = -1$ and $x = 0$ satisfy the equation

(C) $\tan^{-1}\sqrt{x(x+1)}$ and $\sin^{-1}\sqrt{x^2+x+1}$ are defined, if

$$x(x+1) \geq 0 \text{ and } x^2+x+1 \geq 0 \text{ and } x^2+x+1 \leq 1$$

$$\Rightarrow x(x+1) \geq 0 \text{ and } x(x+1) \leq 0$$

Clearly, equality holds

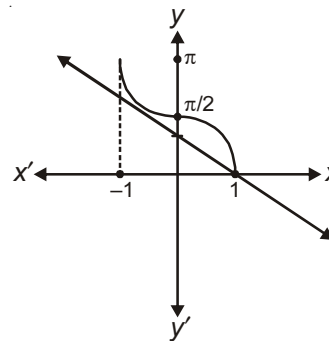
$$\therefore x(x+1) = 0$$

$$\Rightarrow x = 0, -1$$

Both, the values satisfies the given equation.

Hence, number of solutions is 2.

(D) $x = \sin\frac{2\pi}{5}$, Only one solution.



4. Match the following.

Column I (Identities)

Column II (Domain of validity)

(A) $2\sin^{-1}x = \sin^{-1}(2x\sqrt{1-x^2})$

(p) $[0, 1]$

(B) $2\sin^{-1}x = \pi - \sin^{-1}(2x\sqrt{1-x^2})$

(q) $[-1, 0]$

(C) If $2\tan^{-1}x - \sin^{-1}\left(\frac{2x}{1+x^2}\right) = 0$,

(r) $\left[-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right]$

then x may belongs to

(D) $2\cos^{-1}x = 2\pi - \cos^{-1}(2x^2 - 1)$

(s) $\left[\frac{1}{\sqrt{2}}, 1\right]$

Sol. Answer A(r), B(s), C(p, q, r, s), D(q)

(A) Since, $\frac{-\pi}{2} \leq \sin^{-1}(2x\sqrt{1-x^2}) \leq \frac{\pi}{2}$

therefore, $\frac{-\pi}{2} \leq 2\sin^{-1} x \leq \frac{\pi}{2}$

$\Rightarrow \frac{-\pi}{4} \leq \sin^{-1} x \leq \frac{\pi}{4}$

$\Rightarrow \frac{-1}{\sqrt{2}} \leq x \leq \frac{1}{\sqrt{2}}$

(B) Since, $\frac{-\pi}{2} \leq \sin^{-1}(2x\sqrt{1-x^2}) \leq \frac{\pi}{2}$

therefore, $\frac{\pi}{2} \leq \pi - \sin^{-1}(2x\sqrt{1-x^2}) \leq \frac{3\pi}{2}$

$\Rightarrow \frac{\pi}{2} \leq 2\sin^{-1} x \leq \frac{3\pi}{2}$

$\Rightarrow \frac{\pi}{4} < \sin^{-1} x < \frac{3\pi}{4}$

But, maximum value of $\sin^{-1} x$ is $\frac{\pi}{2}$

$\therefore \frac{\pi}{4} \leq \sin^{-1} x \leq \frac{\pi}{2}$

$\Rightarrow \frac{1}{\sqrt{2}} \leq x \leq 1$

(C) Here the number of value of x is zero.

(D) Since, $0 \leq \cos^{-1}(2x^2 - 1) \leq \pi$

therefore, $\pi \leq 2\pi - \cos^{-1}(2x^2 - 1) \leq 2\pi$

$\Rightarrow \pi \leq 2\cos^{-1} x \leq 2\pi$

$\Rightarrow \frac{\pi}{2} \leq \cos^{-1} x \leq \pi$

$\Rightarrow -1 \leq x \leq 0$

5. Match the following.

Column I (Expressions)

Column II (Value)

(A) The value of k if the equation

(p) $\frac{3\pi}{2}$

$$kx + \sin^{-1}(x^2 - 8x + 17) + \cos^{-1}(x^2 - 8x + 17) = \frac{9\pi}{2}$$

has at least one solution is

(B) $\sin^{-1}\frac{4}{5} + \sin^{-1}\frac{8}{17} + \sin^{-1}\frac{13}{85}$

(q) $\frac{\pi}{2}$

(C) $\sin^{-1}\frac{4}{5} + \sin^{-1}\frac{15}{17} + \sin^{-1}\frac{77}{85}$

(r) $2\cos^{-1}\frac{36}{85}$

(D) $\sin^{-1}\frac{3}{5} + \sin^{-1}\frac{8}{17} + \sin^{-1}\frac{77}{85}$

(s) π

Sol. Answer A(s), B(q), C(s), D(r)

(A) $kx + \sin^{-1}((x-4)^2 + 1) + \cos^{-1}((x-4)^2 + 1) = \frac{9\pi}{2}$

Hence the equation is valid only

When $x = 4$

$$\Rightarrow 4k + \frac{\pi}{2} = \frac{9\pi}{2}$$

$$\Rightarrow k = \pi$$

(B) $\sin^{-1}\frac{4}{5} + \sin^{-1}\frac{8}{17} + \sin^{-1}\frac{13}{85}$

$$= \sin^{-1}\left[\frac{4}{5}\sqrt{1-\left(\frac{8}{17}\right)^2} + \frac{8}{17}\sqrt{1-\left(\frac{4}{5}\right)^2}\right] + \sin^{-1}\frac{13}{85}$$

$$= \sin^{-1}\left[\frac{60}{85} + \frac{24}{85}\right] + \sin^{-1}\frac{13}{85}$$

$$= \sin^{-1}\frac{84}{85} + \sin^{-1}\frac{13}{85}$$

$$= \frac{\pi}{2}$$

$$(C) \sin^{-1} \frac{4}{5} + \sin^{-1} \frac{15}{17} + \sin^{-1} \frac{77}{85}$$

$$= \pi - \sin^{-1} \left[\frac{4}{5} \sqrt{1 - \left(\frac{15}{17} \right)^2} + \frac{15}{17} \sqrt{1 - \left(\frac{4}{5} \right)^2} \right] + \sin^{-1} \frac{77}{85}$$

$$= \pi - \sin^{-1} \frac{77}{85} + \sin^{-1} \frac{77}{85}$$

$$= \pi$$

$$(D) \sin^{-1} \frac{3}{5} + \sin^{-1} \frac{8}{17} + \sin^{-1} \frac{77}{85}$$

$$= \sin^{-1} \left[\frac{3}{5} \sqrt{1 - \left(\frac{8}{17} \right)^2} + \frac{8}{17} \sqrt{1 - \left(\frac{3}{5} \right)^2} \right] + \sin^{-1} \frac{77}{85}$$

$$= \sin^{-1} \left[\frac{45}{85} + \frac{32}{85} \right] + \sin^{-1} \frac{77}{85}$$

$$= \sin^{-1} \frac{77}{85} + \sin^{-1} \frac{77}{85}$$

$$= 2 \sin^{-1} \frac{77}{85} = 2 \cos^{-1} \frac{36}{85}$$

6. Match the following

Column I

Column II

$$(A) \frac{1}{2^2} \tan \frac{\pi}{2^2} + \frac{1}{2^3} \tan \frac{\pi}{2^3} + \dots \text{ to } \infty$$

$$(p) \frac{\pi}{2}$$

$$(B) \tan^{-1} \frac{1}{3+3.1+1^2} + \tan^{-1} \frac{1}{3+3.2+2^2} + \dots \text{ to } \infty$$

$$(q) \frac{\pi}{4}$$

$$(C) \tan^{-1} \frac{1}{3} + \tan^{-1} \frac{1}{7} + \tan^{-1} \frac{1}{13} + \dots \text{ to } \infty$$

$$(r) \frac{1}{\pi}$$

$$(D) \tan^{-1} \frac{1}{2.1^2} + \tan^{-1} \frac{1}{2.2^2} + \dots \text{ to } \infty$$

$$(s) \frac{\pi}{2} - \tan^{-1} 2$$

Sol. Answer A(r), B(s), C(q), D(q)

(A) For $0 < \theta < \frac{\pi}{4}$

$$\tan \theta \approx \theta$$

$$\Rightarrow \frac{1}{2^2} + \frac{1}{2^3} \left[\frac{\pi}{2^3} \right] + \frac{1}{2^4} \left[\frac{\pi}{2^4} \right] + \dots \text{to } \infty$$

$$= \frac{1}{4} + \frac{\pi}{2^6} + \frac{\pi}{2^8} + \frac{\pi}{2^{10}} + \dots \text{to } \infty$$

$$= \frac{1}{4} + \frac{\pi}{32} \left[\frac{4}{4-1} \right] = \frac{\pi}{24} + \frac{1}{4} = \frac{\pi+6}{24} \approx \frac{1}{\pi}$$

(B) $\tan^{-1} \left(\frac{1}{3+3r+r^2} \right) = \tan^{-1} \frac{r+2-r-1}{1+(r+1)(r+2)}$

$$= \sum_{r=1}^{\infty} [\tan^{-1}(r+2) - \tan^{-1}(r+1)]$$

$$= \tan^{-1}3 - \tan^{-1}2 + \tan^{-1}4 - \tan^{-1}\sqrt{3} + \dots \tan^{-1}\infty$$

$$= \frac{\pi}{2} - \tan^{-1}2$$

(C) For $\tan^{-1} \left(\frac{1}{3} \right) + \tan^{-1} \left(\frac{1}{7} \right) + \tan^{-1} \left(\frac{1}{13} \right) + \dots$ upto n term

$$= \tan^{-1} \left(\frac{2-1}{1+1.2} \right) + \tan^{-1} \left(\frac{3-2}{1+2.3} \right) + \tan^{-1} \left(\frac{4-3}{1+3.4} \right) + \dots + \tan^{-1} \left(\frac{(n+1)-n}{1+n(n+1)} \right)$$

$$= (\tan^{-1}2 - \tan^{-1}1) + (\tan^{-1}3 - \tan^{-1}2) + (\tan^{-1}4 - \tan^{-1}3) + \dots + (\tan^{-1}(n+1) - \tan^{-1}(n))$$

$$= \tan^{-1}(n+1) - \tan^{-1}1$$

Thus, for $\tan^{-1} \left(\frac{1}{3} \right) + \tan^{-1} \left(\frac{1}{7} \right) + \tan^{-1} \left(\frac{1}{13} \right) + \dots$ upto ∞

$$= \lim_{n \rightarrow \infty} (\tan^{-1}(n+1) - \tan^{-1}1)$$

$$= \frac{\pi}{2} - \frac{\pi}{4}$$

$$= \frac{\pi}{4}$$

(D) $\tan^{-1} \frac{1}{2 \cdot 1^2} + \tan^{-1} \frac{1}{2 \cdot 2^2} + \dots$ to ∞

$$T_n = \tan^{-1} \frac{1}{2n^2} = \tan^{-1} \left(\frac{2n+1-(2n-1)}{1+(2n-1)(2n+1)} \right) = \tan^{-1}(2n+1) - \tan^{-1}(2n-1)$$

7. Match the equations with the number of real solutions

Column I

Column II

(A) $\tan\left(\frac{\pi}{4} + \frac{1}{2}\cos^{-1}x\right) + \tan\left(\frac{\pi}{4} - \frac{1}{2}\cos^{-1}x\right) = 1$

(p) 0

(B) $\tan^{-1}\left(\frac{1}{2x+1}\right) + \tan^{-1}\left(\frac{1}{4x+1}\right) = \tan^{-1}\left(\frac{2}{x^2}\right)$

(q) 2

(C) $\tan^{-1}\left(x + \frac{2}{x}\right) - \tan^{-1}\left(\frac{4}{x}\right) = \tan^{-1}\left(x - \frac{2}{x}\right)$

(r) 3

(D) $\tan^{-1}(1-x) + \tan^{-1}(1+x) = \tan^{-1}(2x)$

(s) 1

Sol. Answer A(p), B(r), C(q), D(s)

8. Match the following

Column I

Column II

(A) The value of $\tan\left\{2\tan^{-1}\left(\frac{1}{5}\right) - \frac{\pi}{4}\right\}$

(p) $\frac{7}{17}$

is not equal to

(B) The value of $\tan\left[\cos^{-1}\left(\frac{4}{5}\right) + \tan^{-1}\left(\frac{2}{3}\right)\right]$

(q) $\frac{17}{6}$

is not equal to

(C) The number of solution of

(r) $\frac{1}{2}$

$$\sin^{-1}\left(\frac{2x}{1+x^2}\right) + \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right) = \pi \quad \text{in} \quad \left(\frac{1}{2}, \frac{3}{2}\right)$$

is not equal to

(D) The value of x for which

(s) $-\frac{1}{2}$

$\sin(\cot^{-1}(1+x)) = \cos(\tan^{-1}x)$ is not equal to

(t) 1

Sol. Answer A(q, r, s, t), B(p, r, s, t), C(p, q, r, s, t), D(p, q, r, t)

(A) We have,

$$\begin{aligned} & \tan \left\{ 2 \tan^{-1} \left(\frac{1}{5} \right) - \frac{\pi}{4} \right\} \\ &= \frac{\tan 2 \left(\tan^{-1} \frac{1}{5} \right) - \tan \frac{\pi}{4}}{1 + \tan \left(2 \tan^{-1} \frac{1}{5} \right) \tan \frac{\pi}{4}} \\ &= \frac{\frac{5}{12} - 1}{1 + \frac{5}{12}} \\ &= -\frac{7}{17} \end{aligned}$$

(B) We have,

$$\begin{aligned} & \tan \left[\cos^{-1} \left(\frac{4}{5} \right) + \tan^{-1} \left(\frac{2}{3} \right) \right] \\ &= \tan \left[\tan^{-1} \frac{3}{4} + \tan^{-1} \frac{2}{3} \right] \\ &= \frac{\frac{3}{4} + \frac{2}{3}}{1 - \frac{3}{4} \cdot \frac{2}{3}} \\ &= \frac{17}{6} \end{aligned}$$

(C) We have,

$$\text{If } x \in \left(\frac{1}{2}, 1 \right] \text{ then } x = 1$$

$$\text{If } x \in \left(1, \frac{3}{2} \right] \text{ then infinite solution exist.}$$

(D) We have,

$$\sin(\cot^{-1}(1+x)) = \cos(\tan^{-1} x)$$

$$\Rightarrow \frac{1}{\sqrt{1+(1+x)^2}} = \frac{1}{\sqrt{1+x^2}}$$

$$\Rightarrow 1+x^2 = 1+(1+x)^2$$

$$\Rightarrow 2x+1=0$$

$$\Rightarrow x = -\frac{1}{2}$$

9. Let (x, y) be such that $\tan^{-1}(ax) + \cot^{-1}\left(\frac{y}{x}\right) + \tan^{-1}\left(b\left(\frac{x}{y}\right)\right) = \frac{\pi}{2}$

Column I

- (A) If $a = 1, b = 1$, then (x, y)
 (B) If $a = 0, b = 0$, then (x, y)
 (C) If $a = 0, b = 1$, then (x, y)
 (D) If $a = 1, b = 0$, then (x, y)

Column II

- (p) Lies on line $y = x$
 (q) Lies on parabola $y^2 = x$
 (r) Lies on parabola $y = x^2$
 (s) Lies on curve $y^2 = x^2(1 + 2y)$
 (t) Does not exist

Sol. Answer A(s), B(t), C(q), D(r)

$$\tan^{-1} ax + \cot^{-1}\left(\frac{y}{x}\right) + \tan^{-1}\left(b\frac{x}{y}\right) = \frac{\pi}{2}$$

(A) $\tan^{-1} x + \cot^{-1}\left(\frac{y}{x}\right) + \tan^{-1}\left(\frac{x}{y}\right) = \frac{\pi}{2}$

$$2\tan^{-1}\left(\frac{x}{y}\right) = \frac{\pi}{2} - \tan^{-1} x = \cot^{-1} x = \tan^{-1} \frac{1}{x}$$

$$\tan \frac{\theta}{2} = \frac{x}{y}, \tan \theta = \frac{1}{x}$$

$$\frac{1}{x} = \frac{2\frac{x}{y}}{1 - \frac{x^2}{y^2}} \Rightarrow y^2 = x^2(1 + 2y)$$

(B) (x, y) doesn't exist.

(C) $a = 0, b = 1$

$$\cot^{-1}\left(\frac{y}{x}\right) + \tan^{-1}\left(\frac{x}{y}\right) = \frac{\pi}{2}$$

$$2\tan^{-1}\left(\frac{x}{y}\right) = \frac{\pi}{2}$$

$$\Rightarrow y = x$$

(D) $a = 1, b = 0$

$$\tan^{-1} x + \cot^{-1}\left(\frac{y}{x}\right) = \frac{\pi}{2}$$

$$\tan^{-1} x = \tan^{-1}\left(\frac{y}{x}\right)$$

$$x^2 = y$$

10. Let (x, y) be such that $\sin^{-1}(ax) + \cos^{-1}(y) + \cos^{-1}(bxy) = \frac{\pi}{2}$. Match the statements in Column I with statements in Column II. [IIT-JEE 2007]

Column I

- (A) If $a = 1$ and $b = 0$, then (x, y)
 (B) If $a = 1$ and $b = 1$, then (x, y)
 (C) If $a = 1$ and $b = 2$, then (x, y)
 (D) If $a = 2$ and $b = 2$, then (x, y)

Column II

- (p) Lies on the circle $x^2 + y^2 = 1$
 (q) Lies on $(x^2 - 1)(y^2 - 1) = 0$
 (r) Lies on $y = x$
 (s) Lies on $(4x^2 - 1)(y^2 - 1) = 0$

Sol. Answer A(p), B(q), C(p), D(s)

11. Match Column I with Column II. [JEE(Advanced)-2013]

Column I

(A) $\left\{ \frac{1}{y^2} \left(\frac{\cos(\tan^{-1} y) + y \sin(\tan^{-1} y)}{\cot(\sin^{-1} y) + \tan(\sin^{-1} y)} \right)^2 + y^4 \right\}^{1/2}$

takes value

- (B) If $\cos x + \cos y + \cos z = 0 = \sin x + \sin y + \sin z$ then

possible value of $\cos \frac{x-y}{2}$ is

- (C) If $\cos\left(\frac{\pi}{4} - x\right) \cos 2x + \sin x \sin 2x \sec x =$

$\cos x \sin 2x \sec x + \cos\left(\frac{\pi}{4} + x\right) \cos 2x$ then

possible value of $\sec x$ is

- (D) If $\cot(\sin^{-1} \sqrt{1-x^2}) = \sin(\tan^{-1}(x\sqrt{6}))$, $x \neq 0$,

then possible value of x is

Column II

(p) $\frac{1}{2}\sqrt{\frac{5}{3}}$

(q) $\sqrt{2}$

(r) $\frac{1}{2}$

(s) 1

Sol. Answer A(s), B(r), C(q), D(p)

(A) $\left\{ \frac{1}{y^2} \left(\frac{\cos(\tan^{-1} y) + y \sin(\tan^{-1} y)}{\cot(\sin^{-1} y) + \tan(\sin^{-1} y)} \right)^2 + y^4 \right\}^{1/2}$

$$= \left\{ \frac{1}{y^2} \left(\frac{\frac{1}{\sqrt{1+y^2}} + \frac{y^2}{\sqrt{1+y^2}}}{\frac{\sqrt{1-y^2}}{y} + \frac{y}{\sqrt{1-y^2}}} \right)^2 + y^4 \right\}^{1/2}$$

$$= \left\{ \frac{1}{y^2} \left(\frac{y\sqrt{1+y^2}\sqrt{1-y^2}}{1} \right)^2 + y^4 \right\}^{1/2}$$

$$= \{1 - y^4 + y^4\}^{1/2}$$

$$= 1$$

(B) If $\cos x + \cos y + \cos z = \sin x + \sin y + \sin z = 0$

then possible value of $x - y$ is $\pm \frac{2\pi}{3}$

$$\text{i.e., } \cos\left(\frac{x-y}{2}\right) = \frac{1}{2}$$

(C) Given equation can be written is

$$\cos 2x \left\{ \cos\left(\frac{\pi}{4} - x\right) - \cos\left(\frac{\pi}{4} + x\right) \right\} = \sin 2x(1 - \tan x)$$

$$\Rightarrow \cos 2x \cdot 2 \sin \frac{\pi}{4} \cdot \sin x = \sin 2x(1 - \tan x)$$

$$\Rightarrow \frac{1}{\sqrt{2}} \cos 2x = \cos x - \sin x$$

$$\Rightarrow \frac{1}{\sqrt{2}} (\cos x + \sin x) = 1$$

$$\Rightarrow x = \frac{\pi}{4}$$

$$\text{So, } \sec x = \sqrt{2}$$

(D) Given equation is

$$\frac{x}{\sqrt{1-x^2}} = \frac{\sqrt{6}x}{\sqrt{1+6x^2}}$$

$$\text{either } x = 0$$

$$\text{or } 1 + 6x^2 = 6 - 6x^2$$

$$\Rightarrow x = \pm \frac{1}{2} \sqrt{\frac{5}{3}}$$

SECTION - F

Integer Answer Type Questions

1. If $f(x) = \sin^{-1}x$ and $\lim_{x \rightarrow 1/2+} f(3x - 4x^3) = a - 3 \lim_{x \rightarrow 1/2+} f(x)$, then $[a]$ is equal to {where $[.]$ denotes G.I.F.}

Sol. Answer (3)

$$\sin^{-1}(3x - 4x^3) = \pi - 3\sin^{-1}x \text{ if } \frac{1}{2} < x < 1$$

$$\therefore \lim_{x \rightarrow 1/2+} f(3x - 4x^3) = \lim_{x \rightarrow 1/2+} (\pi - 3\sin^{-1}x) = \pi - 3 \lim_{x \rightarrow 1/2+} \sin^{-1}x$$

$$\therefore a = \pi$$

$$\therefore [a] = 3$$

2. If $\cos^{-1}(4x^3 - 3x) = a + b \cos^{-1}x$ for $-1 < x < -\frac{1}{2}$, then $[a + b + 4]$ is equal to {where $[.]$ denotes G.I.F.}

Sol. Answer (0)

$$\cos^{-1}(4x^3 - 3x) = \cos^{-1}(\cos 3) = 3\theta - 2\pi = -2\pi + 3 \cos^{-1}x$$

$$\therefore [a + b + 4] = [-2\pi + 3 + 4] = 0$$

3. If $f(x) = \cos^{-1}(4x^3 - 3x)$ and $\lim_{x \rightarrow 1/2+} f'(x) = a$ and $\lim_{x \rightarrow 1/2-} f'(x) = b$, then $a + b + 3$ is equal to

Sol. Answer (3)

$$f(x) = \cos^{-1}(4x^3 - 3x) = \cos^{-1}(\cos 3\theta)$$

$$= \begin{cases} 3\theta & \text{if } 0 < \theta < \frac{\pi}{3} \\ 2\pi - 3\theta & \text{if } \frac{\pi}{3} < \theta < \frac{\pi}{2} \end{cases}$$

$$= \begin{cases} 3\cos^{-1}x & ; \frac{1}{2} < x < 1 \\ 2\pi - 3\cos^{-1}x & ; 0 < x < \frac{1}{2} \end{cases}$$

$$\therefore f'(x) = \begin{cases} -\frac{3}{\sqrt{1-x^2}} & ; \frac{1}{2} < x < 1 \\ \frac{3}{\sqrt{1-x^2}} & ; 0 < x < \frac{1}{2} \end{cases}$$

$$a = \lim_{x \rightarrow 1/2+} f'(x) = -2\sqrt{3}$$

$$b = \lim_{x \rightarrow 1/2-} f'(x) = 2\sqrt{3}$$

$$\therefore a + b + 3 = 3$$

4. If $\tan^{-1}(2\cos^2\theta + 4\cos\theta + 3) + \cot^{-1}(6^{\csc^2\phi} + 3) = \frac{\pi}{2}$ has solution for some θ and ϕ then the value of $\cos\theta + \csc^2\phi$ is equal to_____.

Sol. Answer (2)

The equation holds if

$$2\cos^2\theta + 4\cos\theta + 3 = 6^{\csc^2\phi} + 3$$

Now, LHS

$$2(\cos\theta + 1)^2 + 1 \leq 9$$

and RHS

$$6^{\csc^2\phi} + 3 \geq 9 \quad \therefore \csc^2\phi \geq 1$$

$$\text{Hence, } \cos\theta = 1 \text{ and } \csc^2\phi = 1$$

$$\cos\theta + \csc^2\phi = 2$$

5. If $n \in \mathbb{N}$ and the set of equations, $(\sin^{-1}y)^2 + (\cos^{-1}x) = \frac{n\pi^2}{4}$ and $(\sin^{-1}y)^2 - (\cos^{-1}x) = \frac{\pi^2}{16}$ is consistent, then n can be equal to_____.

Sol. Answer (1)

$$\text{Adding we get, } (\sin^{-1}y)^2 = \left(\frac{4n+1}{32}\right)\pi^2$$

$$\Rightarrow 0 \leq \frac{4n+1}{32}\pi^2 \leq \frac{\pi^2}{4} \Rightarrow -\frac{1}{4} \leq n \leq \frac{7}{4}$$

$$\text{Also, } \cos^{-1}x = \frac{4n-1}{32}\pi^2$$

$$0 \leq \frac{4n-1}{32}\pi^2 \leq \pi \Rightarrow \frac{1}{4} \leq n \leq \frac{8}{\pi} + 1$$

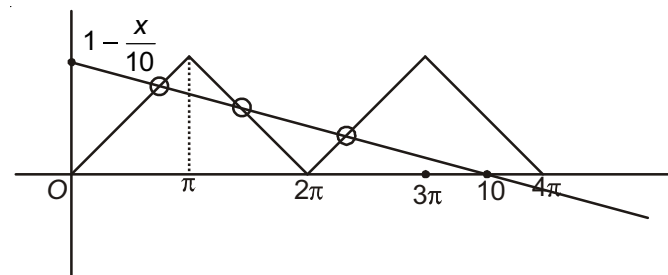
Hence $n = 1$.

6. Let $f: [0, 4\pi] \rightarrow [0, \pi]$ be defined by $f(x) = \cos^{-1}(\cos x)$. The number of points $x \in [0, 4\pi]$ satisfying the equation $f(x) = \frac{10-x}{10}$ is

[JEE(Advanced)2014]

Sol. Answer (3)

$$f: [0, 4\pi] \rightarrow [0, \pi], f(x) = \cos^{-1} \cos x$$



$$f(x) = 1 - \frac{x}{10}$$

7. The number of real solutions of the equation

$$\sin^{-1}\left(\sum_{i=1}^{\infty} x^{i+1} - x \sum_{i=1}^{\infty} \left(\frac{x}{2}\right)^i\right) = \frac{\pi}{2} - \cos^{-1}\left(\sum_{i=1}^{\infty} \left(-\frac{x}{2}\right)^i - \sum_{i=1}^{\infty} (-x)^i\right) \text{ lying in the interval } \left(-\frac{1}{2}, \frac{1}{2}\right) \text{ is } \underline{\hspace{2cm}}.$$

(Here, the inverse trigonometric functions $\sin^{-1}x$ and $\cos^{-1}x$ assume values in $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ and $[0, \pi]$, respectively.)

[JEE(Advanced)2018]

Sol. Answer (2)

$$\sum_{i=1}^{\infty} x^{i+1} - x \sum_{i=1}^{\infty} \left(\frac{x}{2}\right)^i = \sum_{i=1}^{\infty} \left(-\frac{x}{2}\right)^i - \sum_{i=1}^{\infty} (-x)^i$$

$$\Rightarrow \frac{x^2}{1-x} - x \left(\frac{x}{2-x}\right) = \frac{-x}{1+\frac{x}{2}} - \frac{(-x)}{1+x}$$

$$\Rightarrow x = 0 \text{ or } \frac{-x}{x-1} + \frac{x}{x-2} = \frac{-1}{x+2} + \frac{1}{x+1}$$

$$\Rightarrow x(x+2)(x+1) = (x-1)(x-2)$$

$$\Rightarrow x^3 + 2x^2 + 5x - 2 = 0$$

For $f'(x)$, $D < 0$ hence only one real root.

$$\text{Now } f(0) < 0, f\left(\frac{1}{2}\right) > 0$$

$$\text{One root in } \left(0, \frac{1}{2}\right)$$

Hence, total two roots.

SECTION - G

Multiple True-False Type Questions

1. STATEMENT-1 : Solution set of the equation $\sin^{-1}(x^2 + 4x + 3) + \cos^{-1}(x^2 + 6x + 8) = \frac{\pi}{2}$ is $\left\{-\frac{5}{2}\right\}$.

STATEMENT-2 : $\sin^{-1}(\cos(\sin^{-1}x)) + \cos^{-1}(\sin(\cos^{-1}x))$ is equal to π .

STATEMENT-3 : If x_1, x_2, x_3 all lie in $(0, 1)$ and $\sin^{-1}x_1, \sin^{-1}x_2, \sin^{-1}x_3$ are the roots of the equation

$$x^3 - 3x^2 + kx - 1 = 0, k \in \mathbb{R}, \text{ then the value of } \frac{1}{3}(\operatorname{cosec}1)(x_1 + x_2 + x_3) \text{ is } 1.$$

(1) T T F

(2) T F T

(3) T F F

(4) F T T

Sol. Answer (2)

$$\text{Statement-1 : } \sin^{-1}(x^2 + 4x + 3) = \sin^{-1}(x^2 + 6x + 8)$$

$$\Rightarrow x^2 + 6x + 8 = x^2 + 4x + 3$$

$$\Rightarrow x = -\frac{5}{2}$$

$$\therefore x^2 + 4x + 3 = (x+2)^2 - 1 \in [-1, 1] \text{ at } x = -\frac{5}{2}$$

$$\text{and } x^2 + 6x + 8 = (x + 3)^2 - 1 \in [-1, 1] \text{ at } x = -\frac{5}{2}$$

$$\therefore x = -\frac{5}{2}$$

Statement-2 : $-1 \leq \cos(\sin^{-1}x) \leq 1$ and $-1 \leq \sin(\cos^{-1}x) \leq 1$

$$\sin^{-1}\{\cos(\sin^{-1}x)\} + \cos^{-1}[\sin(\cos^{-1}x)] = \frac{\pi}{2}$$

2. STATEMENT-1 : If $\tan^2(\sin^{-1}x) > 1$ then $x \in \left(-1, -\frac{1}{\sqrt{2}}\right) \cup \left(\frac{1}{\sqrt{2}}, 1\right)$.

STATEMENT-2 : The number of positive integral solution of $\tan^{-1}\frac{1}{y} + \cot^{-1}\left(\frac{1}{x}\right) = \cot^{-1}\left(\frac{1}{3}\right)$, where $\frac{x}{y} < 1$, is 2.

STATEMENT-3 : If $\sin^{-1}x = -\cos^{-1}\sqrt{1-x^2}$ and $\sin^{-1}y = \cos^{-1}\sqrt{1-y^2}$, then the exact range of $(\tan^{-1}x + \tan^{-1}y)$ is $\left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$.

(1) T T F

(2) F T F

(3) T T T

(4) F F T

Sol. Answer (3)

Statement-1 : $\tan^2(\sin^{-1}x) > 1 \Rightarrow \sin^{-1}x > \frac{\pi}{4}$ or $\sin^{-1}x < -\frac{\pi}{4}$

$$\Rightarrow x \in \left[-1, -\frac{1}{\sqrt{2}}\right) \cup \left(\frac{1}{\sqrt{2}}, 1\right]$$

Statement -2 : $\tan^{-1}\frac{1}{y} + \tan^{-1}x = \cot^{-1}\left(\frac{1}{3}\right) = \tan^{-1}3$

$$\Rightarrow \tan^{-1}\left(\frac{x + \frac{1}{y}}{1 - \frac{x}{y}}\right) = \tan^{-1}3$$

$$\Rightarrow \frac{x + \frac{1}{y}}{1 - \frac{x}{y}} = 3$$

$$yx + 1 = 3(y - x)$$

$$yx + 1 = 3y - 3x$$

$$xy + 3x = 3y - 1$$

$$x(y + 3) = 3y - 1$$

$$x = \frac{3y - 1}{y + 3} = \frac{3y + 9 - 10}{y + 3} = 3 - \frac{10}{y + 3}$$

Both x and y are integers hence

$$y + 3 = 1, -1, 5, -5, 10, -10, 2, -2$$

$$\Rightarrow y = -2, -4, 2, -8, 7, -13, -1, -5$$

But for x to be positive $y + 3 = -1, -5, -10, 5, 10, 2, -2$

$$y = -4, -8, -13, 2, 7, -1, -5$$

But y is positive $\Rightarrow y = 2, 7 \Rightarrow x = 1, 2$

Hence only two positive values are possible.

Statement-3 : $-1 \leq x \leq 0, 0 \leq y \leq 1$

Let $k = \tan^{-1}x + \tan^{-1}y$

$$k_{\min} = \tan^{-1}(-1) + \tan^{-1}(0) = -\frac{\pi}{4}$$

$$k_{\max} = \tan^{-1}(0) + \tan^{-1}(1) = \frac{\pi}{4}$$

Hence range is $\left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$

SECTION - H

Aakash Challengers Questions

1. Among all the values of $n, n \in N$, that satisfy $\cot^{-1} \frac{n}{\pi} > \frac{\pi}{4}$, the largest value is

Sol. $\cot^{-1} \frac{n}{\pi} > \frac{\pi}{4}$

$$\Rightarrow \frac{n}{\pi} < \cot \frac{\pi}{4} = 1$$

$$\Rightarrow n < \pi$$

$\therefore$ The largest value of n is 3

2. Let $\lambda = \sec^2(\tan^{-1}2) + \operatorname{cosec}^2(\cot^{-1}3) + 2$, then the value of $17\lambda^2 + 7$ is

Sol. $\lambda = \sec^2(\tan^{-1}2) + \operatorname{cosec}^2(\cot^{-1}3) + 2$

$$= 1 + \tan^2(\tan^{-1}2) + 1 + \cot^2(\cot^{-1}3) + 2$$

$$= 4 + (2)^2 + (3)^2$$

$$= 17$$

$$\therefore 17\lambda^2 + 7 = 17^3 + 7 = 4920$$

3. The value of $\cot^{-1}(\cot(-10)) + \cos^{-1}(\cos 10)$ is given by $a + b\pi$, a, b being rational numbers, then $(b - a)$ equals

Sol. $3\pi < 10 < 4\pi$

$$\Rightarrow -\pi < 10 - 4\pi < 0$$

$$\Rightarrow 0 < 4\pi - 10 < \pi$$

$$\therefore \cos^{-1}(\cos 10) = 4\pi - 10$$

$$\text{Next, } -4\pi < -10 - 3\pi$$

$$\Rightarrow 0 < 4\pi - 10 < \pi$$

$$\cot^{-1}(\cot(-10)) = 4\pi - 10$$

$$\cot^{-1}(\cot(-10)) + \cos^{-1}(\cos 10) = 8\pi - 20$$

$$a = -20, b = 8$$

$$b - a = 28$$

4. The value of $13 + \frac{36}{\pi} \left\{ \cos^{-1}\left(\sqrt{\frac{2}{3}}\right) - \cos^{-1}\left(\frac{\sqrt{6}+1}{2\sqrt{3}}\right) \right\}$ is

Sol. $\frac{\sqrt{6}+1}{2\sqrt{3}} = \frac{1}{\sqrt{2}} + \frac{1}{2\sqrt{3}}$

$$= \sqrt{\frac{2}{3}} \cdot \frac{\sqrt{3}}{2} + \sqrt{1 - \frac{2}{3}} \sqrt{1 - \frac{3}{4}}$$

$$\therefore \cos^{-1}\left(\frac{\sqrt{6}+1}{2\sqrt{3}}\right) = \cos^{-1}\sqrt{\frac{2}{3}} - \cos^{-1}\frac{\sqrt{3}}{2}$$

$$\therefore 13 + \frac{36}{\pi} \left\{ \cos^{-1}\sqrt{\frac{2}{3}} - \cos^{-1}\left(\frac{\sqrt{6}+1}{2\sqrt{3}}\right) \right\}$$

$$= 13 + \frac{36}{\pi} \left\{ \cos^{-1}\sqrt{\frac{2}{3}} - \left(\cos^{-1}\sqrt{\frac{2}{3}} - \cos^{-1}\frac{\sqrt{3}}{2} \right) \right\}$$

$$= 13 + \frac{36}{\pi} \left\{ \cos^{-1}\frac{\sqrt{3}}{2} \right\}$$

$$= 13 + \frac{36}{\pi} \left(\frac{\pi}{6} \right)$$

$$= 13 + 6$$

$$= 19$$

5. The solution set of $\sin^{-1}(\sin 5) > x^2 - 4x$ can be expressed in the form $|x - \alpha| < \sqrt{a - b\pi}$, a, b being rational numbers, then the value of $(\alpha + a + b)^2$ is

Sol. $x^2 - 4x < \sin^{-1}(\sin 5) = 5 - 2\pi$

$$\Rightarrow x^2 - 4x + 4 < 9 - 2\pi$$

$$\Rightarrow (x - 2)^2 < 9 - 2\pi$$

$$\Rightarrow |x - 2| < \sqrt{9 - 2\pi}$$

$$\therefore \alpha = 2, a = 9, b = 2$$

$$\therefore (\alpha + a + b)^2 = 13^2 = 169$$

6. Let $\lambda = \tan\left(\frac{3\pi}{4} - \frac{1}{4}\sin^{-1}\left(-\frac{4}{5}\right)\right)$, then $(\lambda^2 - \lambda + 22)^2$ is

Sol. Let $\frac{1}{4}\sin^{-1}\left(\frac{4}{5}\right) = \theta$ $\theta \in \left(-\frac{\pi}{8}, \frac{\pi}{8}\right)$

$$\Rightarrow \sin^{-1}\left(\frac{4}{5}\right) = 4\theta$$

$$\Rightarrow \sin 4\theta = \frac{4}{5}$$

$$\Rightarrow \frac{2 \tan 2\theta}{1 + \tan^2 2\theta} = \frac{4}{5}$$

$$\Rightarrow 2 \tan^2 2\theta - 5 \tan 2\theta + 2 = 0$$

$$(2 \tan 2\theta - 1)(\tan 2\theta - 2) = 0$$

$$\tan 2\theta = \frac{1}{2}, 2$$

$$\Rightarrow \frac{2 \tan \theta}{1 - \tan^2 \theta} = \frac{1}{2} \text{ or } 2$$

$$\Rightarrow \tan^2 \theta + 4 \tan \theta - 1 = 0 \text{ or } \tan^2 \theta + \tan \theta - 1 = 0$$

$$\Rightarrow \tan \theta = -2 \pm \sqrt{5} \text{ or } \tan \theta = -1 \pm \sqrt{5}$$

$$\text{Since, } \theta \in \left(-\frac{\pi}{8}, \frac{\pi}{8}\right), \tan \theta \in (1 - \sqrt{2}, \sqrt{2} - 1)$$

$$\therefore \tan \theta = \sqrt{5} - 2$$

$$\therefore \lambda = \tan\left(\frac{3\pi}{4} - \frac{1}{4}\sin^{-1}\left(-\frac{4}{5}\right)\right)$$

$$= \tan\left(\frac{3\pi}{4} + \frac{1}{4}\sin^{-1}\frac{4}{5}\right)$$

$$= \tan\left(\frac{3\pi}{4} + \theta\right)$$

$$= \frac{-1 + \tan \theta}{1 + \tan \theta} \left(\because \tan \frac{3\pi}{4} = -1 \right)$$

$$= \frac{(\sqrt{5} - 2) - 1}{(\sqrt{5} - 2) + 1}$$

$$= \frac{\sqrt{5} - 3}{\sqrt{5} - 1} \times \frac{\sqrt{5} + 1}{\sqrt{5} + 1}$$

$$= \frac{1}{2} - \frac{\sqrt{5}}{2}$$

$$\therefore \lambda - \frac{1}{2} = -\frac{\sqrt{5}}{2}$$

Squaring,

$$\lambda^2 - \lambda + \frac{1}{4} = \frac{5}{4}$$

$$\Rightarrow \lambda^2 - \lambda = 1$$

$$\therefore (\lambda^2 - \lambda + 22)^2 = 23^2 = 529$$

7. If M and m are the greatest and least value of the function $f(x) = (\cos^{-1}x)^2 + (\sin^{-1}x)^2$, then the value of

$$\left(\frac{M+m}{m}\right)^3 \text{ is.....}$$

Sol. $f(x) = (\cos^{-1}x)^2 + (\sin^{-1}x)^2$

$$= (\cos^{-1}x + \sin^{-1}x)^2 - 2\cos^{-1}x \sin^{-1}x$$

$$= \frac{\pi^2}{4} - 2\left(\frac{\pi}{2} - \sin^{-1}x\right) \sin^{-1}x$$

$$= \frac{\pi^2}{4} - \pi(\sin^{-1}x) + 2(\sin^{-1}x)^2$$

$$= \frac{\pi^2}{4} + 2\left[(\sin^{-1}x)^2 - 2 \cdot \frac{\pi}{4}(\sin^{-1}x) + \frac{\pi^2}{16} - \frac{\pi^2}{16}\right]$$

$$= \frac{\pi^2}{8} + 2\left[\sin^{-1}x - \frac{\pi}{4}\right]^2$$

$$\therefore \text{Minimum of } f(x) = \frac{\pi^2}{8}, \text{ at } \sin^{-1}x = \frac{\pi}{4}$$

$$\text{and, Maximum of } f(x) = \frac{10\pi^2}{8}, \text{ at } \sin^{-1}x = \frac{-\pi}{2}$$

$$\therefore M = \frac{10\pi^2}{8} \text{ and } m = \frac{\pi^2}{8}$$

$$\frac{M}{m} = 10$$

$$\Rightarrow \left(\frac{M+m}{m}\right)^3 = (10+1)^3 = 11^3 = 1331$$

8. Let $k = \sin^{-1}\left\{\cot\left(\sin^{-1}\sqrt{\frac{2-\sqrt{3}}{4}} + \cos^{-1}\frac{\sqrt{12}}{4} + \sec^{-1}\sqrt{2}\right)\right\}$, then the value of $(k^4 - 4)^4$ is

Sol. $\cos^{-1}\frac{\sqrt{12}}{4} = \cos^{-1}\frac{\sqrt{3}}{2} = \frac{\pi}{6}, \sec^{-1}\sqrt{2} = \frac{\pi}{4}$

$$\text{Let } \sin^{-1} \sqrt{\frac{2-\sqrt{3}}{4}} = \theta$$

$$\begin{aligned} \text{Then } k &= \sin^{-1} \left(\cot \left(\frac{\pi}{6} + \frac{\pi}{4} + \theta \right) \right) \\ &= \sin^{-1} \left\{ \cot \left(\frac{5\pi}{12} + \theta \right) \right\} \\ &= \sin^{-1} \left\{ \tan \left(\frac{\pi}{2} - \frac{5\pi}{12} - \theta \right) \right\} \\ &= \sin^{-1} \tan \left(\frac{\pi}{12} - \theta \right) \end{aligned} \quad \dots(1)$$

$$\text{Now, } \tan \left(\frac{\pi}{12} - \theta \right) = \frac{\tan \left(\frac{\pi}{12} \right) - \tan \theta}{1 + \tan \left(\frac{\pi}{12} \right) \tan \theta} \quad \dots(2)$$

$$\begin{aligned} \tan \left(\frac{\pi}{12} \right) &= \tan \left(\frac{\pi}{3} - \frac{\pi}{4} \right) \\ &= \frac{\sqrt{3} - 1}{\sqrt{3} + 1} \\ &= 2 - \sqrt{3} \end{aligned}$$

$$\begin{aligned} \tan \theta &= \frac{\sin \theta}{\sqrt{1 - \sin^2 \theta}} \\ &= \frac{\sqrt{\frac{2-\sqrt{3}}{4}}}{\sqrt{1 - \left(\frac{2-\sqrt{3}}{4} \right)}} \\ &= \frac{\frac{1}{2} \sqrt{2-\sqrt{3}}}{\frac{1}{2} \sqrt{2+\sqrt{3}}} \\ &= \sqrt{\frac{(2-\sqrt{3})(2-\sqrt{3})}{(2+\sqrt{3})(2-\sqrt{3})}} \\ &= 2 - \sqrt{3} \end{aligned}$$

$$\therefore \tan \left(\frac{\pi}{12} - \theta \right) = 0$$

$$\therefore k = \sin^{-1}(0) = 0$$

$$\therefore (k^4 - 4)^4 = 4^4 = 256.$$

9. Let $f(n) = \sum_{r=1}^n \cot^{-1}(r^2 + r + 1)$. Also let $\lim_{n \rightarrow \infty} f(n) = \pi A$, then the value of $4^5 A + 4^3 + 4^2 + 4 + 1$ is

Sol. $f(n) = \sum_{r=1}^n \cot^{-1}(r^2 + r + 1)$

$$= \sum_{r=1}^n \tan^{-1}\left(\frac{1}{r^2 + r + 1}\right)$$

$$= \sum_{r=1}^n \{\tan^{-1}(r+1) - \tan^{-1}r\}$$

$$= \tan^{-1}(n+1) - \tan^{-1}(1)$$

$$\lim_{n \rightarrow \infty} f(n) = \lim_{n \rightarrow \infty} \{\tan^{-1}(n+1) - \tan^{-1}(1)\}$$

$$= \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4} = \pi A$$

$$\Rightarrow A = \frac{1}{4}$$

$$\therefore 4^5 A + 4^3 + 4^2 + 4 + 1 = 4^4 + 4^3 + 4^2 + 4 + 1 = 341$$

10. If $ax + b \sec(\tan^{-1}x) = c$ and $ay + b \sec(\tan^{-1}y) = c$, then $\frac{x+y}{1-xy}$ is equal to _____.

Sol. Let $\tan^{-1}x = \alpha$ and $\tan^{-1}y = \beta \Rightarrow \tan \alpha = x$ and $\tan \beta = y$.

The given system of equation is $a \tan \alpha + b \sec \alpha = c$ and $a \tan \beta + b \sec \beta = c$

$\therefore \alpha$ and β are roots of $a \tan \theta + b \sec \theta = c$

$$\Rightarrow b \sec^2 \theta = (c - a \tan \theta)^2$$

$$\Rightarrow (a^2 - b^2) \tan^2 \theta - 2ac \tan \theta + c^2 - b^2 = 0$$

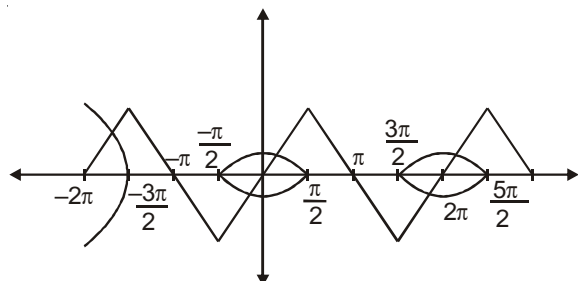
$$\tan \alpha + \tan \beta = \frac{2ac}{a^2 - b^2} \text{ and } \tan \alpha \tan \beta = \frac{c^2 - b^2}{a^2 - b^2}$$

$$\therefore x + y = \frac{2ac}{a^2 - b^2} \text{ and } xy = \frac{c^2 - b^2}{a^2 - b^2} \Rightarrow 1 - xy = \frac{a^2 - c^2}{a^2 - b^2}$$

$$\therefore \frac{x+y}{1-xy} = \frac{2ac}{a^2 - c^2}$$

11. The number of ordered pairs (x, y) satisfying $|y| = \cos x$ and $y = \sin^{-1}(\sin x)$, $x \in (-2\pi, 3\pi)$ is equal to _____.

Sol.



The graphs of $|y| = \cos x$ and $y = \sin^{-1}(\sin x)$ intersect at five points in $[-2\pi, 3\pi]$.

12. The total number of solutions of the equation $\sin^{-1}\left(\frac{3}{5}x\right) + \sin^{-1}\left(\frac{4}{5}x\right) = \sin^{-1}x$

(1) 1

(2) 0

(3) 2

(4) 3

Sol. Answer (4)

$$\sin^{-1}\left(\frac{3x}{5}\right) + \sin^{-1}\left(\frac{4x}{5}\right) = \sin^{-1}x$$

Taking sines of both hand sides, we get

$$\frac{3}{5}x\sqrt{1-\frac{16}{25}x^2} + \frac{4}{5}x\sqrt{1-\frac{9}{25}x^2} = x$$

$$x(3\sqrt{25-16x^2} + 4\sqrt{25-9x^2}) = 25x$$

One root is obvious $x_1 = 0$. It remains to solve the equation $3\sqrt{25-16x^2} + 4\sqrt{25-9x^2} = 25$

Put $x^2 = t$

Solving, we get

$$t^2 = 1 \Rightarrow t = \pm 1$$

Now, it can be verified that they satisfy the original equation.

13. The value(s) of θ satisfying the equation $\theta = \tan^{-1}(2\tan^2\theta) - \frac{1}{2}\left(\sin^{-1}\left(\frac{3\sin 2\theta}{5+4\cos 2\theta}\right)\right)$ is

(1) $n\pi + \frac{\pi}{4}$ (2) $n\pi + \tan^{-1}(-2)$ (3) $n\pi + \frac{\pi}{3}$ (4) $n\pi + \frac{\pi}{6}$

Sol. Answer (1, 2)

We have,

$$\frac{3\sin 2\theta}{5+4\cos 2\theta} = \frac{3(2\sin\theta\cos\theta)}{5+4(2\cos^2\theta-1)}$$

$$= \frac{6\sin\theta\cos\theta}{8\cos^2\theta+1}$$

$$= \frac{6\tan\theta}{8+\sec^2\theta} = \frac{6\tan\theta}{9+\tan^2\theta} \quad \{\text{let } \tan\theta = 3\tan\phi\}$$

$$= \frac{2\tan\phi}{9+\tan^2\phi} = \sin 2\phi$$

$$\Rightarrow \theta = \tan^{-1}(2\tan^2\theta) - \frac{1}{2}\sin^{-1}(\sin 2\phi)$$

$$= \tan^{-1}(2\tan^2\theta) - \phi$$

$$= \tan^{-1}(2\tan^2\theta) - \tan^{-1}\left(\frac{1}{3}\tan\theta\right)$$

$$= \tan^{-1}\left[\frac{2\tan^2\theta - \left(\frac{1}{3}\right)\tan\theta}{1 + 2\tan^2\theta\left(\frac{1}{3}\right)\tan\theta}\right]$$

$$\tan \theta = \frac{2 \tan^2 \theta - \left(\frac{1}{3}\right) \tan \theta}{1 + \left(\frac{2}{3}\right) \tan^3 \theta}$$

$$\Rightarrow \tan \theta = 0 \text{ or } \frac{2 \tan \theta - \frac{1}{3}}{1 + \frac{2}{3} \tan^3 \theta} = 1$$

$$\Rightarrow \tan \theta = 0 \Rightarrow \theta = n\pi \text{ where } n \text{ is any integer and } \frac{2 \tan \theta - \frac{1}{3}}{1 + \frac{2}{3} \tan^3 \theta} = 1 \Rightarrow (\tan \theta - 1)^2 (\tan \theta + 2) = 0$$

$$\Rightarrow \theta = n\pi + \tan^{-1}(-2)$$

14. $2 \tan^{-1} \left\{ \tan \frac{\alpha}{2} \tan \left(\frac{\pi}{4} - \frac{\beta}{2} \right) \right\}$ is equal to $\alpha, \beta \in \left(0, \frac{\pi}{2} \right)$

(1) $\tan^{-1} \left(\frac{\cos \alpha + \sin \beta}{\sin \alpha \cos \beta} \right)$ (2) $\tan^{-1} \left(\frac{\sin \alpha \cos \beta}{\cos \alpha + \sin \beta} \right)$ (3) Both (1) & (2) (4) $\tan^{-1} \left(\frac{\tan \alpha \tan \beta}{\tan \alpha + \tan \beta} \right)$

Sol. Answer (2)

$$\begin{aligned} \text{LHS} &= \tan^{-1} \frac{2 \tan \frac{\alpha}{2} \tan \left(\frac{\pi}{4} - \frac{\beta}{2} \right)}{1 - \tan^2 \frac{\alpha}{2} \tan^2 \left(\frac{\pi}{4} - \frac{\beta}{2} \right)} \\ &= \tan^{-1} \frac{2 \tan \frac{\alpha}{2} \left(\frac{1 - \tan \frac{\beta}{2}}{1 + \tan \frac{\beta}{2}} \right)}{1 - \tan^2 \frac{\alpha}{2} \left[\frac{1 - \tan \left(\frac{\beta}{2} \right)}{1 + \tan \left(\frac{\beta}{2} \right)} \right]^2} \\ &= \tan^{-1} \frac{2 \tan \frac{\alpha}{2} \left(1 - \tan^2 \frac{\beta}{2} \right)}{\left(1 + \tan \frac{\beta}{2} \right)^2 - \tan^2 \frac{\alpha}{2} \left(1 - \tan \frac{\beta}{2} \right)^2} \\ &= \tan^{-1} \frac{2 \tan \frac{\alpha}{2} \left(1 - \tan^2 \frac{\beta}{2} \right)}{\left(1 + \tan^2 \frac{\beta}{2} \right) \left(1 - \tan^2 \frac{\alpha}{2} \right) + 2 \tan \frac{\beta}{2} \left(1 + \tan^2 \frac{\alpha}{2} \right)} \\ &= \tan^{-1} \frac{2 \tan \left(\frac{\alpha}{2} \right) \left(1 - \tan^2 \frac{\beta}{2} \right)}{\left(1 + \tan^2 \frac{\alpha}{2} \right) \left(1 + \tan^2 \frac{\beta}{2} \right)} = \tan^{-1} \left(\frac{\sin \alpha \cos \beta}{\cos \alpha + \sin \beta} \right) \end{aligned}$$

15. If $x = \operatorname{cosec}(\tan^{-1}(\cos(\cot^{-1}(\sec(\sin^{-1}a))))))$ and $y = \sec(\cot^{-1}(\sin(\tan^{-1}(\operatorname{cosec}(\cos^{-1}a))))))$ then $x^2 + y^2$ is equal to
 (1) $3 - a^2$ (2) $6 - 2a^2$ (3) $3 - 2a^2$ (4) $2a^2$

Sol. Answer (2)

$$x = \operatorname{cosec}\left(\tan^{-1}\left(\cos\left(\cot^{-1}\left(\frac{1}{\sqrt{1-a^2}}\right)\right)\right)\right) = \operatorname{cosec}\left(\tan^{-1}\left(\frac{1}{\sqrt{2-a^2}}\right)\right)$$

$$x = \sqrt{3-a^2}$$

$$x^2 = 3 - a^2$$

$$\text{Now, } y = \sec(\cot^{-1}(\sin(\tan^{-1}(\operatorname{cosec}(\cos^{-1}a))))))$$

$$= \sec\left(\cot^{-1}\left(\sin\left(\tan^{-1}\left(\frac{1}{\sqrt{1-a^2}}\right)\right)\right)\right)$$

$$= \sec\left(\cot^{-1}\left(\frac{1}{\sqrt{2-a^2}}\right)\right)$$

$$= \sqrt{3-a^2}$$

$$y^2 = 3 - a^2$$

$$x^2 + y^2 = 6 - 2a^2$$

16. The total number of ordered pairs of (x, y) satisfying the equation $13 + 12[\tan^{-1}x] = 24[\ln x] + 8[e^x] + 6[\cos^{-1}y]$ is/are

- (1) 0 (2) 1
 (3) 2 (4) Infinitely many

Sol. Answer (1)

The given equation is

$$24[\ln x] + 8[e^x] + 6[\cos^{-1}y] = 13 + 12[\tan^{-1}x]$$

Now, clearly $x > 0$

$$0 < \tan^{-1}x < \frac{\pi}{2}$$

$\Rightarrow [\tan^{-1}x]$ may be 0 or 1

Case-I :

When $[\tan^{-1}x] = 0$

$$24[\ln x] + 8[e^x] + 6[\cos^{-1}y] = 13$$

We can see that LHS is a multiple of 2 while RHS is NOT.

So, no solution.

Case-II :

When $[\tan^{-1}x] = 1$

$$24[\ln x] + 8[e^x] + 6[\cos^{-1}y] = 25$$

Again LHS is a multiple of 2 while RHS is not.

So, again no solution so, this equation has no solution.

17. If $x_1 = \sqrt{3}$ and $x_{n+1} = \frac{x_n}{1 + \sqrt{1 + x_n^2}}, \forall n \in \mathbb{N}$, then $\lim_{n \rightarrow \infty} 2^n x_n$ is equal to

- (1) $\frac{3}{2\pi}$ (2) $\frac{2}{3\pi}$ (3) $\frac{2\pi}{3}$ (4) $\frac{3\pi}{2}$

Sol. Answer (3)

Let $x_n = \tan \theta_n$

$$\tan \theta_{n+1} = x_{n+1} = \frac{x_n}{1 + \sqrt{1 + x_n^2}} = \frac{\tan \theta_n}{1 + \sqrt{\tan^2 \theta_n + 1}} = \frac{\tan \theta_n}{1 + \sec \theta_n} = \frac{\sin \theta_n}{1 + \cos \theta_n} = \tan \frac{\theta_n}{2}$$

$$\theta_{n+1} = \frac{\theta_n}{2}$$

$$\Rightarrow \theta_n = \frac{\theta_{n-1}}{2}$$

$$\text{Now, } \theta_1 = \frac{\pi}{3}$$

$$\theta_n = \frac{\pi}{3 \cdot 2^{n-1}}$$

$$\Rightarrow x_n = \frac{\tan\left(\frac{2\pi}{3 \cdot 2^n}\right)}{\left(\frac{1}{2^n}\right)} = \frac{2\pi}{3}$$

18. Let α be the only real root of the equation $x^{2011} + x^{2010} + x^{2009} + \dots + x^3 + x^2 + x + a_0 = 0$, where a_0 is a positive number less than 1, then $\tan^{-1}(\alpha) + \tan^{-1}\left(\frac{1}{\alpha}\right)$ is equal to

- (1) 0 (2) $\frac{\pi}{2}$ (3) $-\frac{\pi}{2}$ (4) π

Sol. Answer (3)

The given equation is $x^{2011} + x^{2010} + x^{2009} + \dots + x^2 + x + a_0 = 0$

$$\det f(x) = x^{2011} + x^{2010} + x^{2009} + \dots + a_0$$

$$f(0) = a_0 < 1, \text{ positive}$$

$$f(-1) = a_0 - 1 < 0, \text{ negative}$$

So, the root of the equation lies between $(-1, 0)$. That implies the root of the equation is negative.

$$\text{Hence, } \tan^{-1}(\alpha) + \tan^{-1}\left(\frac{1}{\alpha}\right) = -\frac{\pi}{2}$$

